Least Loss of Passive Phase Shifters

Deep Bhattacharjee* [1, 2, *] Sanjeevan Singha Roy [3, †] and Ushashi Bhattacharya [4]

[1] *Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India*
[2] *Formerly, University of Calcutta (Heramba Chandra College), 23/49 Gariahat Road, Golpark, Kolkata 700029, West Bengal, India*
[3] *Formerly, Birla Institute of Technology (BIT) Mesra, India*
[4] *National Taiwan University, Taipei, Taiwan*

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A passive device that shifts the phase of every frequency in a band $[a, b]$ by the same angle θ must lose energy. We find the least loss. Without zeros the mean loss is at least $\theta\,(K(\kappa) - E(\kappa))/\kappa$ nepers, $\kappa = (b-a)/(b+a)$, and one explicit device has the least loss at every frequency at once. With zeros a device with one zero does the same; for a right angle its mean loss is $\frac{\pi}{2}(\sqrt{b} - \sqrt{a})/(\sqrt{b} + \sqrt{a})$. If a delay τ is added, the best device with zeros exists and is unique, and its loss falls exponentially in τ, at a rate between a/e and $\pi ab/(b-a)$. Without zeros the loss stays above a fixed floor.

1. INTRODUCTION

Does it cost energy to shift the phase of a wave? At one frequency it does not. A delay line of length τ multiplies a wave of angular frequency ω by $e^{i\omega\tau}$, and a suitable τ gives any phase without loss.[1] The question has content when the same shift is asked of every frequency in a band. A delay cannot do that, since its phase grows with frequency. A system that does it moves the oscillations of a pulse under a fixed envelope (Figure 2).

The question was raised in an earlier paper by one of us with S. N. Thakur [1, 2], where a 1° shift of a 5 MHz wave was given a delay of 555 ps and an energy loss $\Delta E = hf\Delta t$. The delay is right and the formula is not (Section 2): it has the units of an action, and a fixed phase shift changes neither the frequency nor the energy of a wave. The idea behind it, that a phase shift over a band has an energy price, is correct, and this paper finds the price exactly.

1.1. Main results

A passive system is a transfer function H, analytic and bounded by 1 in the upper half-plane $\mathbb{C}_+$, with real impulse response. A pure delay is $e^{i\omega\tau}$, the phase $\varphi(\omega)$ of H is a lag when it is positive, and $\alpha(\omega) = -\log|H(\omega)| \geq 0$ is the attenuation in nepers.[2] We call H *zero-free* when it has no zeros in $\mathbb{C}_+$; minimum-phase systems and Foster's lossless resonances [13] are of this kind.

Let H be zero-free and passive, continuous and nonzero near the band $[a, b]$, $0 < a < b$, with phase $\varphi(\omega) = \theta + \tau\omega$ on the band, $\theta \geq 0$, $\tau \in \mathbb{R}$. Then

$$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\mathrm{d}\omega \;\geq\; \theta\,\Lambda, \qquad \Lambda = \frac{K(\kappa) - E(\kappa)}{\kappa}, \tag{1.1}$$

where $\kappa = (b-a)/(b+a)$ and K, E are the complete elliptic integrals. An explicit system $H_\star$ attains the bound (Theorems 6.2 and 8.2).

The delay τ does not enter: a system may delay the envelope freely, and pays only for the offset θ between carrier and envelope. The constant satisfies

$$\begin{gathered}\frac{\pi}{4}\,\kappa \;<\; \Lambda \;\leq\; \frac{\pi}{4}\,\frac{\kappa}{1-\kappa^2}, \\ \Lambda \sim \tfrac{1}{2}\log\frac{4b}{a} - 1 \quad (a/b \to 0),\end{gathered} \tag{1.2}$$

so a shift of one degree costs at least 0.0415 dB on an octave and 0.143 dB on a decade (Table II), and on a narrow band about $0.0137\,\Delta f/f$ of the power (Corollary 6.7).

The bound holds frequency by frequency. Every zero-free system is $e^{i\tau z}H_\star$ plus terms, one for each part of its spectrum off the band, and each raises the attenuation at every point of the band (Theorem 9.2). So one explicit system has the least attenuation at every frequency at once, for a lag and for a lead with $\tau \leq 0$ (Theorems 9.4 and 9.6). A lossless resonance outside the band only adds loss (Remark 6.4).

Zeros lower the loss, and we find by how much. For $0 < \theta < \pi$ there is a passive system $H^\sharp_\theta$ with one zero, on the imaginary axis, such that every passive system with $\arg H \equiv \theta$ on the band has $\alpha \geq \alpha^\sharp{}_\theta$ at every frequency of the band (Theorem 11.1). For a right angle it is algebraic,

$$H^\sharp_{\pi/2}(z) = -\frac{z^2 + ab}{R(z) + i(a+b)z}, \qquad R(z)^2 = (z^2 - a^2)(z^2 - b^2),$$

its mean attenuation is $\frac{\pi}{2}(\sqrt{b} - \sqrt{a})/(\sqrt{b} + \sqrt{a})$, and no passive system with this phase passes more than the

* d.bhattacharjee@erl-forschung.de; * Corresponding author. Further addresses: itsdeep@live.com, egspl@proton.me (GitHub-related contact; GitHub: creelie).
† sanjeevan9905@gmail.com
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[1] The factor $e^{i\omega\tau}$ has period $2\pi/\omega$ in τ, so a delay of at most one period gives every phase; at 5 MHz one degree takes 555.6 ps (Section 2).

[2] One neper of amplitude is $20/\ln 10 \approx 8.686$ dB, and the power transmission is $|H|^2 = e^{-2\alpha}$.

fraction a/b of the power at $\omega = \sqrt{ab}$: on an octave the loss there is at least 3.01 dB.

The last part of the paper adds a delay to the phase. Without zeros the least mean loss for the phase $-s + \tau\omega$ is found in closed form for every slope τ (Theorem 12.2), and it never falls below a fixed positive floor, however long the delay (Proposition 12.5). With zeros allowed, the optimal system exists and is unique for every phase, and there is a simple test for it (Theorem 12.9). Its loss falls exponentially with the delay: the exponent lies between $-\pi ab/(b-a)$ and $-a/e$ (Theorem 12.11 and Corollary 12.12). We also treat several channels (Theorems 12.13 and 12.14) and several bands (Theorem 12.16).

1.2. Method

The proof rests on one identity. If $H = \pm e^{ih}$ with h a Herglotz function, then

$$\int_a^b \alpha \, \mathrm{d}\omega = \frac{1}{2}\int_a^b \varphi(\omega)\, p(\omega)\, \mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W \, \mathrm{d}\mu, \qquad (1.3)$$

where $S = [-b,-a] \cup [a,b]$, μ is the Herglotz measure of h,

$$p(\omega) = \frac{a^2 + b^2 - 2\omega^2}{\sqrt{(\omega^2 - a^2)(b^2 - \omega^2)}},$$

$$W = \frac{(\rho - 1)^2}{\rho}, \qquad \rho(z) = \sqrt{\frac{z^2 - a^2}{z^2 - b^2}},$$

and W is positive off the band (Figure 3). It follows from the Kramers–Kronig relations [8–10]. Since $\int_a^b \omega p \, \mathrm{d}\omega = 0$ a delay drops out, and since $\int_a^b p \, \mathrm{d}\omega > 0$ a positive constant phase forces loss. With zeros we pass to $\zeta = \log F$, $F = e^{-i\theta}(H - e^{i\theta})/(H - e^{-i\theta})$, where the zeros disappear and the phase condition puts ζ on a fixed curve over the band; a comparison argument in Hardy spaces then gives the pointwise bound. With a delay, the problem with zeros becomes convex once it is written in terms of the boundary values, and the exponential decay comes from a smoothed sign function of exponential type.

Bode's gain–phase relations [12], Fano's matching bound [14], Rozanov's absorber bound [16], the Herglotz bounds of Gustafsson, Sjöberg, Bernland and Luger [17, 18] and Miller's delay–bandwidth bound [19] are of the same family. We did not find a bound on the loss forced by a frequency-independent phase shift in this literature, nor in the work on broadband phase shifters, where constant phase *differences* between lossless channels are built [15]. Other work of the authors, on gravity and on geometry, is in [3–7].

1.3. Plan

Section 2 checks the equations of [1], and Section 3 separates a constant phase factor, a delay and a phase that changes in time. Sections 4 and 5 set up passive systems and the test function, Sections 6 and 7 prove the main theorem and evaluate Λ, Sections 8 and 9 give the extremal and pointwise results without zeros, and Sections 10 and 11 treat zeros. Section 12 adds a delay, several channels and several bands, Section 13 applies the results to waves, and Section 14 lists what is open. Figure 1 shows how the results depend on each other. The appendices derive the facts on elliptic integrals, the integrals of the weight and the extremal profiles, and describe the machine checks: the finite and algebraic steps are certified in Lean 4, and a script recomputes every number. Every statement is proved in the text.

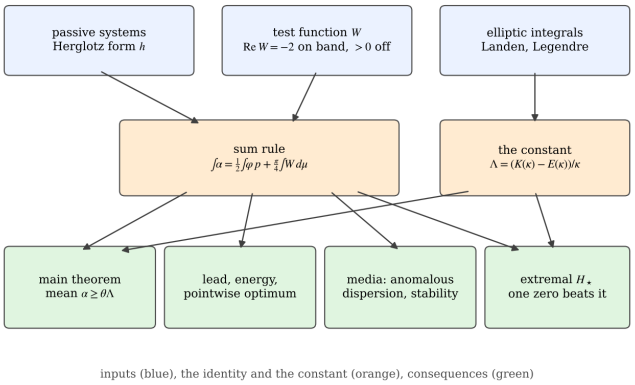


FIG. 1. How the results depend on each other. The sum rule (Theorem 6.1) is built from the test function W (Section 5); the constant Λ comes from the moments of the weight (Section 7); every later statement follows from these two or from the extremal system of Theorem 8.2.

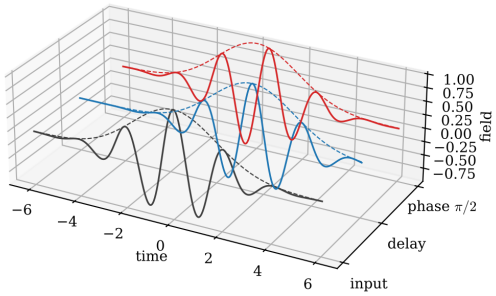


FIG. 2. A pulse (grey), the same pulse after a delay (blue), and after a constant phase shift of $\pi/2$ on its whole band (red). Dashed curves are the envelopes. The delay moves the envelope and the oscillations together; the constant phase shift moves the oscillations under a fixed envelope. Only the second costs energy in a passive system.

2. THE EQUATIONS OF THE EARLIER PAPER

We first go through the formulas of [1], since the rest of the paper replaces them. They fall into two groups: kinematic relations between phase, time and frequency, which are correct, and formulas for an energy loss ΔE, which are not.

2.1. Phase, time and frequency

For a sinusoid of frequency f, a delay Δt produces the phase

$$\Phi^\circ = 360^\circ f\,\Delta t, \tag{2.1}$$

and [1] rewrites this as $\Delta t = \Phi^\circ/(360^\circ f)$ and $f = \Phi^\circ/(360^\circ \Delta t)$. These are correct, with the understanding that the last one gives the frequency at which a *given* delay produces a *given* phase; it does not describe a change of frequency. The delay that produces one degree is

$$T(\text{deg}) = \frac{1}{360 f},$$

which at $f = 5\,\text{MHz}$ is 555.6 ps, as stated in [1]. Two of the phrasings there should be read with care. The equation "1° phase shift $= T/360$" equates an angle with a time; what is meant is that a delay of $T/360$ produces a phase of 1°, where $T = 1/f$ is the period. The name "time distortion" for Δt suggests something more than a delay; the quantity in (2.1) is just the delay. Finally, the statement that $T(\text{deg}) \propto 1/f$ is true of the delay needed for one degree, and it is the reason why a delay cannot produce the same phase at two frequencies.

2.2. The energy formulas

The paper [1] proposes three formulas for an "infinitesimal loss of wave energy",

$$\Delta E = hf\Delta t, \qquad \Delta E = \frac{2\pi h f_1}{360}\,T(\text{deg}), \qquad \Delta E = \frac{2\pi h}{360}\,\frac{T(\text{deg})}{\Delta t}, \tag{2.2}$$

with h Planck's constant, and evaluates the first one for $f = 4{,}986{,}111.11\,\text{Hz}$ and $\Delta t = 555\,\text{ps}$. Four things go wrong.

Units. Planck's constant is an action, measured in J s; hf is an energy and $hf\Delta t$ is again an action. The same holds for the other two formulas in (2.2), in which $T(\text{deg})/\Delta t$ is a pure number and $f_1 T(\text{deg})$ is the number 1/360. None of the three expressions is an energy.[3]

[3] If $T(\text{deg})$ is read as the angle $1^\circ = \pi/180$ rather than as a time, the second formula has units of energy, but it then reads $\Delta E = (\pi/180)\,hf_1\,(2\pi/360)$, a fixed multiple of hf_1 without any connection to the delay.

Arithmetic. The product evaluated in [1] is

$$(6.626\times10^{-34})(4.986\times10^{6})(555\times10^{-12}) = 1.834\times10^{-36},$$

in J s, not 1.848×10^{-27} J. The stated value is larger by a factor of about 10^9, and it would amount to 56% of the energy $E_1 = hf_1 = 3.313\times10^{-27}$ J of a quantum at 5 MHz, which can hardly be called infinitesimal.

A fixed phase shift does not change the frequency. The second example of [1] assigns to a 1° phase shift the new frequency $f_2 = f_1 - f_1/360 = 4{,}986{,}111.11\,\text{Hz}$. Multiplying a wave by a constant $e^{i\theta}$, or delaying it, leaves its frequency unchanged (Proposition 3.1). A frequency change of $f_1/360$ occurs when the phase *slips* by one degree in every period, that is, when the phase grows at the rate $\dot\theta = 2\pi f_1/360$ (Proposition 3.2). In that case the energy of a quantum changes by

$$E_1 - E_2 = \frac{hf_1}{360} = 9.20\times10^{-30}\ \text{J},$$

and this is the only energy difference of the size suggested by the example. The paper also gives $E_2 = hf_2 \approx 3.313\times10^{-27}$ J; the correct value is 3.304×10^{-27} J, and the difference from E_1 is the number just computed, not 1.848×10^{-27} J.

What hf measures. The energy hf is that of one quantum. The energy of a classical wave is fixed by its amplitude, which the formulas (2.2) do not contain. A loss caused by a passive element is a fraction of the incident energy, whether the wave is classical or quantised, and the quantity to compute is therefore the transmission $|H|^2$, or, per quantum, the probability of transmission.

2.3. What remains

Table I lists the corrections. The kinematic part of [1] stands. The energy part asks a good question in the wrong variables: the phase shift that has a price is the one applied uniformly over a band, the price is a fraction of the energy rather than a multiple of h, and its size is governed by the bandwidth. Sections 4–13 compute it.

3. PHASE, DELAY AND FREQUENCY

3.1. Conventions

A signal is a real function $s(t)$ of finite energy, with Fourier transform $\hat s(\omega) = \int_{\mathbb{R}} s(t)e^{i\omega t}\,\mathrm{d}t$, so that $\hat s(-\omega) = \overline{\hat s(\omega)}$. A linear time-invariant system acts by $\widehat{s_{\text{out}}} = H\hat s_{\text{in}}$. With this convention a delay by τ has $H(\omega) = e^{i\omega\tau}$, so a positive phase of H is a *lag*, and causality means that H extends analytically to the upper half-plane $\mathbb{C}_+ = \{\text{Im}\, z > 0\}$ (Section 4). The analytic signal of s is

$$\psi(t) = \frac{1}{\pi}\int_0^\infty \hat s(\omega)e^{-i\omega t}\,\mathrm{d}\omega, \qquad s = \text{Re}\,\psi.$$

TABLE I. The formulas of [1] and their status.

Formula in [1]	Status
$\Phi^\circ = 360^\circ f\Delta t$ and its inversions	correct
$T(\text{deg}) = 1/(360f)$, 555 ps at 5 MHz	correct
"1° phase shift = $T/360$"	angle equated with time; a delay $T/360$ gives 1°
$\Delta E = hf\Delta t$	an action, in J s; its value is 1.83×10^{-36}, not 1.848×10^{-27}
$\Delta E = (2\pi hf_1/360)T(\text{deg})$	an action; equals $2\pi h/360^2$
$\Delta E = (2\pi h/360)\,T(\text{deg})/\Delta t$	an action
$f_2 = f_1 - f_1/360$ after a 1° shift	only if the phase slips 1° per period
$E_2 \approx 3.313\times10^{-27}$ J	3.304×10^{-27} J; $E_1 - E_2 = 9.20\times10^{-30}$ J

For a narrowband pulse we write $\psi(t) = A(t)e^{-i(\omega_0 t+\phi)}$ with a slowly varying envelope $A \geq 0$, a carrier frequency ω_0 and a carrier–envelope phase ϕ. The phase ϕ locates the crests of the carrier relative to the peak of the envelope; it is the quantity controlled in frequency combs [23, 24].[4]

Three different operations go under the name of a phase shift. We treat them in turn.

3.2. A constant phase factor

The *ideal phase shifter* by θ multiplies every positive frequency by $e^{i\theta}$ and, to keep the signal real, every negative frequency by $e^{-i\theta}$:

$$H_\theta(\omega) = e^{i\theta\,\operatorname{sgn}\omega}, \qquad \psi_{\text{out}} = e^{i\theta}\psi. \tag{3.1}$$

It is not causal.[5]

Proposition 3.1. *The ideal phase shifter preserves the energy $\int s^2\,\mathrm{d}t$, the power spectrum $|\hat{s}|^2$ and hence every frequency present in the signal. For a single quantised mode of frequency ω the operator $U = e^{-i\theta N}$, N the number operator, commutes with the Hamiltonian $\hbar\omega(N+\frac{1}{2})$, so the probability distribution of the energy is unchanged. If $\sin\theta \neq 0$, the ideal phase shifter is not causal.*

Proof. The first statements follow from $|H_\theta| = 1$ and Parseval's identity, and from $[N, U] = 0$. For the last, $H_\theta = \cos\theta + i\sin\theta\operatorname{sgn}\omega$, and $i\operatorname{sgn}\omega$ is the transfer function of convolution with the Hilbert kernel $\pm 1/(\pi t)$, which does not vanish for $t < 0$. □

[4] In a mode-locked laser ϕ advances by a fixed amount from one pulse to the next; the rate of this slip is the carrier–envelope offset frequency of the comb, which is measured and locked in [23, 24].

[5] With our Fourier convention the multiplier $i\operatorname{sgn}\omega$ has the kernel $1/(\pi t)$, since $\text{p.v.}\int \operatorname{sgn}\omega\, e^{-i\omega t}\,\mathrm{d}\omega = -2i/t$. Hence $H_\theta = \cos\theta + i\sin\theta\operatorname{sgn}\omega$ has the impulse response $\cos\theta\,\delta(t) + \sin\theta/(\pi t)$, which does not vanish for $t < 0$ unless $\sin\theta = 0$.

So a phase shift, as such, neither changes the frequency nor costs energy. What it costs is causality, and the price of restoring causality in a passive system is the subject of the rest of the paper.

3.3. A delay

A delay τ gives $\psi_{\text{out}}(t) = \psi(t-\tau)$ and the phase $\omega\tau$ at frequency ω, which is the relation (2.1). It is causal and lossless, and it realises any prescribed phase at one frequency. On a band it realises a phase that is proportional to frequency, never a constant one.

3.4. A phase that changes in time

Let a device multiply the analytic signal by $e^{i\theta(t)}$.

Proposition 3.2. *If $\theta(t) = \theta_0 + \Omega t$, then $\widehat{\psi_{\text{out}}}(\omega) = e^{i\theta_0}\hat{\psi}(\omega+\Omega)$: the spectrum is moved down by Ω, and a component of frequency ω leaves with frequency $\omega - \Omega$ and energy $\hbar(\omega-\Omega)$ per quantum. If the device conserves the number of quanta, the energy of the signal changes by the fraction Ω/ω at each frequency, the difference being exchanged with the agent that drives θ.*

Proof. The Fourier transform of $e^{i\Omega t}\psi(t)$ is $\hat{\psi}(\omega+\Omega)$ in our convention. The rest is the relation $E = \hbar\omega$ for each quantum. □

The conservation of the number of quanta is what the Manley–Rowe relations [21] give for a lossless parametric modulator, and the linear phase ramp is the serrodyne frequency translator [22]. A phase that slips by one degree per period has $\Omega = \omega/360$; at 5 MHz this is a shift of 13,888.89 Hz and an energy of $hf/360 = 9.20\times10^{-30}$ J per quantum, the numbers of Section 2. This is the only reading under which a "1° phase shift" changes frequency. It is an active operation, not a passive one.

3.5. Constant phase on a band

The operation that keeps the envelope and moves the carrier is the following.

Proposition 3.3. *Let $0 < a < b$ and let H satisfy $H(-\omega) = \overline{H(\omega)}$ and*

$$H(\omega) = e^{-\alpha(\omega)}e^{i(\theta+\tau\omega)}, \qquad a \leq \omega \leq b. \tag{3.2}$$

If the spectrum of s lies in $[-b,-a]\cup[a,b]$ and $\alpha = 0$ on $[a,b]$, then $\psi_{\text{out}}(t) = e^{i\theta}\psi(t-\tau)$. In particular a pulse $A(t)e^{-i(\omega_0 t+\phi)}$ leaves as $A(t-\tau)e^{-i(\omega_0(t-\tau)+\phi-\theta)}$: the envelope is delayed by τ and the carrier–envelope phase is moved by $-\theta$. Conversely, if $\psi_{\text{out}} = e^{i\theta}\psi(\cdot-\tau)$ for every such signal, then (3.2) holds with $\alpha = 0$.

Proof. On positive frequencies $\widehat{\psi_{\mathrm{out}}} = H\hat{\psi} = e^{i\theta}e^{i\tau\omega}\hat{\psi}$, which is the transform of $e^{i\theta}\psi(t-\tau)$. The converse follows by testing with signals whose spectra are concentrated near each $\omega \in [a,b]$. □

The phase of H on the band is then $\varphi(\omega) = \theta + \tau\omega$. Its slope τ is the group delay; its intercept

$$\theta = \varphi(\omega) - \omega\varphi'(\omega) = \omega\big(\tau_{\mathrm{ph}} - \tau_{\mathrm{gr}}\big), \qquad \tau_{\mathrm{ph}} = \frac{\varphi}{\omega},\ \tau_{\mathrm{gr}} = \varphi',$$

is the carrier–envelope offset, ω times the excess of phase delay over group delay. A delay has $\theta = 0$. The main theorem says that a passive zero-free system can have $\theta > 0$ only by attenuating, on average over the band, at least $\theta\Lambda$ nepers.

4. PASSIVE SYSTEMS AND HERGLOTZ FUNCTIONS

4.1. Passive systems

Definition 4.1. A *passive system* is a function H analytic in $\mathbb{C}_+$ with $|H(z)| \le 1$ there and with the real symmetry $H(-\bar{z}) = \overline{H(z)}$. Its frequency response is the boundary function $H(\omega) = \lim_{\varepsilon\downarrow 0} H(\omega + i\varepsilon)$, which exists for almost every real ω.[6]

The definition encodes three physical requirements. A time-invariant system that never puts out more energy than it receives acts on signals of finite energy by a multiplier of norm at most one; causality makes the multiplier the boundary value of a bounded analytic function in $\mathbb{C}_+$; and a real impulse response gives the symmetry. That these requirements are equivalent to Definition 4.1 is classical [10, 11, 25]. The transmission coefficient of a passive two-port, the reflection coefficient of a passive one-port and the propagation factor of a passive medium (Section 13) are passive systems.

The zeros of a passive system that is not identically zero satisfy the Blaschke condition, and $H = BG$ where B is the Blaschke product of the zeros and G is passive and has no zeros in $\mathbb{C}_+$ [25, Ch. II]. Since the zeros of H come in pairs $\zeta, -\bar{\zeta}$, B can be chosen with the real symmetry.[7] We call H *zero-free* when B is constant. This includes every minimum-phase system, and also the singular inner functions, which are not minimum-phase in the usual sense: the factor $\exp(i\gamma/(\omega_0 - z))$, $\gamma > 0$, is the lossless resonance of a reactance at ω_0 in the sense of Foster [13].

[6] Fatou's theorem: a bounded analytic function in $\mathbb{C}_+$ has nontangential limits at almost every point of $\mathbb{R}$ [25, Ch. I].

[7] With $B_\zeta(z) = (z-\zeta)/(z-\bar{\zeta})$ one has $B_\zeta(-\bar{z}) = \overline{B_{-\bar{\zeta}}(z)}$, so the product $B_\zeta B_{-\bar{\zeta}}$ has the symmetry, and so does the Blaschke product over such pairs, with the usual unimodular convergence factors chosen in pairs.

4.2. The Herglotz function of a zero-free system

A *Herglotz function* is an analytic map h from $\mathbb{C}_+$ to the closed upper half-plane. Every such function has the representation [11, 18, 25]

$$h(z) = c + \beta z + \int_{\mathbb{R}} \Big(\frac{1}{t-z} - \frac{t}{1+t^2}\Big)\,\mathrm{d}\mu(t), \tag{4.1}$$

with $c \in \mathbb{R}$, $\beta \ge 0$ and a positive measure μ with $\int \mathrm{d}\mu/(1+t^2) < \infty$. The measure is recovered by the Stieltjes inversion formula,

$$\mu\big((x_1,x_2)\big) + \tfrac{1}{2}\mu(\{x_1,x_2\}) = \lim_{y\downarrow 0} \frac{1}{\pi}\int_{x_1}^{x_2} \operatorname{Im} h(x+iy)\,\mathrm{d}x, \tag{4.2}$$

and $\beta = \lim_{y\to\infty} h(iy)/(iy)$.

Lemma 4.2. *Let H be a zero-free passive system, $H \not\equiv 0$. There are a sign $\sigma \in \{1,-1\}$ and a unique Herglotz function h with*

$$H = \sigma e^{ih}, \qquad h(iy) \in i\mathbb{R}\ (y>0), \qquad h(-\bar{z}) = -\overline{h(z)}. \tag{4.3}$$

In the representation (4.1) of h one has $c = 0$ and μ is even, so that

$$h(z) = \beta z - \frac{\mu(\{0\})}{z} + \int_{(0,\infty)} \frac{2z}{t^2 - z^2}\,\mathrm{d}\mu(t). \tag{4.4}$$

Proof. On the imaginary axis $H(iy) = \overline{H(iy)}$ is real and, having no zeros, of constant sign σ. Since $\mathbb{C}_+$ is simply connected, σH has an analytic logarithm, which we fix to be real on $i\mathbb{R}_+$. Put $h = -i\log(\sigma H)$. Then $\operatorname{Im} h = -\log|H| \ge 0$, so h is a Herglotz function, and $h(iy) \in i\mathbb{R}$. The function $g(z) = -\overline{h(-\bar{z})}$ is analytic, satisfies $e^{ig} = \overline{\sigma H(-\bar{z})} = \sigma H = e^{ih}$, and agrees with h on $i\mathbb{R}_+$; hence $g = h$. In (4.1), $c = \operatorname{Re} h(i) = 0$; the symmetry and (4.2) show that μ is even, and pairing t with $-t$ in (4.1) gives (4.4). Uniqueness holds because two such logarithms differ by a constant in $2\pi\mathbb{Z}$ that vanishes on $i\mathbb{R}_+$. □

We call

$$\varphi(\omega) = \operatorname{Re} h(\omega + i0), \qquad \alpha(\omega) = \operatorname{Im} h(\omega + i0) = -\log|H(\omega)|$$

the *phase* and the *attenuation* of H. Then $\arg H(\omega) = \varphi(\omega) + \frac{1-\sigma}{2}\pi$ modulo 2π. The phase φ is the argument of σH continued from the imaginary axis, where it vanishes; when H is continuous and positive at $\omega = 0$ and nonzero on $[0,b]$, it is simply the argument of H unwrapped from $\varphi(0) = 0$. The representation (4.4) separates the three ingredients of a zero-free system: the delay βz, the lossless resonances, which are the atoms and the singular part of μ, and the losses, which are the absolutely continuous part of μ, with density α/π.

4.3. The band

Fix $0 < a < b$ and write $S = [-b, -a] \cup [a, b]$.

Definition 4.3. A zero-free passive system H is *regular on the band* if there is an open interval $I \supset [a, b]$, $I \subset (0, \infty)$, such that H extends continuously to $\mathbb{C}_+ \cup I$ and has no zeros on I.

Lemma 4.4. *If H is regular on the band, then h extends continuously to $\mathbb{C}_+ \cup I$, φ and α are continuous on I, and the restriction of μ to $I \cup (-I)$ is $\pi^{-1}\alpha(|t|)\,\mathrm{d}t$. In particular μ has no atoms at $\pm a, \pm b$.*

Proof. Near each point of I the function σH is continuous and nonzero, so its logarithm, normalised as in Lemma 4.2, is continuous up to I. The convergence in (4.2) is then uniform on compact subintervals of I, which gives $\mu = \pi^{-1}\alpha\,\mathrm{d}t$ on I; evenness gives the statement on $-I$. □

4.4. Lossless bands

The following is the form of Foster's reactance theorem [13] that we need.

Proposition 4.5. *Let $H = B\sigma e^{ih}$ be passive, with B a real-symmetric Blaschke product whose zeros stay away from I, and let σe^{ih} be regular on the band. If $|H| = 1$ on I, then the argument of H is real-analytic and nondecreasing on I, and strictly increasing unless H is constant.*

Proof. On I we have $\alpha = 0$, so by Lemma 4.4 μ vanishes on I and (4.4) continues analytically across I, with

$$\varphi'(x) = \beta + \frac{\mu(\{0\})}{x^2} + \int_{(0,\infty)\setminus I} \frac{2(t^2 + x^2)}{(t^2 - x^2)^2}\,\mathrm{d}\mu(t) \;\geq\; 0.$$

Each factor $(z - \zeta)/(z - \bar{\zeta})$ of B, $\operatorname{Im}\zeta > 0$, has argument with derivative $2\operatorname{Im}\zeta/\,|x - \zeta|^2 > 0$ on the real line, and the series for B converges with its derivatives on I. The sum is strictly positive unless $\beta = 0$, $\mu = 0$ and B is constant. □

Corollary 4.6. *Under the hypotheses of Proposition 4.5, a passive system that is lossless on I and has constant argument on $[a, b]$ is a constant, $H \equiv \pm 1$.*

So constant phase on a band is impossible without loss. The main theorem measures how much loss.

5. A TEST FUNCTION FOR THE BAND

5.1. The functions ρ and W

Let

$$\rho(z) = \left(\frac{z - a}{z - b}\right)^{1/2}\left(\frac{z + a}{z + b}\right)^{1/2}, \qquad z \in \mathbb{C}_+,$$

with principal square roots. The map $z \mapsto (z-a)/(z-b)$ is a real Möbius transformation of determinant $a - b < 0$ and sends $\mathbb{C}_+$ onto the lower half-plane; $z \mapsto (z+a)/(z+b)$ has determinant $b - a > 0$ and sends $\mathbb{C}_+$ onto itself. Neither meets the negative real axis, so ρ is analytic and zero-free in $\mathbb{C}_+$.[8] Put

$$W(z) = \rho(z) - 2 + \frac{1}{\rho(z)} = \frac{(\rho(z) - 1)^2}{\rho(z)}, \tag{5.1}$$

and, for $a < t < b$,

$$r(t) = \sqrt{(t^2 - a^2)(b^2 - t^2)},$$
$$p(t) = \frac{a^2 + b^2 - 2t^2}{r(t)} = \sqrt{\frac{b^2 - t^2}{t^2 - a^2}} - \sqrt{\frac{t^2 - a^2}{b^2 - t^2}}. \tag{5.2}$$

Lemma 5.1. *(i) $\rho(-\bar{z}) = \overline{\rho(z)}$, and $\rho(z) = 1 + \frac{b^2 - a^2}{2z^2} + O(z^{-4})$, so $W(z) = O(z^{-4})$ as $z \to \infty$ in $\overline{\mathbb{C}_+}$.*

(ii) ρ and W extend continuously to $\overline{\mathbb{C}_+}$ minus the four points $\pm a, \pm b$, near which $|W(z)| = O(|z \mp a|^{-1/2})$, $O(|z \mp b|^{-1/2})$.

(iii) On $\mathbb{R} \setminus S$, ρ is real and positive, and $W = (\rho - 1)^2/\rho$ is real and strictly positive.

(iv) For $a < t < b$, $\rho(t + i0) = -i\,s(t)$ with $s(t) = \sqrt{(t^2 - a^2)/(b^2 - t^2)}$, hence

$$W(t) = -2 + i\,p(t), \qquad W(-t) = -2 - i\,p(t).$$

Proof. (i) The symmetry holds for each factor. Since $\rho^2 = (z^2 - a^2)/(z^2 - b^2) = 1 + (b^2 - a^2)/(z^2 - b^2)$ and $\rho \to 1$, the expansion follows, and $W = (\rho - 1)^2/\rho$. (ii) Near b the factor $(z - a)/(z - b)$ behaves like a constant over $z - b$, so $\rho \sim c(z - b)^{-1/2}$ and $W \sim \rho$; near a, $\rho \sim c(z - a)^{1/2}$ and $W \sim 1/\rho$. The points $-a, -b$ follow by symmetry. (iii) For real $t \notin S$ both quotients are positive. (iv) For $a < t < b$ and $\varepsilon \downarrow 0$ the number $(t - a)/(t - b + i\varepsilon)$ has negative real part and negative imaginary part, so its principal square root tends to $-i\sqrt{(t - a)/(b - t)}$, while $\sqrt{(t + a)/(t + b)}$ is positive. Hence $\rho = -is$ and $W = -is - 2 + i/s = -2 + i(1/s - s) = -2 + ip$. The value at $-t$ follows from (i). □

Figure 3 shows $\operatorname{Re} W$: a plateau at -2 over the band, positive values elsewhere, and integrable spikes at the band edges.

5.2. The kernel identity

Lemma 5.2. *For $s > 0$, $s \neq a, b$,*

$$\frac{1}{\pi}\,\mathrm{p.v.}\int_a^b p(x)\,\frac{2x}{x^2 - s^2}\,\mathrm{d}x = \begin{cases} -2, & a < s < b, \\ W(s), & s \notin [a, b]. \end{cases} \tag{5.3}$$

[8] The principal square root is analytic and zero-free on $\mathbb{C} \setminus (-\infty, 0]$, and an open half-plane bounded by the real axis does not meet $(-\infty, 0]$.

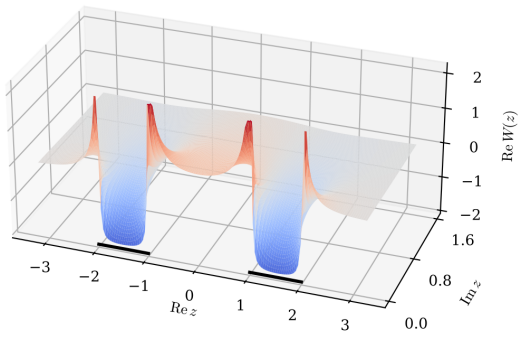


FIG. 3. The real part of the test function W over the upper half-plane for $a = 1$, $b = 2$ (values clipped at ± 2.2). On the real axis it equals -2 on the band (black segments) and is strictly positive elsewhere.

Proof. Apply Cauchy's theorem to $W(z)/(z-s)$ on the boundary of the half-disc of radius R in $\mathbb{C}_+$, indented by semicircles of radius ε above $s, \pm a, \pm b$. By Lemma 5.1(ii) the small semicircles at $\pm a, \pm b$ contribute $O(\varepsilon^{1/2})$; by Lemma 5.1(i) the large one contributes $O(R^{-4})$; the one at s contributes $-\pi i W(s)$ in the limit, W being continuous at s. Therefore

$$\mathrm{p.\,v.} \int_{\mathbb{R}} \frac{W(x)}{x-s}\,\mathrm{d}x = \pi i\, W(s).$$

Taking imaginary parts and using that $\operatorname{Im} W$ vanishes off S and equals $\pm p(|x|)$ on $\pm[a,b]$ (Lemma 5.1(iii),(iv)),

$$\begin{aligned}\pi \operatorname{Re} W(s) &= \mathrm{p.\,v.} \int_a^b p(x)\Big(\frac{1}{x-s} + \frac{1}{x+s}\Big)\,\mathrm{d}x \\ &= \mathrm{p.\,v.} \int_a^b p(x)\frac{2x}{x^2-s^2}\,\mathrm{d}x.\end{aligned}$$

By Lemma 5.1, $\operatorname{Re} W(s) = -2$ for $a < s < b$ and $\operatorname{Re} W(s) = W(s)$ otherwise. □

5.3. Moments of the weight

The weight p is positive on (a,c) and negative on (c,b), where $c^2 = (a^2+b^2)/2$. Under $\sigma = x^2$ it becomes antisymmetric, and this gives all the sign information we need.

Lemma 5.3. *Let $A = a^2$, $B = b^2$ and $\varpi(\sigma) = (A+B-2\sigma)/\sqrt{(\sigma-A)(B-\sigma)}$, so that $x\,p(x)\,\mathrm{d}x = \frac{1}{2}\varpi(x^2)\,\mathrm{d}(x^2)$.*

(i) $\varpi(A+B-\sigma) = -\varpi(\sigma)$, and $\int_a^b x\,p(x)\,\mathrm{d}x = 0$.

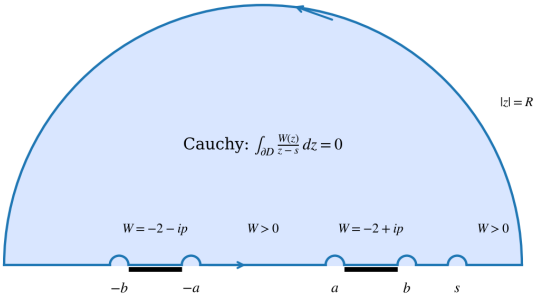


FIG. 4. The contour of the proof of Lemma 5.2: a half-disc of radius R in $\mathbb{C}_+$, indented above the branch points $\pm a, \pm b$ of W and above the pole s. The branch cuts of W are the two bands $\pm[a,b]$ (thick); the indentations at the branch points contribute nothing in the limit, the one at s contributes $-\pi i W(s)$.

(ii) If f is continuous and nonincreasing on $[A,B]$, then $\int_a^b x\,p(x) f(x^2)\,\mathrm{d}x \geq 0$, with equality only for constant f.

(iii) $J := \int_a^b p(x)\,\mathrm{d}x > 0$ and $\int_a^b \frac{p(x)}{x}\,\mathrm{d}x = \frac{\pi(b-a)^2}{2ab}$.

(iv) $\int_a^b x^{2j+1} p(x)\,\mathrm{d}x < 0$ for $j \geq 1$, and $\int_a^b x^3 p(x)\,\mathrm{d}x = -\frac{\pi}{8}(b^2-a^2)^2$.

Proof. (i) The antisymmetry is visible, and it gives $\int_A^B \varpi = 0$. (ii) With $\sigma_m = (A+B)/2$,

$$\int_a^b xpf(x^2)\,\mathrm{d}x = \frac{1}{2}\int_A^{\sigma_m} \varpi(\sigma)\big(f(\sigma) - f(A+B-\sigma)\big)\,\mathrm{d}\sigma,$$

and both factors are nonnegative for $\sigma < \sigma_m$; the integral vanishes only if $f(\sigma) = f(A+B-\sigma)$ throughout, which for a monotone f means f constant. (iii) Apply (ii) to $f(\sigma) = \sigma^{-1/2}$ and $f(\sigma) = \sigma^{-1}$. The value follows from the elementary integrals

$$\begin{aligned}\int_A^B \frac{\mathrm{d}\sigma}{\sqrt{(\sigma-A)(B-\sigma)}} &= \pi, \\ \int_A^B \frac{\mathrm{d}\sigma}{\sigma\sqrt{(\sigma-A)(B-\sigma)}} &= \frac{\pi}{\sqrt{AB}},\end{aligned}$$

which give $\frac{1}{2}\big((A+B)\pi/\sqrt{AB} - 2\pi\big) = \pi(b-a)^2/(2ab)$. (iv) Apply (ii) to $f(\sigma) = -\sigma^j$. For $j = 1$ use also $\int_A^B \sigma\,\mathrm{d}\sigma/\sqrt{\cdot} = \pi(A+B)/2$ and $\int_A^B \sigma^2\,\mathrm{d}\sigma/\sqrt{\cdot} = \pi(3A^2 + 2AB + 3B^2)/8$. □

The first moment vanishing is the reason why delays are free; the positivity of J is the reason why a constant phase lag is not.

6. THE SUM RULE AND THE MAIN THEOREM

6.1. The sum rule

Theorem 6.1. *Let $H = \sigma e^{ih}$ be a zero-free passive system that is regular on the band, with phase φ, attenuation α and Herglotz measure μ. Then*

$$\int_a^b \alpha(\omega)\,\mathrm{d}\omega = \frac{1}{2}\int_a^b \varphi(\omega)\,p(\omega)\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W(t)\,\mathrm{d}\mu(t), \tag{6.1}$$

and every term is finite. The last integral is nonnegative, and it vanishes if and only if $\mu(\mathbb{R}\setminus S)=0$.

The proof uses one fact about the Hilbert transform $\mathcal{H}f(x) = \pi^{-1}\,\mathrm{p.v.}\int f(t)\,\mathrm{d}t/(x-t)$ on the line, the antisymmetry

$$\int_{\mathbb{R}} f\,\mathcal{H}g\,\mathrm{d}x = -\int_{\mathbb{R}} (\mathcal{H}f)\,g\,\mathrm{d}x, \qquad f\in L^q,\ g\in L^{q'}, \tag{6.2}$$

valid for $1<q<\infty$ by M. Riesz's theorem [29, 30].[9]

Proof. Finiteness. φ is continuous on $[a,b]$ and p is integrable. On $I\setminus[a,b]$ the measure is $\pi^{-1}\alpha\,\mathrm{d}t$ with α bounded (Lemma 4.4), and W has integrable singularities at a,b. Away from $I\cup(-I)$, W is bounded and $O(t^{-4})$, while $\int \mathrm{d}\mu/(1+t^2)<\infty$.

Decomposition. Choose $a_1<a$ and $b_1>b$ with $[a_1,b_1]\subset I$, let $I_1=(a_1,b_1)$, and split $\mu=\mu_1+\mu_2$ with μ_1 the restriction of μ to $I_1\cup(-I_1)$. By (4.4),

$$h = \beta z + h_1 + h_2,$$
$$h_j(z) = -\frac{\mu_j(\{0\})}{z} + \int_{(0,\infty)} \frac{2z}{t^2-z^2}\,\mathrm{d}\mu_j(t),$$

and $\varphi = \beta x + \operatorname{Re} h_1 + \operatorname{Re} h_2$ on (a,b).

The delay. $\frac{1}{2}\int_a^b \beta x\,p(x)\,\mathrm{d}x = 0$ by Lemma 5.3(i).

The far part. The measure μ_2 has no mass in $I_1\cup(-I_1)$, so h_2 is continuous up to (a,b) and the kernel $2x/(t^2-x^2)$ is bounded by a constant times $(1+t^2)^{-1}$ for $x\in[a,b]$ and $t\notin I_1$. Fubini's theorem and Lemmas 5.2 and 5.3(iii) give

$$\frac{1}{2}\int_a^b \operatorname{Re} h_2\,p\,\mathrm{d}x = -\frac{\mu(\{0\})}{2}\int_a^b \frac{p}{x}\,\mathrm{d}x - \frac{\pi}{2}\int_{(0,\infty)\setminus I_1} W\,\mathrm{d}\mu$$
$$= -\frac{\pi}{4}\int_{\mathbb{R}\setminus(I_1\cup -I_1)} W\,\mathrm{d}\mu,$$

where we used $W(0) = (b-a)^2/(ab)$,[10] the evenness of W on $\mathbb{R}\setminus S$ and of μ.

The near part. Here $\mu_1 = \psi(t)\,\mathrm{d}t$ with $\psi = \pi^{-1}\alpha(|t|)$ on $I_1\cup(-I_1)$, a bounded function. For almost every $x\in(a,b)$,

$$\operatorname{Re} h_1(x) = \mathrm{p.v.}\int_{I_1} \psi(t)\frac{2x}{t^2-x^2}\,\mathrm{d}t$$
$$= \mathrm{p.v.}\int_{I_1} \psi(t)\Big(\frac{1}{t-x}-\frac{1}{t+x}\Big)\,\mathrm{d}t,$$

the boundary value of the real part of a Cauchy integral with bounded density [29, 31]. We claim that

$$\int_a^b p(x)\,\operatorname{Re} h_1(x)\,\mathrm{d}x$$
$$= \int_{I_1}\psi(t)\Big(\mathrm{p.v.}\int_a^b p(x)\frac{2x}{t^2-x^2}\,\mathrm{d}x\Big)\,\mathrm{d}t. \tag{6.3}$$

For the part $1/(t+x)$ of the kernel this is Fubini's theorem, the kernel being bounded. For the part $1/(t-x)$ it is (6.2) with $f = p\,\mathbf{1}_{[a,b]}\in L^{3/2}$ and $g=\psi\mathbf{1}_{I_1}\in L^3$; the function p lies in L^q for every $q<2$ because its singularities are of order $|x-a|^{-1/2}$ and $|x-b|^{-1/2}$. By Lemma 5.2 the inner integral in (6.3) equals 2π for $t\in(a,b)$ and $-\pi W(t)$ for $t\in I_1\setminus[a,b]$. Hence

$$\frac{1}{2}\int_a^b p\,\operatorname{Re} h_1\,\mathrm{d}x = \pi\int_a^b \psi\,\mathrm{d}t - \frac{\pi}{2}\int_{I_1\setminus[a,b]} W\,\mathrm{d}\mu$$
$$= \int_a^b \alpha\,\mathrm{d}t - \frac{\pi}{4}\int_{(I_1\cup -I_1)\setminus S} W\,\mathrm{d}\mu.$$

Adding the three parts gives (6.1). Since $W>0$ on $\mathbb{R}\setminus S$ and μ has no atoms at $\pm a,\pm b$, the last term vanishes exactly when $\mu(\mathbb{R}\setminus S)=0$. □

The identity (6.1) holds for every phase. Its content is that the band loss is a fixed linear functional of the band phase plus a nonnegative price for everything the system does outside the band: a lossless resonance at $\omega_0\notin[a,b]$ with weight γ in μ costs $\frac{\pi}{2}W(\omega_0)\gamma$,[11] a pole at the origin costs $\frac{\pi}{4}W(0)\mu(\{0\})$, and attenuation outside the band costs its W-weighted total.

6.2. Constant phase lag

Put

$$\Lambda = \frac{J}{2(b-a)}$$
$$= \frac{1}{2(b-a)}\int_a^b \frac{a^2+b^2-2\omega^2}{\sqrt{(\omega^2-a^2)(b^2-\omega^2)}}\,\mathrm{d}\omega > 0. \tag{6.4}$$

Section 7 shows $\Lambda = (K(\kappa)-E(\kappa))/\kappa$.

---

[9] Riesz's theorem is the boundedness of $\mathcal{H}$ on L^q, $1<q<\infty$. For smooth compactly supported f,g the antisymmetry is Fubini's theorem for the truncated kernel $\mathbf{1}_{|x-t|>\varepsilon}/(x-t)$, which is odd under $x\leftrightarrow t$; it extends to $L^q\times L^{q'}$ by density and Riesz's bound.

[10] $\rho(0)=\sqrt{a/b}\sqrt{a/b}=a/b$, so $W(0)=a/b-2+b/a=(b-a)^2/(ab)$. With Lemma 5.3(iii), $\int_a^b p/x\,\mathrm{d}x = \pi(b-a)^2/(2ab) = \frac{\pi}{2}W(0)$, which turns the atom at 0 into $-\frac{\pi}{4}W(0)\mu(\{0\})$.

[11] The measure is even, so the resonance puts mass γ at both $\pm\omega_0$, and $W(-\omega_0)=W(\omega_0)$ off the band; the two atoms give $\frac{\pi}{4}\cdot 2W(\omega_0)\gamma$.

Theorem 6.2 (Main theorem)**.** *Let H be a zero-free passive system, regular on the band $[a,b]$, whose phase on the band is*

$$\varphi(\omega) = \theta + \tau\omega, \qquad a < \omega < b,$$

with $\theta \geq 0$ and $\tau \in \mathbb{R}$. Then

$$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\mathrm{d}\omega \;\geq\; \theta\,\Lambda. \tag{6.5}$$

Equality holds if and only if $\mu(\mathbb{R}\setminus S) = 0$; for every $\theta \geq 0$ it is attained by the system $H_\star$ of Theorem 8.2.

Proof. By Lemma 5.3(i), $\frac{1}{2}\int_a^b (\theta+\tau\omega)p\,\mathrm{d}\omega = \frac{1}{2}\theta J = (b-a)\theta\Lambda$, and Theorem 6.1 gives the inequality and the case of equality. □

Three remarks make the statement concrete.

Remark 6.3 (Constant argument). If $\arg H \equiv \theta$ modulo 2π on the band, then $\varphi = \theta - \frac{1-\sigma}{2}\pi$ modulo 2π, and since φ is continuous it is a constant θ' with $\theta' \equiv \theta \pmod{\pi}$. The theorem applies with θ' when $\theta' > 0$. Which representative occurs is not decided by the band: it depends on how much phase the system has accumulated from the imaginary axis, for instance from zero frequency.

Remark 6.4 (Lossless resonances). The equality case shows that every lossless resonance outside the band, and every atom of μ at zero frequency, raises the loss strictly above the bound. Shaping the phase with reactances, the classical tool of all-pass and phase-equaliser design, therefore works against a constant phase lag. Figure 5 shows the price of a unit of measure off the band, the factor $\frac{\pi}{4}W(t)$ of (6.1).

Remark 6.5 (Passivity everywhere is needed). The constant-phase element $H(z) = (-iz/a)^{-\gamma}$, $0 < \gamma < 1$, familiar from dielectric relaxation [20], has phase $\theta = \gamma\pi/2$ at every positive frequency and attenuation $\alpha = \gamma\log(\omega/a) \geq 0$ on $[a,\infty)$. Over an octave its mean attenuation is $(2\theta/\pi)(2\log 2 - 1) = 0.2459\,\theta$, below the bound $0.2735\,\theta$. There is no contradiction: $|H| > 1$ for $\omega < a$, so H is not passive. The bound depends on passivity at all frequencies, not only on the band.

6.3. Constant phase lead

For a negative intercept the same identity, applied after an inversion of the frequency axis, gives a weighted bound.

Theorem 6.6. *Let H be as in Theorem 6.2 but with $\theta < 0$. Then*

$$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\frac{ab}{\omega^2}\,\mathrm{d}\omega \;\geq\; |\theta|\,\Lambda - \frac{\pi}{4}\,\tau\,(b-a). \tag{6.6}$$

For $\tau = 0$ equality holds if and only if $\beta = 0$ and μ vanishes on $\mathbb{R}\setminus(S\cup\{0\})$; it is attained by $H_\star(-ab/z)$, where $H_\star$ is the extremal system for $|\theta|$.

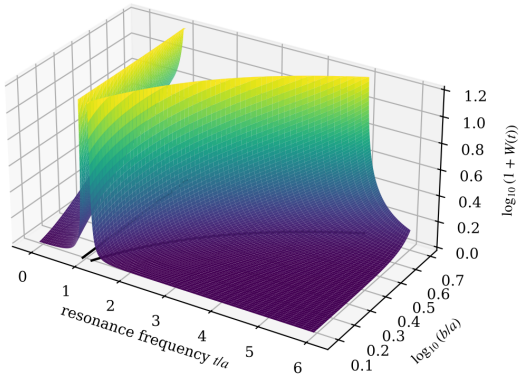


FIG. 5. The cost $\log_{10}(1+W(t))$ of a lossless resonance at frequency t, as a function of t/a and of the band ratio b/a. It vanishes only on the band (the gap between the two sheets), grows without bound at the band edges, and decays like t^{-4} at high frequency (Lemma 5.1(i)); at $t = 0$ it is $W(0) = a/b + b/a - 2 > 0$.

Proof. The map $z \mapsto -ab/z$ preserves $\mathbb{C}_+$, so $\tilde{h}(z) = h(-ab/z)$ is a Herglotz function with the symmetry of (4.3), and $\tilde{H} = \sigma e^{i\tilde{h}}$ is zero-free, passive and regular on the band, since $x \mapsto ab/x$ maps a neighbourhood of $[a,b]$ onto another. For $x \in (a,b)$ the point $-ab/(x+i0)$ lies above $-ab/x \in (-b,-a)$, so by the oddness of φ,

$$\tilde{\varphi}(x) = -\varphi(ab/x), \qquad \tilde{\alpha}(x) = \alpha(ab/x).$$

Substituting $x = ab/\omega$ and using $r(ab/\omega) = ab\,r(\omega)/\omega^2$,

$$\int_a^b f(ab/x)\,p(x)\,\mathrm{d}x = \int_a^b f(\omega)\,p^*(\omega)\,\mathrm{d}\omega,$$
$$p^*(\omega) = \frac{a^2+b^2-2a^2b^2/\omega^2}{r(\omega)},$$

for every integrable f. With $f = 1$ and $f(\omega) = \omega$ this gives $\int_a^b p^* = J$ and $\int_a^b \omega p^* = ab\int_a^b p/x = \pi(b-a)^2/2$ (Lemma 5.3(iii)). Theorem 6.1 for $\tilde{H}$ now reads

$$\int_a^b \alpha(\omega)\frac{ab}{\omega^2}\,\mathrm{d}\omega = -\frac{1}{2}\int_a^b (\theta+\tau\omega)p^*\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W\,\mathrm{d}\tilde{\mu},$$

which is (6.6) after division by $b-a$. The measure $\tilde{\mu}$ is the image of μ under $t \mapsto -ab/t$ with density ab/t^2 on $\mathbb{R}\setminus\{0\}$, plus an atom $ab\beta$ at 0 coming from the term βz; an atom of μ at 0 becomes a linear term of $\tilde{h}$ and costs nothing. This gives the case of equality, and $H_\star(-ab/z)$ has $\beta = 0$ and its measure on $S\cup\{0\}$. □

6.4. The worst frequency

Corollary 6.7. *Let H be zero-free, passive and regular on the band, with $\arg H \equiv \theta$ modulo 2π on $[a,b]$. Then*

$$\max_{[a,b]} \alpha \;\geq\; \operatorname{dist}(\theta, \pi\mathbb{Z})\,\Lambda,$$

so at some frequency of the band the transmitted power is at most $e^{-2\operatorname{dist}(\theta,\pi\mathbb{Z})\Lambda}$. If the phase θ' of Remark 6.3 is positive, also

$$\exp\Big(\frac{1}{b-a}\int_a^b \log|H(\omega)|^2\,\mathrm{d}\omega\Big) \leq e^{-2\theta'\Lambda},$$

that is, the geometric mean of the power transmission over the band is at most $e^{-2\theta'\Lambda}$.

Proof. By Remark 6.3 the phase is a constant $\theta' \equiv \theta \pmod{\pi}$. If $\theta' > 0$ then $\theta' \geq \operatorname{dist}(\theta, \pi\mathbb{Z})$ and Theorem 6.2 bounds the mean, hence the maximum, of α. If $\theta' < 0$ then $|\theta'| \geq \operatorname{dist}(\theta,\pi\mathbb{Z})$, and since $\int_a^b ab\,\omega^{-2}\,\mathrm{d}\omega = b-a$ the left side of (6.6) with $\tau = 0$ is at most $\max\alpha$. The second statement is (6.5) with $\log|H| = -\alpha$. □

The bound by the mean is not sharp for the maximum: Theorem 9.8 replaces Λ by the larger constant $A(m)$, which is attained.

Remark 6.8 (The lead extremal). For the system $H_\star(-ab/z)$ of Theorem 6.6, whose phase is $-|\theta|$ on the band, Theorem 6.1 applied directly gives the plain mean attenuation

$$\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega = |\theta|\,\frac{E(\kappa) - \kappa'^2 K(\kappa)}{\kappa}, \qquad \kappa'^2 = 1-\kappa^2,$$

because its measure off S is the atom $ab\,\tau_\star$ at 0, of cost $\frac{\pi}{4}W(0)ab\tau_\star$, with $\tau_\star$ from Theorem 8.2 and the evaluation of Section 7. For an octave this is $0.2656\,|\theta|$, slightly below $\Lambda = 0.2735$. It is the least plain mean among systems with constant phase lead (Corollary 9.7(iii)).

7. THE CONSTANT

We use the complete elliptic integrals of modulus $k \in [0,1)$,

$$K(k) = \int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}},$$
$$E(k) = \int_0^{\pi/2} \sqrt{1-k^2\sin^2\phi}\;\mathrm{d}\phi,$$

and the moduli attached to the band,

$$k = \sqrt{1-a^2/b^2}, \qquad k' = a/b, \qquad \kappa = \frac{b-a}{b+a} = \frac{1-k'}{1+k'}.$$

The facts on K and E that we use are collected in Appendix A.

Proposition 7.1. *With $I_0 = \int_a^b \mathrm{d}t/r(t)$ and $I_2 = \int_a^b t^2\,\mathrm{d}t/r(t)$,*

$$I_0 = \frac{K(k)}{b}, \qquad I_2 = b\,E(k),$$

and

$$\begin{aligned} J &= (a^2+b^2)I_0 - 2I_2 = b\big[(1+k'^2)K(k) - 2E(k)\big] \\ &= 2(a+b)\big[K(\kappa) - E(\kappa)\big]. \end{aligned} \tag{7.1}$$

Consequently

$$\Lambda = \frac{K(\kappa) - E(\kappa)}{\kappa}. \tag{7.2}$$

Proof. Substitute $t^2 = b^2 - (b^2-a^2)\sin^2\phi$, $0 \leq \phi \leq \pi/2$. Then $b^2 - t^2 = (b^2-a^2)\sin^2\phi$, $t^2 - a^2 = (b^2-a^2)\cos^2\phi$, $t\,\mathrm{d}t = -(b^2-a^2)\sin\phi\cos\phi\,\mathrm{d}\phi$, hence $\mathrm{d}t/r(t) = -\,\mathrm{d}\phi/t$ and $t = b\sqrt{1-k^2\sin^2\phi}$. This gives I_0 and I_2, and the second form of J. The Landen transformation (Appendix A) states

$$K(k) = (1+\kappa)K(\kappa), \qquad E(k) = (1+k')E(\kappa) - k'K(k).$$

Inserting these,[12] and using $b(1+\kappa) = 2b^2/(a+b)$ and $1+k' = (a+b)/b$,

$$\begin{aligned} J &= b(1+\kappa)(1+k')^2K(\kappa) - 2b(1+k')E(\kappa) \\ &= 2(a+b)\big[K(\kappa) - E(\kappa)\big]. \end{aligned}$$

Dividing by $2(b-a) = 2\kappa(a+b)$ gives (7.2). □

Proposition 7.2. *(i) With $c_n = \binom{2n}{n}4^{-n}$,*

$$\Lambda = \frac{\pi}{2}\sum_{n\geq 1}\frac{2n}{2n-1}\,c_n^2\,\kappa^{2n-1} = \frac{\pi}{4}\kappa\Big(1+\frac{3}{8}\kappa^2+\frac{15}{64}\kappa^4+\cdots\Big),$$

a series with positive coefficients;[13] *in particular Λ increases with κ, that is, with b/a.*

(ii) For all $0 < a < b$,

$$\frac{\pi}{4}\,\frac{b-a}{b+a} \;<\; \Lambda \;\leq\; \frac{\pi}{4}\,\frac{\kappa}{1-\kappa^2} = \frac{\pi}{16}\,\frac{b^2-a^2}{ab}.$$

(iii) As $a/b \to 0$, $\Lambda = \log\frac{4}{\kappa'} - 1 + o(1) = \frac{1}{2}\log\frac{4b}{a} - 1 + o(1)$, with $\kappa' = \sqrt{1-\kappa^2} = 2\sqrt{ab}/(a+b)$.

Proof. (i) From $K = \frac{\pi}{2}\sum c_n^2\kappa^{2n}$ and $E = \frac{\pi}{2}\sum c_n^2\kappa^{2n}/(1-2n)$ one gets $K - E = \frac{\pi}{2}\sum_{n\geq 1} c_n^2\kappa^{2n}\,2n/(2n-1)$. The coefficients for $n = 1,2,3$ are $\frac{1}{2}$, $\frac{3}{16}$ and $\frac{15}{128}$, which give the displayed expansion after division by κ. (ii) The lower

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[12] The coefficient of $K(\kappa)$ is $b(1+\kappa)\big[(1+k'^2)+2k'\big] = b(1+\kappa)(1+k')^2$, the term $2k'$ coming from $-2E(k)$.

[13] By (A.1), $K - E = \frac{\pi}{2}\sum_{n\geq 1} c_n^2\big(1 - \frac{1}{1-2n}\big)\kappa^{2n}$ and $1 - \frac{1}{1-2n} = \frac{2n}{2n-1}$; dividing by κ gives (i).

TABLE II. The constant Λ of (7.2), the least mean attenuation per radian of constant phase lag, for several band ratios. The last column is the least worst-frequency power loss for $\theta = 1^\circ$.

b/a	κ	Λ (Np/rad)	dB/deg	$1 - e^{-2\Lambda\pi/180}$
1.01	0.004975	0.003907	0.000592	1.36×10^{-4}
1.1	0.04762	0.03743	0.00567	1.31×10^{-3}
1.5	0.2	0.1595	0.0242	5.55×10^{-3}
2	0.3333	0.2735	0.0415	9.50×10^{-3}
3	0.5	0.4366	0.0662	1.51×10^{-2}
10	0.8182	0.9448	0.143	3.24×10^{-2}
100	0.9802	2.016	0.306	6.80×10^{-2}
1000	0.9980	3.150	0.478	1.04×10^{-1}

bound is the first term. For the upper bound, the ratio of consecutive coefficients $d_n = c_n^2\, 2n/(2n-1)$ is

$$\frac{d_{n+1}}{d_n} = \frac{(2n+1)(2n-1)}{(2n+2)\,2n} < 1,$$

so $d_n \le d_1 = \frac{1}{2}$ and $K - E \le \frac{\pi}{4}\sum_{n\ge1}\kappa^{2n} = \frac{\pi}{4}\kappa^2/(1-\kappa^2)$. Finally $\kappa/(1-\kappa^2) = (b^2-a^2)/(4ab)$. (iii) As $\kappa \to 1$, $K(\kappa) = \log(4/\kappa') + o(1)$ and $E(\kappa) \to 1$ (Appendix A); and $\log(4/\kappa') = \log\big(2(a+b)/\sqrt{ab}\big) = \frac{1}{2}\log(4b/a) + o(1)$. □

For a narrow band with centre $\bar\omega = (a+b)/2$ and width $\Delta\omega = b - a$, $\kappa = \Delta\omega/(2\bar\omega)$, so

$$\Lambda = \frac{\pi}{8}\,\frac{\Delta\omega}{\bar\omega}\Big(1 + O\big((\Delta\omega/\bar\omega)^2\big)\Big). \tag{7.3}$$

The bound is linear in the relative bandwidth, which is what one expects: on an infinitely narrow band a delay achieves any phase without loss. For wide bands it grows only logarithmically in b/a. Table II gives values, also in decibels per degree of phase, which is Λ times $(20/\ln 10)(\pi/180) = 0.15160$, and Figure 6 shows the bound as a surface.

8. SHARPNESS: THE EXTREMAL SYSTEMS

8.1. The square root R

Let R be the branch of $\sqrt{(z^2-a^2)(z^2-b^2)}$ that is analytic in $\mathbb{C}\setminus S$ and satisfies $R(z) = z^2\big(1 + O(z^{-2})\big)$ at infinity. It exists because a loop around one component of S encircles two branch points.[14]

Lemma 8.1. *$R(-z) = R(z)$ and $R(\bar z) = \overline{R(z)}$. On the real axis, $R(t) = \sqrt{(t^2-a^2)(t^2-b^2)} > 0$ for $|t| > b$, $R(t) = -\sqrt{(a^2-t^2)(b^2-t^2)} < 0$ for $|t| < a$, and*

$$R(t+i0) = i\,r(t), \quad R(-t+i0) = -i\,r(t) \qquad (a < t < b).$$

[14] Each of the four factors $(z \mp a)^{1/2}$, $(z \mp b)^{1/2}$ changes sign along a loop around its branch point; a loop around $[a,b]$ changes the sign twice, so the product returns to its value.

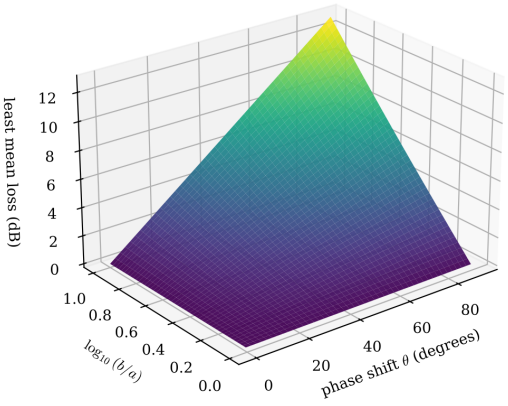


FIG. 6. The least mean loss $\theta\Lambda$, in decibels, over the phase lag θ and the band ratio b/a. The surface is linear in θ and increasing in b/a.

Proof. The symmetries hold because $R(-z)$ and $\overline{R(\bar z)}$ are branches of the same root with the same behaviour at infinity. Near $z = b$, $R(z) = (z-b)^{1/2}G(z)$ with G analytic and $G(b) = \sqrt{2b(b^2-a^2)} > 0$, the root being positive for $z > b$. Passing from $t > b$ to $t < b$ through $\mathbb{C}_+$ turns $(z-b)^{1/2}$ into $i\sqrt{b-t}$, so $R = ir$ on (a,b). Near $z = a$ write $R = (z-a)^{1/2}G_1(z)$ with $(z-a)^{1/2} > 0$ for $t > a$; then G_1 is positive imaginary on (a,b), and passing to $t < a$ through $\mathbb{C}_+$ gives $R = i\sqrt{a-t}\,G_1 < 0$. The values on the negative axis follow from the symmetries. □

8.2. The extremal system for constant phase

Put

$$m^2 = b^2\,\frac{E(k)}{K(k)}, \qquad \tau_\star = \frac{2\theta K(k)}{\pi b} = \frac{4\theta K(\kappa)}{\pi(a+b)}, \tag{8.1}$$

so that $a^2 < m^2 < b^2$, because $m^2 = I_2/I_0$ is an average of t^2 over (a,b) (Proposition 7.1), and define

$$h_\star(z) = \tau_\star \int_0^z \frac{s^2 - m^2}{R(s)}\,\mathrm{d}s, \qquad H_\star = e^{ih_\star}. \tag{8.2}$$

Theorem 8.2. *Let $\theta > 0$.*

(i) The integral (8.2) does not depend on the path in $\mathbb{C}\setminus S$; $h_\star$ is analytic there, odd, real on $\mathbb{R}\setminus S$, and $h_\star(z) = \tau_\star z + O(z^{-1})$ at infinity.

(ii) On the band, $h_\star(x+i0) = \theta + i\alpha_\star(x)$ with

$$\alpha_\star(x) = \tau_\star \int_a^x \frac{m^2 - s^2}{r(s)}\,\mathrm{d}s, \tag{8.3}$$

which is positive on (a,b), vanishes at a and b, and is largest at $x = m$.

(iii) $h_\star$ *is a Herglotz function, and* $H_\star$ *is a zero-free passive system, regular on the band, lossless on* $\mathbb{R}\setminus S$*, with constant phase* θ *on* $[a,b]$ *and*

$$\frac{1}{b-a}\int_a^b \alpha_\star(\omega)\,\mathrm{d}\omega = \theta\Lambda.$$

Proof. (i) The integrand is even, analytic in $\mathbb{C}\setminus S$, and integrable at the four branch points. A loop around $[a,b]$ can be shrunk onto the two sides of the cut, where $R = \pm ir$; its integral is $\pm 2i\int_a^b (s^2-m^2)\,\mathrm{d}s/r(s) = \pm 2i(I_2 - m^2 I_0) = 0$. By evenness the same holds around $[-b,-a]$. Hence $h_\star$ is single-valued, and odd. On $(-a,a)$ and on $|t|>b$ the integrand is real, so $h_\star$ is real there; it is also real along the path from 0 to $\pm a$, and the vanishing of the periods shows that it returns to real values beyond $\pm b$. At infinity, $(s^2-m^2)/R(s) = 1 + O(s^{-2})$, so $h_\star(z) - \tau_\star z$ tends to a constant; it is odd and analytic for $|z| > b$, so its Laurent series has only odd powers and the constant is 0. (ii) For $0 < x < a$, $h_\star(x) = \tau_\star \int_0^x (m^2 - s^2)\,\mathrm{d}s/\sqrt{(a^2-s^2)(b^2-s^2)}$ by Lemma 8.1, and at $x = a$ this is

$$\begin{aligned}
&\tau_\star\Big(\frac{m^2}{b}K(k') - b\big(K(k')-E(k')\big)\Big)\\
&\quad= \frac{\tau_\star b}{K(k)}\big(E(k)K(k') + E(k')K(k) - K(k)K(k')\big)\\
&\quad= \frac{\pi b\,\tau_\star}{2K(k)} = \theta,
\end{aligned}$$

by the substitution $s = a\sin\phi$[15] and Legendre's relation (Appendix A). Continuing from a along the upper side of the cut, where $R = ir$,

$$h_\star(x+i0) = \theta + \tau_\star\int_a^x \frac{s^2-m^2}{ir(s)}\,\mathrm{d}s = \theta + i\alpha_\star(x).$$

The integrand of (8.3) is positive for $s < m$ and negative for $s > m$, and its integral over (a,b) is $I_2 - m^2 I_0 = 0$; so $\alpha_\star$ rises from 0 at a to its maximum at m and falls back to 0 at b. (iii) The function $v = \operatorname{Im}(h_\star - \tau_\star z)$ is harmonic in $\mathbb{C}_+$, continuous on $\overline{\mathbb{C}_+}$, tends to 0 at infinity by (i), vanishes on $\mathbb{R}\setminus S$, and equals $\alpha_\star \geq 0$ on (a,b) and, by the symmetry $h_\star(-\bar z) = -\overline{h_\star(z)}$, also on $(-b,-a)$. By the minimum principle $v \geq 0$, so $\operatorname{Im} h_\star \geq \tau_\star \operatorname{Im} z \geq 0$. Thus $h_\star$ is a Herglotz function with $h_\star(iy) \in i\mathbb{R}$, $H_\star$ is passive and zero-free, and it is regular on the band since $h_\star$ is continuous up to $(0,\infty)$. As $\operatorname{Im} h_\star = 0$ on $\mathbb{R}\setminus S$, (4.2) shows $\mu_\star(\mathbb{R}\setminus S) = 0$, and Theorem 6.2 holds with equality. □

[15] It gives $\mathrm{d}s/\sqrt{(a^2-s^2)(b^2-s^2)} = \mathrm{d}\phi/(b\sqrt{1-k'^2\sin^2\phi})$, and $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi/\sqrt{1-k'^2\sin^2\phi} = (K(k')-E(k'))/k'^2$ with $a^2/k'^2 = b^2$.

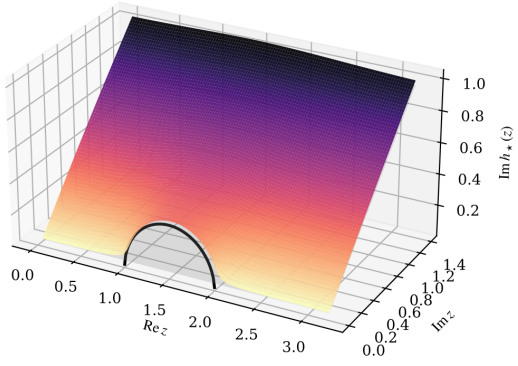


FIG. 7. The surface $\operatorname{Im} h_\star(x+iy)$ over the upper half-plane for an octave and $\theta = 1$. It is positive in $\mathbb{C}_+$, as a Herglotz function must be; on the real axis it vanishes off the band, and over the band its boundary values are the attenuation $\alpha_\star$ (black curve). Far from the band it approaches the plane $\tau_\star y$ (grey) of the delay.

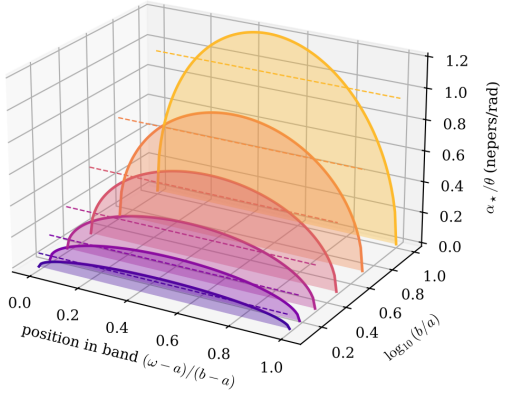


FIG. 8. The attenuation $\alpha_\star/\theta$ of the extremal system across the band, for $b/a = 1.25, 1.5, 2, 3, 5, 10$. Dashed lines mark the mean, Λ. The profile vanishes at both edges.

Remark 8.3. The extremal system is a delay $\tau_\star$ at high frequency, as its asymptotics shows, yet its group delay on the band is zero, its phase being constant there. All of its attenuation is in the band, and none of its response outside the band is reactive: in the language of (4.4) it has $\beta = \tau_\star$ and a measure carried by S. For an octave and $\theta = 1$, $\tau_\star = 0.6864/a$, $m = 1.4988\,a$, and the peak attenuation is $\alpha_\star(m) = 0.3482\,\mathrm{Np}$, against the mean 0.2735. Figure 8 shows $\alpha_\star$ for several band ratios, and Figure 7 the function $\operatorname{Im} h_\star$ over $\mathbb{C}_+$.

8.3. Extremal systems for other phases

The construction extends to phases that are not constant, and this is what we need for systems with zeros.

Proposition 8.4. *Let g be continuously differentiable on $[a,b]$ and put*

$$h_g(z) = \frac{2zR(z)}{\pi}\int_a^b \frac{g(t)}{r(t)\,(z^2-t^2)}\,\mathrm{d}t,$$
$$\beta_g = \frac{2}{\pi}\int_a^b \frac{g(t)}{r(t)}\,\mathrm{d}t.$$

(i) h_g is analytic in $\mathbb{C}\setminus S$, odd, real on $\mathbb{R}\setminus S$, continuous up to S from both sides, and $h_g(z)=\beta_g z+O(z^{-1})$ at infinity.

(ii) For $a<x<b$, $\operatorname{Re} h_g(x+i0)=g(x)$ and

$$\operatorname{Im} h_g(x+i0)=v_g(x)=\frac{2x\,r(x)}{\pi}\,\mathrm{p.\,v.}\int_a^b\frac{g(t)}{r(t)\,(x^2-t^2)}\,\mathrm{d}t.$$

(iii) If $v_g\ge 0$ on (a,b) and $\beta_g\ge 0$, then e^{ih_g} is a zero-free passive system, regular on the band, with phase g and attenuation v_g on the band and no measure outside S, so that $\int_a^b v_g=\frac12\int_a^b g\,p$.

(iv) For constant $g=\theta$, $h_g=h_\star$.

Proof. (i) The Cauchy integral $q(z)=\int_a^b g(t)\,(r(t)(z^2-t^2))^{-1}\,\mathrm{d}t$ is analytic off S and even; $zR(z)q(z)$ is odd and real on $\mathbb{R}\setminus S$ because R is real there (Lemma 8.1). The density g/r has singularities of order $|t-a|^{-1/2}$, $|t-b|^{-1/2}$ at the ends, so q is $O(|z-a|^{-1/2})$, $O(|z-b|^{-1/2})$ there and Hölder continuous up to the open cut [31, Ch. 4]; multiplied by R, which vanishes like the square root, the product is continuous up to S. At infinity $q(z)=z^{-2}\int_a^b g/r+O(z^{-4})$ and $zR(z)=z^3(1+O(z^{-2}))$, which gives the expansion. (ii) Write $2z/(z^2-t^2)=1/(z-t)+1/(z+t)$. By the Plemelj formula, at $x+i0$ with $a<x<b$ the first part of q contributes $\mathrm{p.\,v.}\int g\,(r(t)(x-t))^{-1}\,\mathrm{d}t-i\pi g(x)/r(x)$, so

$$\frac{2}{\pi}xq(x+i0)=\frac{2x}{\pi}\,\mathrm{p.\,v.}\int_a^b\frac{g(t)\,\mathrm{d}t}{r(t)(x^2-t^2)}-\frac{ig(x)}{r(x)}.$$

Multiplying by $R(x+i0)=ir(x)$ gives the real part $g(x)$ and the imaginary part $v_g(x)$. (iii) As in the proof of Theorem 8.2(iii), $\operatorname{Im}(h_g-\beta_g z)$ is harmonic, continuous up to the axis, zero on $\mathbb{R}\setminus S$, equal to $v_g\ge 0$ on S, and tends to 0 at infinity; so $\operatorname{Im} h_g\ge\beta_g\operatorname{Im} z\ge 0$. The last identity is Theorem 6.1. (iv) Let $d=h_g-h_\star$. Both functions have real part θ on the upper side of (a,b), are real on $\mathbb{R}\setminus S$ and satisfy $h(\bar z)=\overline{h(z)}$, so on the lower side the real part is again θ and $d(x\pm i0)=\pm i\,e(x)$ with e real and continuous. Since $R(x\pm i0)=\pm ir(x)$, the function d/R has the same boundary values e/r from both sides of (a,b), and likewise on $(-b,-a)$. It is therefore analytic across the open cuts, by Morera's theorem, and its singularities at $\pm a,\pm b$ are $O(|z-c|^{-1/2})$, hence removable. So d/R is entire. Both functions have the linear term $\tau_\star z$, because $\beta_\theta=2\theta I_0/\pi=\tau_\star$, so $d=O(z^{-1})$ and $d/R\to 0$ at infinity. Thus $d=0$. □

Lemma 8.5. *If $g(t)/t$ is nonincreasing on $[a,b]$, then $v_g\ge 0$. In particular this holds when $g>0$ and g is nonincreasing, and then also $\beta_g>0$.*

Proof. Substitute $\sigma=t^2$ and write $G(\sigma)=g(\sqrt\sigma)/\sqrt\sigma$, which is nonincreasing. Then

$$\frac{g(t)\,\mathrm{d}t}{r(t)}=\frac{G(\sigma)\,\mathrm{d}\sigma}{2\sqrt{(\sigma-a^2)(b^2-\sigma)}},$$

and with $\sigma_0=x^2$,

$$v_g(x)=\frac{x\,r(x)}{\pi}\times\mathrm{p.\,v.}\int_{a^2}^{b^2}\frac{G(\sigma)}{(\sigma_0-\sigma)\sqrt{(\sigma-a^2)(b^2-\sigma)}}\,\mathrm{d}\sigma.$$

Glauert's integral [29],

$$\mathrm{p.\,v.}\int_A^B\frac{\mathrm{d}\sigma}{(\sigma_0-\sigma)\sqrt{(\sigma-A)(B-\sigma)}}=0$$

for $A<\sigma_0<B$, which follows from the substitution $\sigma=\frac{A+B}{2}-\frac{B-A}{2}\cos\phi$ and $\mathrm{p.\,v.}\int_0^\pi \mathrm{d}\phi/(\cos\phi-\cos\phi_0)=0$, allows us to subtract $G(\sigma_0)$:

$$v_g(x)=\frac{x\,r(x)}{\pi}\int_{a^2}^{b^2}\frac{G(\sigma)-G(\sigma_0)}{\sigma_0-\sigma}\,\frac{\mathrm{d}\sigma}{\sqrt{(\sigma-a^2)(b^2-\sigma)}}.$$

For a nonincreasing G the difference quotient is nonnegative, so $v_g\ge 0$. If $g>0$ is nonincreasing, g/t is decreasing, and $\beta_g>0$ is clear. □

9. POINTWISE BOUNDS WITHOUT ZEROS

The sum rule bounds a mean. For zero-free systems with an affine phase on the band there is a stronger statement: one explicit system has the least attenuation at every frequency of the band at once. The proof writes every admissible h as the function h_g of Proposition 8.4 plus terms, one for each unit of measure off the band, each of which raises the attenuation at every point of the band.[16]

[16] The positive reconstruction of Theorem 9.2 and the pointwise comparisons Theorems 9.4, 9.6 and 9.8 were found with OpenAI's ChatGPT, starting from the sum rule of Section 6. The uniqueness step below, which reuses the argument of Proposition 8.4(iv), is ours.

Throughout, $\Sigma = (0,a)\cup(b,\infty)$ is the part of the positive axis off the band, and $\mu_0 = \mu(\{0\})$. Put

$$\tau_\circ = \frac{2K(k)}{\pi b} = \frac{\tau_\star}{\theta}, \qquad h_\circ = \frac{h_\star}{\theta}, \qquad A = \frac{\alpha_\star}{\theta},$$

so that $h_\circ$ is the extremal function of Theorem 8.2 for $\theta = 1$, with linear coefficient $\tau_\circ$ and attenuation $A(x) = \tau_\circ \int_a^x (m^2 - s^2)\,\mathrm{d}s/r(s)$ on the band.

9.1. One unit of measure off the band

For $t \in \Sigma$ and $a < x < b$ put

$$K_0(z) = \frac{R(z)}{abz}, \qquad K_t(z) = \frac{2zR(z)}{R(t)\,(t^2 - z^2)},$$
$$k_0(x) = \frac{r(x)}{abx}, \qquad k_t(x) = \frac{2x\,r(x)}{R(t)\,(t^2 - x^2)}.$$

By Lemma 8.1, $R(t)$ and $t^2 - x^2$ have the same sign for $t \in \Sigma$, so $k_0 > 0$ and $k_t > 0$ on (a,b).

Lemma 9.1. *K_0 and K_t, $t \in \Sigma$, are odd, analytic in $\mathbb{C}\setminus(S\cup\{0,\pm t\})$, real on $\mathbb{R}\setminus S$ away from their poles, and*

$$K_0(z) = \frac{z}{ab} - \frac{1}{z} + \frac{1}{\pi}\int_a^b \frac{2z\,k_0(s)}{s^2 - z^2}\,\mathrm{d}s,$$
$$K_t(z) = -\frac{2z}{R(t)} + \frac{2z}{t^2 - z^2} + \frac{1}{\pi}\int_a^b \frac{2z\,k_t(s)}{s^2 - z^2}\,\mathrm{d}s.$$

On the band $\operatorname{Re} K_0(x+i0) = \operatorname{Re} K_t(x+i0) = 0$, $\operatorname{Im} K_0(x+i0) = k_0(x)$ *and* $\operatorname{Im} K_t(x+i0) = k_t(x)$.

Proof. The parity and reality follow from Lemma 8.1, and the boundary values on the band from $R(x+i0) = ir(x)$. Call D the difference of the two sides of either identity. At infinity $R(z) = z^2 + O(1)$, so $K_0 = z/(ab) + O(z^{-1})$ and $K_t = -2z/R(t) + O(z^{-1})$, and the Cauchy integrals are $O(z^{-1})$; hence $D = O(z^{-1})$. At $z = 0$, $R(0) = -ab$, so $K_0 = -1/z + O(z)$. At $z = \pm t$ the residue of K_t is -1, as is that of $2z/(t^2 - z^2)$. So D is analytic off S. The Cauchy integral of the density k/π, $k = k_0$ or k_t, which is Hölder continuous inside the band and of order $|s-c|^{1/2}$ at its ends c, has the boundary imaginary part k on (a,b) by the Plemelj formula, as in the proof of Proposition 8.4(ii), and is continuous up to S [31, Ch. 4]. Both sides of each identity have imaginary part k on the upper side of the band, so D is real there, and by the symmetry $D(\bar z) = \overline{D(z)}$ it takes the same values from below. So D has no jump across the open band, and by oddness none across $(-b,-a)$; it is bounded near $\pm a, \pm b$, hence entire, and it tends to 0. Thus $D = 0$. □

9.2. The reconstruction

Theorem 9.2. *Let $H = \sigma e^{ih}$ be a zero-free passive system, regular on the band, whose phase $g = \varphi$ is continuously differentiable on $[a,b]$. Then $\int_\Sigma \mathrm{d}\mu(t)/|R(t)| < \infty$ and*

$$h = h_g + \mu_0 K_0 + \int_\Sigma K_t\,\mathrm{d}\mu(t). \tag{9.1}$$

Consequently, for $a < x < b$,

$$\alpha(x) = v_g(x) + \mu_0\,k_0(x) + \int_\Sigma k_t(x)\,\mathrm{d}\mu(t), \tag{9.2}$$
$$\beta = \beta_g + \frac{\mu_0}{ab} - \int_\Sigma \frac{2\,\mathrm{d}\mu(t)}{R(t)}. \tag{9.3}$$

The first identity says that the attenuation on the band is the attenuation v_g forced by the phase alone, plus a strictly positive contribution of every unit of measure off the band. The second shows the price of that measure in another currency: mass at frequencies below the band, where $R < 0$, raises the linear coefficient, and mass above the band lowers it.

Proof. Finiteness. For $t \to \infty$, $|R(t)| \sim t^2$, and $\int \mathrm{d}\mu/(1+t^2) < \infty$. Near a and b the measure has a bounded density by Lemma 4.4, and $|R|^{-1}$ has singularities of order $|t-a|^{-1/2}$, $|t-b|^{-1/2}$. Near 0, $|R| \ge ab/2$.

The right side. Call it $\hat h$. Since $|K_t(z)|$ is bounded by a constant times $|R(t)|^{-1}$ for $t \in \Sigma$ and z in a compact set at positive distance from $S\cup\{0\}\cup\operatorname{supp}\mu$, and for x in a compact subset of (a,b) the same holds near $x + i0$, the integral converges locally uniformly there; $\hat h$ is analytic off $S\cup\{0\}\cup\operatorname{supp}\mu\cup -\operatorname{supp}\mu$ and continuous up to the open band, where by Proposition 8.4(ii) and Lemma 9.1 its real part is g. Integrating the identities of Lemma 9.1 against μ, by Fubini's theorem since every term is absolutely integrable,

$$\hat h(z) = \hat\beta z - \frac{\mu_0}{z} + \int_\Sigma \frac{2z}{t^2 - z^2}\,\mathrm{d}\mu(t) + \frac{1}{\pi}\int_a^b \frac{2z\,\hat v(s)}{s^2 - z^2}\,\mathrm{d}s,$$

with $\hat\beta$ the right side of (9.3) and $\hat v$ that of (9.2). Here we used that $h_g = \beta_g z + \frac{1}{\pi}\int_a^b 2zv_g(s)\,\mathrm{d}s/(s^2 - z^2)$: the difference of the two sides is odd, analytic off S, $O(z^{-1})$ at infinity, continuous up to S, and has no jump across the band, so it is entire and vanishes.

$\hat v$ is bounded. v_g is continuous on $[a,b]$ by Proposition 8.4(i). In $\hat v(x)$ only the mass of μ near the band edges can produce a singularity. For $x = a + \varepsilon$ and $t = a - u$ with $0 < u < \delta$, $k_t(x)$ is at most a constant times $\sqrt{\varepsilon}/(\sqrt{u}\,(u+\varepsilon))$, and the density of μ is bounded there, while

$$\sqrt{\varepsilon}\int_0^\delta \frac{\mathrm{d}u}{\sqrt{u}\,(u+\varepsilon)} \le \pi;$$

the edge b is the same. The rest of μ contributes a bounded amount by the finiteness above.

Uniqueness. By (4.4) and Lemma 4.4, $h = \beta z - \mu_0/z + \int_\Sigma 2z\,\mathrm{d}\mu/(t^2 - z^2) + \frac{1}{\pi}\int_a^b 2z\alpha(s)\,\mathrm{d}s/(s^2 - z^2)$. So

$$q = h - \hat h = (\beta - \hat\beta)z + \frac{1}{\pi}\int_a^b \frac{2z\,(\alpha - \hat v)(s)}{s^2 - z^2}\,\mathrm{d}s,$$

which is analytic off S, odd, $O(z)$ at infinity, and $O(\log|z-c|)$ near each end c of S, because $\alpha-\hat{v}$ is bounded. Both h and $\hat{h}$ are continuous up to the open band with real part g, so, exactly as in the proof of Proposition 8.4(iv), q/R has the same boundary values from both sides of (a,b) and of $(-b,-a)$. It is therefore analytic across the open cuts, its singularities at $\pm a,\pm b$ are $o(|z-c|^{-1})$ and hence removable, and q/R is entire. At infinity $q/R=O(z^{-1})$. Hence $q=0$, which is (9.1); comparing linear coefficients and densities gives $\beta=\hat{\beta}$ and $\alpha=\hat{v}$ almost everywhere, hence everywhere by continuity. □

Corollary 9.3. *Under the hypotheses of Theorem 9.2, for $a<x<b$,*

$$\alpha(x) \;\geq\; v_g(x)+\frac{r(x)}{x}\,\max\{0,-\beta_g\}.$$

Proof. For $0<t<a<x$, $x/(x^2-t^2)>1/x$, so $k_t(x)>2r(x)/(x\,|R(t)|)$. Put

$$C_{\mathrm{in}}=\frac{\mu_0}{ab}+\int_{(0,a)}\frac{2\,\mathrm{d}\mu(t)}{|R(t)|},\qquad C_{\mathrm{out}}=\int_{(b,\infty)}\frac{2\,\mathrm{d}\mu(t)}{R(t)},$$

both nonnegative. By (9.2), $\alpha(x)\geq v_g(x)+\frac{r(x)}{x}C_{\mathrm{in}}+\int_{(b,\infty)}k_t(x)\,\mathrm{d}\mu(t)$, and by (9.3) and $\beta\geq 0$, $C_{\mathrm{in}}\geq C_{\mathrm{out}}-\beta_g\geq-\beta_g$. □

9.3. Affine phase

For $g(t)=t$, Proposition 8.4 gives $\beta_g=\frac{2}{\pi}\int_a^b t\,\mathrm{d}t/r(t)=1$ and $v_g=0$, by the substitution $\sigma=t^2$ and Glauert's integral (proof of Lemma 8.5); hence $h_g=z$. By linearity and Proposition 8.4(iv), the phase $\theta+\tau\omega$ has

$$h_g=\theta h_\circ+\tau z,\qquad v_g=\theta A,\qquad \beta_g=\theta\tau_\circ+\tau. \tag{9.4}$$

The function $R(z)/z=abK_0(z)$ has, by Lemma 9.1, linear coefficient 1, an atom ab at the origin and band attenuation $r(x)/x$.

Theorem 9.4. *Let $\theta\geq 0$, $\tau\in\mathbb{R}$, and $C=\max\{0,-\theta\tau_\circ-\tau\}$. Every zero-free passive system, regular on the band, with phase $\theta+\tau\omega$ on (a,b) satisfies*

$$\alpha(x)\;\geq\;\theta A(x)+C\,\frac{r(x)}{x}\qquad(a\leq x\leq b). \tag{9.5}$$

Equality holds throughout the band for

$$h_{\mathrm{opt}}(z)=\theta h_\circ(z)+\tau z+C\,\frac{R(z)}{z},$$

and equality at a single point of (a,b) forces $h=h_{\mathrm{opt}}$.

Proof. The bound is Corollary 9.3 with (9.4), extended to the ends by continuity. By Lemma 9.1 and (9.4), h_{opt} has the form (4.4) with linear coefficient $\theta\tau_\circ+\tau+C\geq 0$, an atom $Cab\geq 0$ at the origin, and the measure $\pi^{-1}(\theta A+Cr/x)\,\mathrm{d}x\geq 0$ on the band and nothing else. So it is a Herglotz function with the symmetry of (4.3), continuous up to a neighbourhood of $[a,b]$, and $e^{ih_{\mathrm{opt}}}$ is admissible with phase $\theta+\tau\omega$ and attenuation $\theta A+Cr/x$.

Suppose equality at $x_0\in(a,b)$. In the proof of Corollary 9.3 each inequality is then an equality at x_0. Since $k_t(x_0)>0$ for $t>b$, μ vanishes on (b,∞); since the comparison for $0<t<a$ is strict, μ vanishes on $(0,a)$; and $C_{\mathrm{in}}=\mu_0/(ab)$ equals C. By (9.1), $h=\theta h_\circ+\tau z+CabK_0=h_{\mathrm{opt}}$. □

For $\tau\geq-\theta\tau_\circ$, in particular for a pure phase lag, the optimum is $e^{i\tau z}H_\star$, and an additional delay costs nothing. A negative slope beyond $-\theta\tau_\circ=-\tau_\star$ would make the linear coefficient negative; the cheapest compensation, at every frequency of the band at once, is then a pole of h at zero frequency.

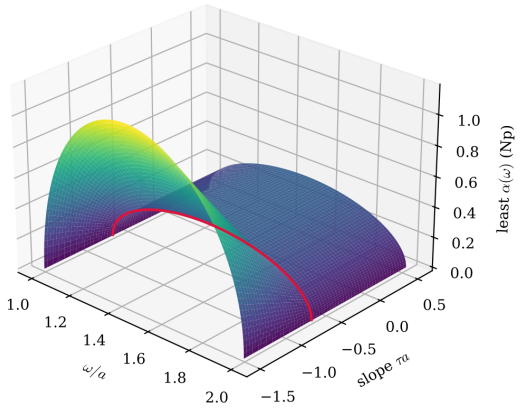


FIG. 9. The least attenuation $A(\omega)+\max\{0,-\tau_\circ-\tau\}\,r(\omega)/\omega$ of Theorem 9.4 at each frequency of an octave, for the phase $1+\tau\omega$, as a function of the slope τ. For $\tau\geq-\tau_\circ$ (red curve at $\tau=-\tau_\circ=-0.6864/a$) it does not depend on τ; below it the pole at zero frequency adds $r(\omega)/\omega$ per unit of slope.

9.4. Constant phase lead

Lemma 9.5. *$h_\circ(-ab/z)=-h_\circ(z)+\tau_\circ R(z)/z$. Consequently*

$$A(ab/x)=\tau_\circ\frac{r(x)}{x}-A(x)>0\qquad(a<x<b).$$

Proof. The function $f(z)=h_\circ(-ab/z)$ is a Herglotz function with the symmetry of (4.3), regular on the band, as in the proof of Theorem 6.6; its phase on the band is -1 and its attenuation is $A(ab/x)$. Since $h_\circ$ is analytic at 0 with $h_\circ(0)=0$, f has linear coefficient 0; since $h_\circ(w)=\tau_\circ w+O(w^{-1})$, $f(z)=-\tau_\circ ab/z+O(z)$ near 0, an atom $\tau_\circ ab$; and f is real on Σ, so μ has no other mass there. Theorem 9.2 with $g=-1$, for which

$h_g = -h_\circ$, gives $f = -h_\circ + \tau_\circ abK_0$. Taking imaginary parts on the band gives the second identity; $A(ab/x) > 0$ by Theorem 8.2(ii). □

Theorem 9.6. *Let $s > 0$ and $\tau \le 0$. Every zero-free passive system, regular on the band, with phase $-s + \tau\omega$ on (a, b) satisfies*

$$\alpha(x) \;\ge\; s\,A(ab/x) - \tau\,\frac{r(x)}{x} \qquad (a \le x \le b),$$

with equality throughout the band for $h = -sh_\circ + \tau z + (s\tau_\circ - \tau)R/z$, and equality at one point of (a, b) forces this h. For $\tau = 0$ the optimum is $H_\star(-ab/z)$ with $H_\star$ the extremal system for $\theta = s$.

Proof. Here $v_g = -sA$ and $\beta_g = -s\tau_\circ + \tau < 0$, so Corollary 9.3 and Lemma 9.5 give $\alpha \ge -sA + (s\tau_\circ - \tau)r/x = sA(ab/x) - \tau r/x$. The stated h has linear coefficient 0, an atom $(s\tau_\circ - \tau)ab > 0$ at the origin and band density $\pi^{-1}(sA(ab/x) - \tau r/x) > 0$, so it is admissible. Equality at one point forces h as in the proof of Theorem 9.4. For $\tau = 0$, Lemma 9.5 identifies h with $sh_\circ(-ab/z)$. □

9.5. Consequences

Corollary 9.7. *(i) For phase $\theta + \tau\omega$ with $\theta \ge 0$, $\tau \in \mathbb{R}$, or with $\theta < 0$, $\tau \le 0$, the optimal system of Theorems 9.4 and 9.6 minimises $\int_a^b w\,\alpha$ for every weight $w \ge 0$, every L^p norm of α on the band, $1 \le p \le \infty$, and every functional of α that increases with α; for $\int w\alpha$ with $w \ne 0$ it is the only minimiser.*

(ii) For phase $\theta \ge 0$, the least value of $\max_{[a,b]} \alpha$ is $\theta A(m) = \alpha_\star(m)$; for phase $-s < 0$ it is $sA(m)$, attained at $x = ab/m$.

(iii) For phase $-s < 0$ the least mean attenuation over the band is $s\Lambda_-$, with

$$\Lambda_- = \frac{E(\kappa) - \kappa'^2 K(\kappa)}{\kappa}, \qquad \kappa'^2 = 1 - \kappa^2.$$

(iv) For phase $\theta + \tau\omega$ with $\theta \ge 0$ and $\tau < -\tau_\star$, the least mean attenuation is

$$\theta\Lambda + \frac{\pi}{4}\,(b - a)\,(-\tau - \tau_\star).$$

Proof. (i) follows from the pointwise bounds and their equality cases; a weight $w \ge 0$ that is not almost everywhere zero is positive on a set of positive measure, where any other admissible system has strictly larger attenuation, by the equality cases and continuity. (ii) A is largest at m (Theorem 8.2(ii)). (iii) is the value computed in Remark 6.8 for $H_\star(-ab/z)$. (iv) Integrate (9.5) with $C = -\tau - \tau_\star$, using

$$\int_a^b \frac{r(x)}{x}\,\mathrm{d}x = \frac{\pi}{4}(b - a)^2,$$

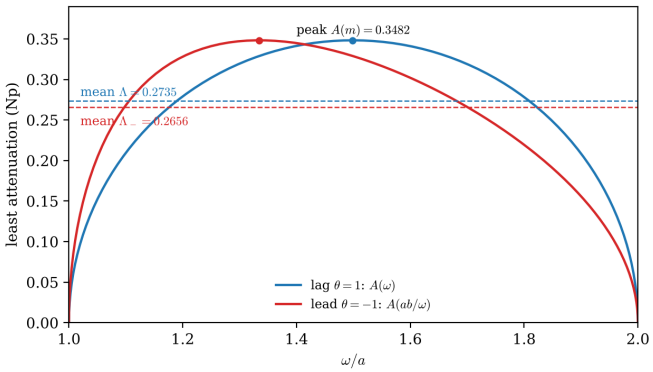


FIG. 10. The least attenuation at each frequency of an octave for a unit lag (Theorem 9.4) and a unit lead (Theorem 9.6). The lead profile is the lag profile reflected by $\omega \mapsto ab/\omega$; the two have the same peak $A(m)$, at m and at ab/m, and different means.

which follows from $\sigma = x^2$, $A = a^2$, $B = b^2$ and the substitution $\sigma = c - d\cos\phi$, $c = \frac{A+B}{2}$, $d = \frac{B-A}{2}$:

$$\begin{aligned}\int_A^B \frac{\sqrt{(\sigma - A)(B - \sigma)}}{\sigma}\,\mathrm{d}\sigma &= d^2 \int_0^\pi \frac{\sin^2\phi\;\mathrm{d}\phi}{c - d\cos\phi} \\ &= \pi\big(c - \sqrt{c^2 - d^2}\big) \\ &= \frac{\pi}{2}\big(\sqrt{B} - \sqrt{A}\big)^2,\end{aligned}$$

since $c^2 - d^2 = AB$. □

For an octave $A(m) = 0.34824$, so the least worst-frequency attenuation for a lag θ is $1.2731\,\theta\Lambda$, against the mean $\theta\Lambda = 0.27353\,\theta$; and $\Lambda_- = 0.26560$. So the sharp worst-frequency bound for a lag and the sharp plain mean for a lead are both attained by the extremal system or by its inversion.

9.6. A prescribed argument

Theorem 9.8. *Let $0 < \vartheta < \pi$. Among zero-free passive systems regular on the band with $\arg H \equiv \vartheta$ modulo 2π on $[a, b]$, the least mean attenuation is $\min\{\vartheta\Lambda, (\pi - \vartheta)\Lambda_-\}$ and the least value of $\max_{[a,b]} \alpha$ is $\min\{\vartheta, \pi - \vartheta\}A(m)$.*

Proof. By Remark 6.3 the phase is a constant $\theta' \equiv \vartheta \pmod\pi$. If $\theta' > 0$ then $\theta' \ge \vartheta$, and by Theorem 6.2 and Corollary 9.7(ii) both costs are at least $\vartheta\Lambda$ and $\vartheta A(m)$, with equality for $H_\star$ with $\theta = \vartheta$. If $\theta' < 0$ then $|\theta'| \ge \pi - \vartheta$, and Corollary 9.7(ii), (iii) give $(\pi - \vartheta)\Lambda_-$ and $(\pi - \vartheta)A(m)$, attained by $-H_\star(-ab/z)$ with $\theta = \pi - \vartheta$. □

So the mean is least on the lagging branch for $\vartheta < \vartheta_c = \pi\Lambda_-/(\Lambda + \Lambda_-)$ and on the leading branch beyond it, while the worst frequency changes branch at $\pi/2$. For an octave $\vartheta_c = 1.5477$, just below $\pi/2$: the two costs prefer different systems for $\vartheta_c < \vartheta < \pi/2$. Theorem 9.8 sharpens Corollary 6.7, whose constant Λ becomes $A(m)$.

10. SYSTEMS WITH ZEROS

The main theorem concerns zero-free systems. A passive system may also have zeros in $\mathbb{C}_+$, and a zero contributes phase without contributing loss on the real axis. This section shows that zeros do help, and by how much one zero helps.

10.1. The bound with an inner factor

Proposition 10.1. *Let $H = BG$ be passive, where B is a real-symmetric Blaschke product whose zeros stay away from $I \cup (-I)$ and G is zero-free and regular on the band. Let φ be a continuous argument of H on $[a,b]$ and let ψ be the continuous argument of B on $[a,b]$ for which $\varphi - \psi$ is the phase of G in the sense of Lemma 4.2. Then $\alpha = \alpha_G$ on the band, and*

$$\begin{aligned}\int_a^b \alpha \,\mathrm{d}\omega &= \frac{1}{2}\int_a^b (\varphi-\psi)\,p\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W\,\mathrm{d}\mu_G \\ &\geq \frac{1}{2}\int_a^b (\varphi-\psi)\,p\,\mathrm{d}\omega.\end{aligned} \tag{10.1}$$

The function ψ is real-analytic and strictly increasing on $[a,b]$.

Proof. The Blaschke product is analytic across I and unimodular there, so $\log|H| = \log|G|$ on the band, and $\arg H = \arg B + \arg G$ modulo 2π. The continuous arguments on $[a,b]$ therefore differ by a constant in $2\pi\mathbb{Z}$, which fixes ψ. Theorem 6.1 applied to G gives (10.1). The monotonicity of ψ is the computation in the proof of Proposition 4.5.[17] □

An increasing ψ behaves like a delay distributed over frequency. A pure delay is free (Lemma 5.3(i)), but a phase that rises steeply at low frequencies and flattens on the band is not a delay: its weighted integral $\int \psi p$ is positive, and (10.1) then allows less loss than (6.5). The simplest instance is a single zero on the imaginary axis.

10.2. One zero

For $\eta > 0$ let

$$B_\eta(z) = \frac{z - i\eta}{z + i\eta}.$$

It is passive, unimodular on $\mathbb{R}$, has the real symmetry, and the argument of $-B_\eta$ on $(0,\infty)$ is

$$\arg\big(-B_\eta(\omega)\big) = \pi - 2\arctan\frac{\eta}{\omega} = 2\arctan\frac{\omega}{\eta} \in (0,\pi). \tag{10.2}$$

For $0 < \theta < \pi$ put $\eta_\theta = b\cot(\theta/2)$, so that $2\arctan(b/\eta_\theta) = \theta$.

Proposition 10.2. *Let $0 < \theta < \pi$ and $\eta \geq \eta_\theta$, and put*

$$g_\eta(\omega) = \theta - 2\arctan\frac{\omega}{\eta}, \qquad H_\eta = -B_\eta\, e^{ih_{g_\eta}},$$

with h_g as in Proposition 8.4. Then H_η is passive and regular on the band, has exactly one zero in $\mathbb{C}_+$, at $i\eta$, has argument θ on $[a,b]$, and

$$\begin{aligned}\frac{1}{b-a}\int_a^b \alpha_\eta\,\mathrm{d}\omega &= \theta\Lambda - \delta(\eta), \\ \delta(\eta) &= \frac{1}{b-a}\int_a^b \arctan\frac{\omega}{\eta}\,p(\omega)\,\mathrm{d}\omega.\end{aligned} \tag{10.3}$$

The reduction δ is strictly positive and strictly decreasing in η, and

$$\delta(\eta) = \frac{\pi(b^2-a^2)^2}{24(b-a)\eta^3} + O(\eta^{-5}) \qquad (\eta\to\infty). \tag{10.4}$$

Proof. On $[a,b]$ the function g_η is decreasing, and $g_\eta(b) \geq 0$ because $\eta \geq \eta_\theta$. Hence $g_\eta \geq 0$, $g_\eta > 0$ on $[a,b)$, and $g_\eta(t)/t$ is decreasing. By Lemma 8.5, $v_{g_\eta} \geq 0$ and $\beta_{g_\eta} > 0$, so by Proposition 8.4(iii) $e^{ih_{g_\eta}}$ is a zero-free passive system, regular on the band, with phase g_η and attenuation v_{g_η}, and with no measure outside S. Multiplying by $-B_\eta$ keeps passivity and the real symmetry, adds the single zero $i\eta$, leaves the modulus on $\mathbb{R}$ unchanged, and adds the argument (10.2). The argument of H_η on the band is therefore $g_\eta + 2\arctan(\omega/\eta) = \theta$, and Proposition 8.4(iii) gives

$$\int_a^b \alpha_\eta = \frac{1}{2}\int_a^b g_\eta\,p = \frac{\theta}{2}J - \int_a^b \arctan\frac{\omega}{\eta}\,p\,\mathrm{d}\omega,$$

which is (10.3) since $J = 2(b-a)\Lambda$.

For the sign, write $\arctan(x/\eta) = x\,f(x^2)$ with $f(\sigma) = \arctan(\sqrt{\sigma}/\eta)/\sqrt{\sigma}$. Since $u \mapsto \arctan(u)/u$ is strictly decreasing on $(0,\infty)$,[18] f is strictly decreasing, and Lemma 5.3(ii) gives $\delta(\eta) > 0$. Similarly

$$\delta'(\eta) = -\frac{1}{b-a}\int_a^b \frac{x\,p(x)}{\eta^2+x^2}\,\mathrm{d}x < 0,$$

by Lemma 5.3(ii) with the strictly decreasing $f(\sigma) = 1/(\eta^2+\sigma)$. Finally $\arctan u = u - u^3/3 + O(u^5)$ uniformly on $[0, b/\eta]$, and Lemma 5.3(i),(iv) give $\int_a^b x\,p = 0$ and $\int_a^b x^3 p = -\frac{\pi}{8}(b^2-a^2)^2$, which yields (10.4). □

Since δ decreases, the best choice within the family is $\eta = \eta_\theta$, where the phase g_η of the zero-free factor reaches zero at the upper band edge. Figure 11 shows the mean loss of H_η, relative to $\theta\Lambda$, over θ and η/η_θ.

[17] For a single factor $(z-\zeta)/(z-\bar\zeta)$ with $\zeta = \xi + i\eta$, $\eta > 0$, the argument on the real line is $\arg(x-\zeta) - \arg(x-\bar\zeta)$, and $\frac{\mathrm{d}}{\mathrm{d}x}\arg(x-\xi\mp i\eta) = \pm\eta/((x-\xi)^2+\eta^2)$; the difference is $2\eta/|x-\zeta|^2$.

[18] Its derivative is $\big(u/(1+u^2) - \arctan u\big)/u^2$, and $\arctan u > u/(1+u^2)$ for $u > 0$ because both vanish at 0 and $\frac{\mathrm{d}}{\mathrm{d}u}\arctan u = \frac{1}{1+u^2} > \frac{1-u^2}{(1+u^2)^2} = \frac{\mathrm{d}}{\mathrm{d}u}\frac{u}{1+u^2}$.

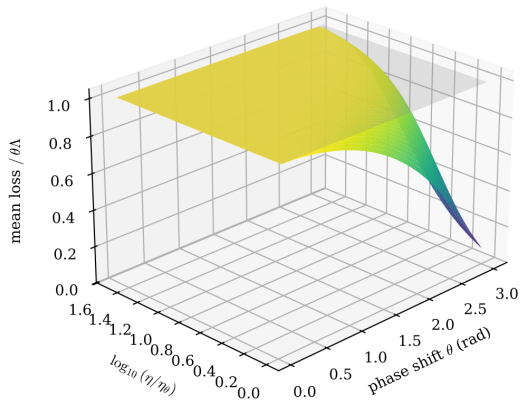


FIG. 11. The mean attenuation of the one-zero system H_η of Proposition 10.2, divided by the zero-free bound $\theta\Lambda$, for $a = 1$, $b = 2$, over the phase θ and $\log_{10}(\eta/\eta_\theta)$. The grey plane is the level 1. The ratio is below 1 everywhere and tends to 1 as the zero moves to infinity.

10.3. One zero beats every zero-free system

The statement of Theorem 6.2 fixes the continuous phase φ. A measurement only gives the argument modulo 2π, and a zero-free system with $\arg H \equiv \theta$ may realise θ through a lead $\theta - \pi$, by Remark 6.3. The next corollary takes this into account.

Corollary 10.3. *Let $0 < \theta < a\pi/(a+b)$. Among zero-free passive systems regular on the band with $\arg H \equiv \theta$ modulo 2π on $[a, b]$, the least mean attenuation over the band is $\theta\Lambda$, attained by $H_\star$. The system H_{η_θ} of Proposition 10.2, which has one zero, has the same argument and a strictly smaller mean attenuation.*

Proof. By Remark 6.3 the phase of such a system is a constant θ' with $\theta' \equiv \theta \pmod{\pi}$. If $\theta' \geq \theta$, Theorem 6.2 gives mean attenuation at least $\theta'\Lambda \geq \theta\Lambda$. Otherwise $\theta' \leq \theta - \pi < 0$, and Theorem 6.6 with $\tau = 0$ bounds the ab/ω^2-weighted mean by $|\theta'|\,\Lambda \geq (\pi - \theta)\Lambda$. Since $ab/\omega^2 \leq b/a$ on $[a, b]$, the plain mean is at least $(a/b)(\pi - \theta)\Lambda$, which exceeds $\theta\Lambda$ exactly when $\theta < a\pi/(a + b)$. The value $\theta\Lambda$ is attained by $H_\star$ (Theorem 8.2), and Proposition 10.2 gives the second statement. □

For larger θ the lead branch competes. The system $-H_\star(-ab/z)$, built from the extremal for $\pi - \theta$, is zero-free, has $\sigma = -1$ and phase $\theta - \pi$, hence argument θ modulo 2π, and by Remark 6.8 its mean attenuation is $(\pi - \theta)\,(E(\kappa) - \kappa'^2 K(\kappa))/\kappa$. Table III compares the three systems for an octave.

The one-zero system is the best of the three for $\theta \leq \pi/2$; by $\theta = 2$ the lead branch is better. For $\theta = 0.5$ the zero lowers the loss by 1.7%, for $\theta = 1$ by 7.2%.

TABLE III. Mean attenuation over the octave $[1, 2]$ of three systems with argument θ modulo 2π: the lag extremal $H_\star$ (the least value among zero-free systems with phase θ), the lead extremal $-H_\star(-ab/z)$ built for $\pi - \theta$, and the one-zero system H_{η_θ}. For $\theta < \pi/3$ the first column is the least value among all zero-free systems (Corollary 10.3).

θ	$\theta\Lambda$	$(\pi - \theta)\,0.26560$	H_{η_θ}
0.5	0.13677	0.70160	0.13443
1	0.27353	0.56880	0.25394
$\pi/2$	0.42966	0.41720	0.34659
2	0.54707	0.30320	0.35977
2.5	0.68383	0.17040	0.27677

10.4. The problem

Problem 10.4. Determine

$$\Lambda_{\mathrm{all}}(\theta) = \inf\Big\{\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega\Big\},$$

the infimum over all passive systems H, with or without zeros, that are continuous and nonzero near the band and satisfy $\arg H \equiv \theta$ modulo 2π on $[a, b]$. Is $\Lambda_{\mathrm{all}}(\theta) > 0$ for $\theta \notin \pi\mathbb{Z}$?

Corollary 4.6 shows that the infimum is not attained by a lossless system, but it does not exclude that it is zero. By (10.1), the question is how large $\int_a^b \psi\,p$ can be made by the argument ψ of a Blaschke product, while the remaining phase $\theta - \psi$ is still the phase of a zero-free passive system. The difficulty is the second condition: by Proposition 8.4 it requires $v_g \geq 0$ and $\beta_g \geq 0$ for $g = \theta - \psi$, and $\beta_g \geq 0$ fails as soon as g is too negative on the band.[19] The next section solves the problem. The infimum is positive and attained, the minimiser has exactly one zero, and it is the least lossy system at every frequency of the band at once.

11. THE LEAST LOSS WITH ZEROS

This section solves Problem 10.4. The answer is stronger than the problem asks. For every phase $\theta \notin \pi\mathbb{Z}$ there is one passive system whose attenuation is the least possible at every frequency of the band simultaneously, among all passive systems with that argument, with or without zeros. It has exactly one zero, on the imaginary axis, and it is lossless off the band. For $\theta = \pi/2$ it is algebraic, and every quantity is elementary.

[19] For instance, two zeros chosen so that $g < 0$ on the whole band give $\beta_g = \frac{2}{\pi}\int g/r < 0$; the resulting function e^{ih_g} grows like $e^{-\beta_g y}$ along $i\mathbb{R}_+$ and is not passive.

11.1. Statement

We call a passive system H *admissible for* θ if it is regular on the band in the sense of Definition 4.3 (the definition makes sense for any passive system) and $\arg H \equiv \theta \pmod{2\pi}$ on $[a,b]$. The admissible systems are exactly those of Problem 10.4. Since $-H$ is passive whenever H is, and $\arg(-H) = \arg H + \pi$, it suffices to treat $0 < \theta < \pi$. We use r, h_g, v_g, β_g, B_η and δ as in Sections 8 and 10.

Theorem 11.1. *Let $0 < \theta < \pi$. The equation*

$$\int_a^b \frac{2\arctan(t/\eta)}{r(t)}\,\mathrm{d}t = \theta \int_a^b \frac{\mathrm{d}t}{r(t)} \tag{11.1}$$

has exactly one root $\eta_\star \in (0, \eta_\theta)$. Put

$$g_\sharp(\omega) = \theta - 2\arctan\frac{\omega}{\eta_\star}, \qquad H_\theta^\sharp = -B_{\eta_\star} e^{ih_{g_\sharp}},$$
$$\alpha^\sharp{}_\theta = v_{g_\sharp} \quad \text{on } [a,b].$$

(i) $H_\theta^\sharp$ is admissible for θ. It has exactly one zero in $\mathbb{C}_+$, at $i\eta_\star$, it is lossless on $\mathbb{R}\setminus S$, it satisfies $H_\theta^\sharp(0) = 1$ and $H_\theta^\sharp(iy) \to -1$ as $y \to \infty$, and its attenuation $\alpha^\sharp{}_\theta$ is positive on (a,b) and vanishes at a and b.

(ii) Every system H admissible for θ satisfies

$$\alpha(\omega) \ge \alpha^\sharp{}_\theta(\omega) \qquad (a \le \omega \le b). \tag{11.2}$$

If equality holds on a set of positive measure, then $H = H_\theta^\sharp$.

(iii) Consequently

$$\Lambda_{\mathrm{all}}(\theta) = \frac{1}{b-a}\int_a^b \alpha^\sharp{}_\theta\,\mathrm{d}\omega = \theta\Lambda - \delta(\eta_\star) > 0, \tag{11.3}$$

the infimum is attained only by $H_\theta^\sharp$, and it is strictly smaller than the mean attenuation of every system of Proposition 10.2. For $\pi < \theta < 2\pi$, $\Lambda_{\mathrm{all}}(\theta) = \Lambda_{\mathrm{all}}(\theta-\pi)$, attained only by $-H_{\theta-\pi}^\sharp$.

The inequality (11.2) is pointwise. A single system is optimal for the mean, for every weighted mean, for the worst frequency and for the geometric mean at the same time (Corollary 11.8). The zero-free theory has the same feature for an affine phase (Theorem 9.4), although its sum rule bounds only the mean; there the constraint $\beta \ge 0$ ties the frequencies of the band together, and zeros remove it (Remark 11.5).

For a right angle everything is explicit.

Theorem 11.2. *Let $\theta = \pi/2$. Then $\eta_\star = \sqrt{ab}$, and*

$$H_{\pi/2}^\sharp(z) = -\frac{z^2 + ab}{R(z) + i(a+b)z}, \tag{11.4}$$

TABLE IV. The least mean attenuation over the octave $[1,2]$ of passive systems with argument θ modulo 2π, with or without zeros (Theorem 11.1), the position of the zero of the optimal system relative to $\eta_\theta = b\cot(\theta/2)$ of Proposition 10.2, the ratio to the zero-free value $\theta\Lambda$, and the least possible worst-frequency attenuation $\max \alpha^\sharp{}_\theta$.

θ	$\eta_\star/\eta_\theta$	$\Lambda_{\mathrm{all}}(\theta)$	$\Lambda_{\mathrm{all}}/\theta\Lambda$	$\max\alpha^\sharp{}_\theta$
0.5	0.72576	0.13090	0.9571	0.16677
1	0.71853	0.22860	0.8357	0.29204
$\pi/2$	0.70711	0.26951	0.6272	0.34657
2	0.69843	0.24357	0.4452	0.31540
2.5	0.69050	0.15941	0.2331	0.20805

with R as in Lemma 8.1. Its attenuation is

$$\begin{aligned}\alpha^\sharp{}_{\pi/2}(\omega) &= \log\frac{(a+b)\,\omega + r(\omega)}{\omega^2 + ab} \\ &= 2\,\mathrm{artanh}\sqrt{\frac{(\omega-a)(b-\omega)}{(\omega+a)(\omega+b)}},\end{aligned} \tag{11.5}$$

and

$$\int_a^b \alpha^\sharp{}_{\pi/2}\,\mathrm{d}\omega = \frac{\pi}{2}\bigl(\sqrt{b}-\sqrt{a}\bigr)^2, \quad \Lambda_{\mathrm{all}}\Bigl(\frac{\pi}{2}\Bigr) = \frac{\pi}{2}\,\frac{\sqrt{b}-\sqrt{a}}{\sqrt{b}+\sqrt{a}}. \tag{11.6}$$

The profile $\alpha^\sharp{}_{\pi/2}$ is invariant under $\omega \mapsto ab/\omega$ and has its only maximum at the geometric centre $\omega_0 = \sqrt{ab}$, where $\alpha^\sharp{}_{\pi/2}(\omega_0) = \frac{1}{2}\log(b/a)$. Hence every passive system with $\arg H \equiv \pi/2 \pmod{\pi}$ on $[a,b]$ that is regular on the band satisfies

$$|H(\omega)| \le \frac{\omega^2 + ab}{(a+b)\,\omega + r(\omega)}, \qquad \bigl|H(\sqrt{ab})\bigr|^2 \le \frac{a}{b}. \tag{11.7}$$

For an octave, a passive system whose phase is a right angle at every frequency of the band transmits at most half of the power at the geometric centre: its loss there is at least 3.01 dB, and for a decade at least 10 dB.[20] The sum rule of the zero-free theory gives no bound of this pointwise kind. The mean attenuation of $H_{\pi/2}^\sharp$ over an octave is $\frac{\pi}{2}(3-2\sqrt{2}) = 0.26951\,\mathrm{Np}$, which is 62.7% of the zero-free value $\frac{\pi}{2}\Lambda = 0.42966$. For a narrow band the ratio of the two minimal means tends to $\sin\theta/\theta$ (Corollary 11.9), which is $2/\pi$ at a right angle.

Table IV gives the octave values, and Figure 12 the optimal profiles. Comparing with Table III: the optimal system beats the one-zero system H_{η_θ} for every θ, by 2.6% at $\theta = 0.5$ and 22% at $\theta = \pi/2$, and for $\theta \ge 2$ it also beats the lead branch. Its zero sits closer to the origin than η_θ, so that the phase $g_\sharp$ of its zero-free factor is negative on part of the band, which Lemma 8.5 does not allow. At $\theta = \pi/2$ the ratio $\eta_\star/\eta_\theta$ is $\sqrt{a/b}$, by Theorem 11.2.

[20] The power transmission is $|H|^2 \le a/b$, so the loss is at least $10\log_{10}(b/a)$ dB: $10\log_{10}2 = 3.0103$ and $10\log_{10}10 = 10$.

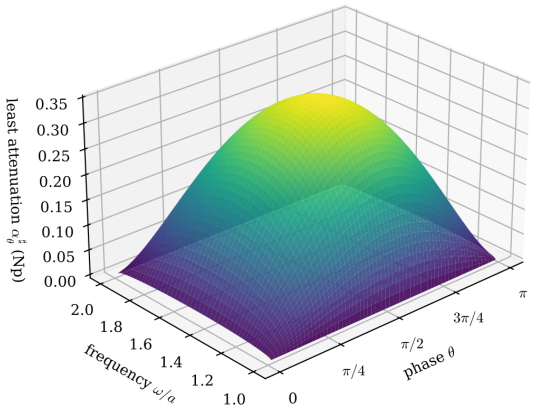


FIG. 12. The least attenuation $\alpha^\sharp{}_\theta(\omega)$ at each frequency of the octave, over the phase $\theta \in (0,\pi)$ (Theorem 11.1). No passive system with argument θ on the band lies below the surface at any frequency. The black curve is the closed form (11.5) at $\theta = \pi/2$, with its peak $\frac{1}{2}\log 2$ at $\omega = \sqrt{2}$.

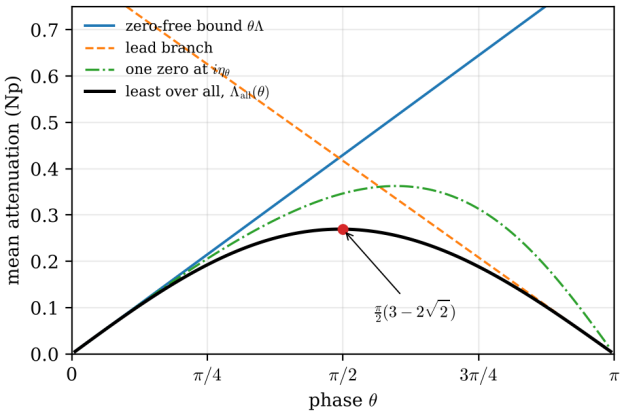


FIG. 13. Mean attenuation over the octave against the phase θ: the zero-free bound $\theta\Lambda$ (Theorem 6.2), the lead branch of Table III, the one-zero system H_{η_θ} (Proposition 10.2), and the exact minimum $\Lambda_{\rm all}(\theta)$ over all passive systems (Theorem 11.1). The dot marks the closed form $\frac{\pi}{2}(3-2\sqrt{2})$ at $\theta = \pi/2$.

11.2. A change of variables

The phase condition $\arg H \equiv \theta$ is not linear in $\log H$ once zeros are allowed, since $\log H$ is then not defined. A Möbius transformation of the values removes this difficulty.

Lemma 11.3. *Let $0 < \theta < \pi$, let H be a passive system that is not constant, and put*

$$F = e^{-i\theta}\,\frac{H - e^{i\theta}}{H - e^{-i\theta}}, \qquad H = \frac{F - e^{i\theta}}{e^{i\theta}F - 1}. \tag{11.8}$$

(i) F is analytic in $\mathbb{C}_+$ with values in $\mathbb{C}_+$, and $F(-\bar z) = 1/\overline{F(z)}$. The function $\zeta = \log F$, with $0 < \operatorname{Im}\zeta < \pi$, satisfies $\zeta(-\bar z) = -\overline{\zeta(z)}$, and

$$\zeta(z) = \frac{1}{\pi}\int_0^\infty g(t)\,\frac{2z}{t^2 - z^2}\,\mathrm{d}t, \tag{11.9}$$

where $g(t) = \operatorname{Im}\zeta(t + i0) \in [0,\pi]$ for almost every $t > 0$. In particular $\operatorname{Re}\zeta = 0$ on $i\mathbb{R}_+$.

(ii) Conversely, if F is analytic in $\mathbb{C}_+$ with values in $\mathbb{C}_+$ and $F(-\bar z) = 1/\overline{F(z)}$, then (11.8) defines a passive system.

(iii) $|H| = 1$ at a point $t \in \mathbb{R}$ where H is continuous if and only if $F(t)$ is real there, and then $g(t) \in \{0,\pi\}$. The zeros of H are the points where $F = e^{i\theta}$.

(iv) Let $H(\omega) = e^{i\theta-\alpha}$ with $\alpha > 0$ at a point ω of the band, and write $\zeta = u + iv$ there. (If $\alpha = 0$, then $F = 0$.) Then

$$\frac{1}{F} = \cos\theta - i\sin\theta\coth\frac{\alpha}{2}, \qquad e^u\cos\theta = \cos v, \tag{11.10}$$

$u < 0$, v lies strictly between θ and $\pi/2$ (or $v = \pi/2$ if $\theta = \pi/2$), and

$$\alpha = A_\theta(u), \qquad A_\theta(u) = 2\,\operatorname{artanh}\frac{e^u\sin\theta}{\sqrt{1 - e^{2u}\cos^2\theta}}, \tag{11.11}$$

where A_θ is a strictly increasing bijection of $(-\infty,0)$ onto $(0,\infty)$. Conversely, a value $1/F$ of the form (11.10) with $\alpha > 0$ corresponds to $H = e^{i\theta-\alpha}$.

Proof. (i) Since H is not constant, $|H| < 1$ in $\mathbb{C}_+$ by the maximum principle, so $H \neq e^{\pm i\theta}$ there. The Möbius map $w \mapsto e^{-i\theta}(w - e^{i\theta})/(w - e^{-i\theta})$ sends the unit circle into the real line[21] and 0 to $e^{i\theta} \in \mathbb{C}_+$, so it maps the open disc onto $\mathbb{C}_+$. The symmetry $H(-\bar z) = \overline{H(z)}$ gives $F(-\bar z) = e^{-i\theta}(\bar H - e^{i\theta})/(\bar H - e^{-i\theta}) = 1/\overline{F(z)}$, hence the symmetry of ζ. The function ζ is a Herglotz function with bounded imaginary part, so in (4.1) $\beta = 0$ and $\mathrm{d}\mu = \pi^{-1}g\,\mathrm{d}t$ with g the boundary values of $\operatorname{Im}\zeta$ [25, Ch. I]. On $i\mathbb{R}_+$ the symmetry reads $\zeta = -\bar\zeta$, so $\operatorname{Re}\zeta = 0$ there and $c = \operatorname{Re}\zeta(i) = 0$. The symmetry also makes g even, and folding the integral over $\mathbb{R}$ onto $(0,\infty)$ with $\frac{1}{t-z} - \frac{1}{-t-z} = \frac{2z}{t^2-z^2}$ gives (11.9); the terms $t/(1+t^2)$ cancel. (ii) The inverse Möbius map in (11.8) sends $\mathbb{C}_+$ into the open unit disc, and the symmetry of H follows by reversing the computation in (i). (iii) The Möbius map sends the unit circle onto the real line, and $H = 0$ exactly when $F = e^{-i\theta}(-e^{i\theta})/(-e^{-i\theta}) = e^{i\theta}$. (iv) Substituting $H = e^{i\theta-\alpha}$ in $1/F = e^{i\theta}(H - e^{-i\theta})/(H - e^{i\theta})$ gives

$$\frac{1}{F} = \frac{(e^{-\alpha} - 1)\cos\theta + i(e^{-\alpha} + 1)\sin\theta}{e^{-\alpha} - 1},$$

[21] For $w = e^{i\phi}$, $e^{i\phi} - e^{i\theta} = 2ie^{i(\phi+\theta)/2}\sin\frac{\phi-\theta}{2}$ and $e^{i\phi} - e^{-i\theta} = 2ie^{i(\phi-\theta)/2}\sin\frac{\phi+\theta}{2}$, so the image is $\sin\frac{\phi-\theta}{2}/\sin\frac{\phi+\theta}{2} \in \mathbb{R}$.

which is the first identity of (11.10). Since $1/F = e^{-u}(\cos v - i\sin v)$, the real parts give $e^u\cos\theta = \cos v$, and the imaginary parts $\tanh(\alpha/2) = e^u\sin\theta/\sin v$. As $\coth(\alpha/2) > 1$, $|1/F| > 1$, so $u < 0$, and then $|\cos v| < |\cos\theta|$ with the same sign, which places v between θ and $\pi/2$. Finally $\sin v = \sqrt{1 - e^{2u}\cos^2\theta}$ gives (11.11). The quotient $e^u\sin\theta/\sqrt{1-e^{2u}\cos^2\theta}$ increases with u, tends to 0 as $u \to -\infty$ and to 1 as $u \to 0$. □

So the band condition places the boundary value of ζ on the curve

$$\Gamma_\theta = \{u + iv :\ u < 0,\ e^u\cos\theta = \cos v\},$$

with v between θ and $\pi/2$, and the attenuation is an increasing function of u alone. Off the band, g is free in $[0,\pi]$; a lossless stretch has $g \in \{0,\pi\}$. In the variable ζ the zeros of H have disappeared.

11.3. The comparison theorem

We use two facts about Hardy spaces of the unit disc $\mathbb{D}$ [26, Chs. 3, 4]. (a) If f is analytic in $\mathbb{D}$ and $|\arg f| \le \lambda < \pi/2$, then $f \in H^q$ for every $q < \pi/(2\lambda)$; this follows from the case $\lambda = \pi/2$, $q < 1$, applied to $f^{\pi/(2\lambda)}$. (b) If $\operatorname{Im} f$ is bounded, then $f \in H^p$ for every $p < \infty$, by the theorem of M. Riesz. A function in H^1 is the Poisson integral of its boundary values, and so is its imaginary part. Through the Cayley map $w = (z-i)/(z+i)$ these statements apply to functions in $\mathbb{C}_+$, and for Y in H^1 of the disc they give

$$Y(z) = \operatorname{Re} Y(i) + \frac{1}{\pi}\int_{\mathbb{R}} \operatorname{Im} Y(t)\Big(\frac{1}{t-z} - \frac{t}{1+t^2}\Big)\,\mathrm{d}t, \tag{11.12}$$

with $\int |Y(t)|\,\mathrm{d}t/(1+t^2) < \infty$.[22]

Proposition 11.4. *Let $0 < \theta < \pi$ and let $H^\sharp$ be admissible for θ, with*

$$g_\sharp = \pi \text{ on } (0,a), \qquad g_\sharp = 0 \text{ on } (b,\infty) \tag{11.13}$$

for the function $g_\sharp$ of Lemma 11.3 belonging to $H^\sharp$. Then every H admissible for θ satisfies $\alpha \ge \alpha^\sharp$ on $[a,b]$, where $\alpha^\sharp$ is the attenuation of $H^\sharp$. If $\alpha = \alpha^\sharp$ on a set of positive measure, then $H = H^\sharp$. In particular at most one admissible system satisfies (11.13).

Proof. Let $\zeta = u + iv$ and $\zeta_\sharp = u_\sharp + iv_\sharp$ be the functions of Lemma 11.3 for H and $H^\sharp$, and $d = \zeta - \zeta_\sharp$. The function F of H is continuous up to (a,b), so $u = \log|F|$ is continuous there with values in $[-\infty, 0)$, and $u = -\infty$ exactly on the closed set Z where $H = e^{i\theta}$. Since $\operatorname{Im}\zeta$ is bounded, ζ has finite boundary values almost everywhere (fact (b) above), so Z has measure zero. The attenuation $\alpha^\sharp$ of $H^\sharp$ is positive on (a,b), since $u_\sharp$ is finite there, $H^\sharp$ being regular on the band. By Lemma 11.3(iv) it suffices to show $u \ge u_\sharp$ almost everywhere on (a,b): then, by continuity, everywhere, so that Z is empty and $\alpha = A_\theta(u) \ge A_\theta(u_\sharp) = \alpha^\sharp$.

Step 1: the chord. The curve Γ_θ is the graph of $V(u) = \arccos(e^u\cos\theta)$, and $V'(u) = -\cot V(u)$, where $\cot V$ has the sign $\epsilon = \operatorname{sign}\cos\theta$ and $|\cot V| \le |\cot\theta|$ because V lies between θ and $\pi/2$. By the mean value theorem, at each $x \in (a,b)\setminus Z$

$$d(x) = \big(u(x) - u_\sharp(x)\big)\big(1 - i\,m(x)\big)$$

with $m(x) = \cot V(\xi)$ for some ξ between $u(x)$ and $u_\sharp(x)$ (and $m = 0$ where $u = u_\sharp$). Put $\psi = \arctan m$. Then $|\psi| \le \lambda$, with $\lambda = |\pi/2 - \theta| < \pi/2$, ψ is measurable, and $1 - im = \sqrt{1+m^2}\,e^{-i\psi}$.

Step 2: a rotating factor. Define

$$\Xi(z) = \frac{1}{\pi}\int_a^b \psi(t)\,\frac{2t}{t^2 - z^2}\,\mathrm{d}t, \qquad X = e^\Xi.$$

The density $\frac{1}{\pi}\psi(|t|)\operatorname{sign} t$ on S is bounded, so $\operatorname{Im}\Xi$ is its Poisson integral: $|\operatorname{Im}\Xi| \le \lambda$ in $\mathbb{C}_+$, $\operatorname{Im}\Xi = \psi$ almost everywhere on (a,b), $\operatorname{Im}\Xi = 0$ on $\mathbb{R}\setminus S$, and $\Xi(-\bar z) = \overline{\Xi(z)}$. Hence X is real and positive on $i\mathbb{R}_+$ and on $\mathbb{R}\setminus S$, $X \to 1$ at infinity, and $|\arg X| \le \lambda$. If $\theta = \pi/2$, then $X \equiv 1$.

Step 3: the product. Let $Y = Xd$. By (a), $X \in H^q$ for some $q > 1$; by (b), $d \in H^p$ for every $p < \infty$, since $|\operatorname{Im} d| \le \pi$. By Hölder's inequality $Y \in H^1$. At almost every $x \in (a,b)$,

$$Y(x) = |X(x)|\,\sqrt{1+m(x)^2}\,\big(u(x) - u_\sharp(x)\big) \in \mathbb{R}, \tag{11.14}$$

since the factor $e^{i\psi}$ of X cancels $e^{-i\psi}$. On $\mathbb{R}\setminus S$, $\operatorname{Im} Y = X\,(g - g_\sharp)$, which by (11.13) and $0 \le g \le \pi$ is ≤ 0 on $(0,a)$ and ≥ 0 on (b,∞). Both X and d have the symmetries $X(-\bar z) = \overline{X(z)}$, $d(-\bar z) = -\overline{d(z)}$, so $Y(-\bar z) = -\overline{Y(z)}$; thus $\operatorname{Im} Y$ is even on $\mathbb{R}$ and Y is purely imaginary on $i\mathbb{R}_+$, so that $\operatorname{Re} Y(i) = 0$.

Step 4: the sign. By (11.12), folded onto $(0,\infty)$ as in Lemma 11.3, and since $\operatorname{Im} Y = 0$ on (a,b),

$$Y(z) = \frac{1}{\pi}\int_{(0,a)\cup(b,\infty)} \operatorname{Im} Y(t)\,\frac{2z}{t^2 - z^2}\,\mathrm{d}t.$$

The right side is analytic across (a,b), so for $a < x < b$

$$Y(x) = \frac{1}{\pi}\int_0^a \big(-\operatorname{Im} Y(t)\big)\frac{2x}{x^2 - t^2}\,\mathrm{d}t + \frac{1}{\pi}\int_b^\infty \operatorname{Im} Y(t)\,\frac{2x}{t^2 - x^2}\,\mathrm{d}t \ \ge\ 0, \tag{11.15}$$

both integrands being nonnegative. With (11.14) this gives $u \ge u_\sharp$ almost everywhere on (a,b).

Equality. If $\alpha = \alpha^\sharp$ on a set of positive measure, pick a point of it at which (11.14) holds. There $Y = 0$, so

[22] At $z = i$ the integral is $\frac{i}{\pi}\int \operatorname{Im} Y(t)\,\mathrm{d}t/(1+t^2)$, which is purely imaginary, so the real constant is $\operatorname{Re} Y(i)$. The imaginary part of the kernel is the Poisson kernel $y/((t-x)^2 + y^2)$.

both integrals in (11.15) vanish, and since $X>0$ and the kernels do not vanish, $g=g_\sharp$ almost everywhere off the band. Then $\operatorname{Im} Y=0$ almost everywhere on $\mathbb{R}$, so $Y\equiv 0$ by (11.12), hence $d\equiv 0$ and $H=H^\sharp$. □

Remark 11.5. For zero-free systems the analogue of ζ is ih, and the constraint $\operatorname{Re} h=\theta$ on the band is linear; there the coefficient $\beta\ge 0$ of the linear term of h acts as a single global constraint, which Corollary 9.3 turns into a pointwise price $r(x)/x$ per unit of $-\beta_g$. In the variable ζ the linear term is absent, because $\operatorname{Im}\zeta$ is bounded, and the band condition is a curve rather than a line. The proof above replaces linearity by the observation that the chords of Γ_θ have slopes in a fixed sector, which the factor X rotates to the real axis. The constraint $\beta\ge 0$ is what zeros remove: in the factorisation $H_\theta^\sharp=-B_{\eta_\star}e^{ih_{g_\sharp}}$ the zero-free factor has $\beta_{g_\sharp}=0$ (Lemma 11.7).

11.4. The right angle

Proof of Theorem 11.2. Let $Q(z)=(z+a)(z+b)$ and

$$\zeta_0(z)=\tfrac12\big(\log(z-a)+\log(z-b)-\log(z+a)-\log(z+b)\big)$$

with the arguments in $(0,\pi)$. For $z\in\mathbb{C}_+$ the angle $\arg(z-c)-\arg(z+c)$ lies in $(0,\pi)$ for $c=a,b$,[23] so $0<\operatorname{Im}\zeta_0<\pi$. On $(0,a)$, $\operatorname{Im}\zeta_0=\frac12(\pi+\pi)=\pi$; on (b,∞) it is 0; on (a,b) it is $\pi/2$, and $\zeta_0(-\bar z)=-\overline{\zeta_0(z)}$. So ζ_0 has the form (11.9), with $g_0=\pi$ on $(0,a)$, $\pi/2$ on (a,b), 0 on (b,∞). Since $e^{2\zeta_0}=(z-a)(z-b)/Q(z)=R^2/Q^2$ and both e^{ζ_0} and R/Q tend to 1 at infinity, $F_0=e^{\zeta_0}=R/Q$. By Lemma 11.3(ii), (11.8) with $\theta=\pi/2$ defines a passive system

$$K=\frac{F_0-i}{iF_0-1}=\frac{R-iQ}{iR-Q}=-\frac{z^2+ab}{R+i(a+b)z},$$

where the last step uses $R^2=(z^2-a^2)(z^2-b^2)$ and $(z^2-a^2)(z^2-b^2)+(a+b)^2z^2=(z^2+ab)^2$.[24] On the band $F_0=ir/Q$ is purely imaginary, so $\operatorname{Re}(1/F_0)=0=\cos(\pi/2)$ and, by (11.10), $K=e^{i\pi/2-\alpha}$ with $\tanh(\alpha/2)=e^u=r/Q$, which is the second form of (11.5). The first form is $\log\frac{1+s}{1-s}$ with $s=r/Q$, because $(Q+r)(\omega^2+ab)=(Q-r)\big((a+b)\omega+r\big)$ on the band.[25] The system K is regular on the band, with $K\to e^{i\pi/2}$ at the band edges, and its function g_0 satisfies (11.13). By Proposition 11.4, K is the unique admissible system with this property. It has exactly one zero, a simple one at $i\sqrt{ab}$, where $z^2+ab=0$ and $R+i(a+b)z=-2(a+b)\sqrt{ab}\neq 0$;[26] moreover $K(0)=-ab/R(0)=1$ and $K(iy)\to -1$. So Lemma 11.7 below, applied to K, gives $K=H^\sharp_{\pi/2}$ and $\eta_\star=\sqrt{ab}$. (Directly: the substitution $t=ab/s$ maps $\mathrm{d}t/r(t)$ to $\mathrm{d}s/r(s)$ and $\arctan(t/\sqrt{ab})$ to $\frac\pi2-\arctan(s/\sqrt{ab})$, so the left side of (11.1) at $\eta=\sqrt{ab}$ is $\frac\pi2\int\mathrm{d}t/r$.)

For (11.6), write $K=-B_{\sqrt{ab}}\,e^{ih}$, where $h=h_{g_\sharp}$. By Proposition 8.4(iii) and (4.4),

$$h(z)=\frac1\pi\int_a^b \alpha^\sharp{}_{\pi/2}(t)\,\frac{2z}{t^2-z^2}\,\mathrm{d}t,$$

so

$$\lim_{y\to\infty} iy\,h(iy)=-\frac2\pi\int_a^b\alpha^\sharp{}_{\pi/2}\,\mathrm{d}t.$$

On the other hand $R+i(a+b)z=z^2\big(1+i(a+b)/z+O(z^{-2})\big)$ and $-B_\eta(z)=-1+2i\eta/z+O(z^{-2})$, so

$$e^{ih}=\frac{K}{-B_{\sqrt{ab}}}=1-\frac{i\,(a+b-2\sqrt{ab})}{z}+O(z^{-2}),$$

whence $zh(z)\to-(\sqrt b-\sqrt a)^2$, which is (11.6).

Finally, $q(\omega)=(\omega-a)(b-\omega)/((\omega+a)(\omega+b))$ satisfies $q(ab/\omega)=q(\omega)$, and $\log q$ is strictly concave on (a,b), since $(\log q)''=\frac{1}{(\omega+a)^2}-\frac{1}{(\omega-a)^2}+\frac{1}{(\omega+b)^2}-\frac{1}{(b-\omega)^2}<0$. So its only maximum is at the fixed point $\sqrt{ab}$, where $\sqrt q=(\sqrt b-\sqrt a)/(\sqrt b+\sqrt a)$ and $2\operatorname{artanh}\sqrt q=\log(\sqrt b/\sqrt a)$. The bound (11.7) is $|H|=e^{-\alpha}\le e^{-\alpha^\sharp{}_{\pi/2}}$, from Theorem 11.1(ii), applied to H or $-H$. □

11.5. Existence for every phase

For $\theta\neq\pi/2$ the optimal system is not algebraic, and we construct it by a conformal map. In the variable $\sigma=z^2$, the first quadrant becomes $\mathbb{C}_+$, and the boundary conditions of Lemma 11.3 for a system with (11.13) become conditions on $G=1/F$: G is real and below -1 on $(0,a^2)$, real and above 1 on (b^2,∞), unimodular on $(-\infty,0)$ (the imaginary axis of z, where $\operatorname{Re}\zeta=0$), and $\operatorname{Re} G=\cos\theta$ on (a^2,b^2), by (11.10). This describes a slit domain (Figure 14).

Proposition 11.6. *Let $0<\theta<\pi$. There is a system $H^\sharp$, admissible for θ, with (11.13), lossless on $\mathbb{R}\setminus S$, with exactly one zero in $\mathbb{C}_+$, which is simple and lies on $i\mathbb{R}_+$, and with $H^\sharp(0)=1$, $H^\sharp(iy)\to-1$ as $y\to\infty$, and attenuation positive on (a,b) and zero at a and b.*

Proof. For $T>\sin\theta$ let $\Sigma_T=\{\cos\theta-is:\ s\ge T\}$ and

$$D_T=\{G:\ \operatorname{Im}G<0,\ |G|>1\}\setminus\Sigma_T.$$

[23] It is the angle at z of the triangle with vertices $-c$, c, z.

[24] Cross-multiplying, $(R-iQ)(R+i(a+b)z)=R^2+(a+b)zQ-iR(z^2+ab)$, since $(a+b)z-Q=-(z^2+ab)$, while $-(z^2+ab)(iR-Q)=(z^2+ab)Q-iR(z^2+ab)$, and $R^2=Q\,(z-a)(z-b)=Q(z^2+ab-(a+b)z)$.

[25] Write $c=(a+b)\omega$ and $m=\omega^2+ab$, so $Q=m+c$. The two sides are $m^2+cm+rm$ and $mc+mr+c^2-r^2$, and $c^2-m^2=(a^2+b^2)\omega^2-\omega^4-a^2b^2=r^2$.

[26] On $i\mathbb{R}_+$, $R(iy)=-\sqrt{(y^2+a^2)(y^2+b^2)}$, being real, nonvanishing and $\sim-y^2$.

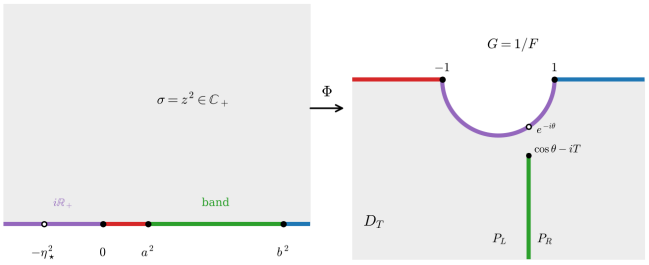


FIG. 14. The construction of Proposition 11.6. Left: the plane of $\sigma = z^2$, with the boundary arcs coloured. Right: the domain D_T of $G = 1/F$, the part of the lower half-plane outside the unit disc, cut along the vertical ray $\operatorname{Re} G = \cos\theta$ below the tip $\cos\theta - iT$. The map Φ sends each arc on the left to the arc of the same colour; the two ends of the band go to the two ends of the slit at infinity. The zero of the optimal system is the preimage of $e^{-i\theta}$.

The ray Σ_T does not meet the closed unit disc, since $|\cos\theta - is| > 1$ for $s > \sin\theta$. The domain D_T is simply connected and its boundary is locally connected, so a conformal map Φ of $\mathbb{C}_+$ onto D_T extends to a homeomorphism of $\mathbb{R} \cup \{\infty\}$ onto the prime ends of D_T [27, Ch. 2]. Traversed with D_T on the left, the boundary consists of the lower unit semicircle C from 1 to -1, the ray $(-\infty, -1]$, the left side of Σ_T up to the tip and its right side down again, and the ray $[1, \infty)$. It has two prime ends at infinity: P_L, between $(-\infty, -1]$ and the left side of the slit, and P_R, between the right side and $[1, \infty)$. We normalise Φ by $\Phi(\infty) = 1$, $\Phi(0) = -1$ and $\Phi(a^2) = P_L$, and let $s(T) > a^2$ be the preimage of P_R.

Choice of T. Let $\lambda(T)$ be the extremal distance in D_T between C and Σ_T [28, Ch. 4]. By conformal invariance it equals the extremal distance in $\mathbb{C}_+$ between $(-\infty, 0)$ and $[a^2, s(T)]$, which is $L(s(T)/a^2)$ for a continuous strictly decreasing bijection L of $(1, \infty)$ onto $(0, \infty)$.[27] The function λ is continuous in T: for T' near T a diffeomorphism $G \mapsto G - i(T' - T)\chi(G)$, with χ a smooth bump equal to 1 near the tip and supported in a disc around it that avoids the unit circle and the real axis, maps D_T onto $D_{T'}$ arc by arc, with dilatation $1 + O(|T' - T|)$, and extremal distances change at most by that factor.[28] As $T \to \infty$, every curve in D_T from C to Σ_T crosses the annulus $1 < |G| < T$, so $\lambda(T) \geq \frac{1}{2\pi} \log T \to \infty$. As $T \downarrow \sin\theta$, the tip $\cos\theta - iT$ lies at the distance $\delta = T - \sin\theta$ below the point $e^{-i\theta}$ of C, and for $\delta < \rho < \rho_0$ an arc of the circle $\left|G - e^{-i\theta}\right| = \rho$ of length less than $2\pi\rho$ joins C to Σ_T inside D_T; this gives $\lambda(T) \leq 2\pi / \log(\rho_0/\delta) \to 0$. Hence $s(T)/a^2$ is continuous and takes every value in $(1, \infty)$, and we fix T with $s(T) = b^2$.

The system. On the open first quadrant put $F(z) = 1/\Phi(z^2)$, and on the open second quadrant $F(z) = 1/\overline{F(-\bar{z})}$. On $i\mathbb{R}_+$, $z^2 < 0$ is mapped to C, where $|\Phi| = 1$, so the two definitions have the same continuous boundary values there, and F is analytic in $\mathbb{C}_+$ by Morera's theorem. F maps $\mathbb{C}_+$ into $\mathbb{C}_+$, since $\operatorname{Im}\Phi < 0$, and $F(-\bar{z}) = 1/\overline{F(z)}$. Define $H^\sharp$ from F by (11.8). By Lemma 11.3(ii) it is a passive system. On $(0, a)$, $\Phi(t^2) < -1$, so $F \in (-1, 0)$ and $g_\sharp = \pi$; on (b, ∞), $F \in (0, 1)$ and $g_\sharp = 0$; in both cases $\left|H^\sharp\right| = 1$. On (a, b), $\Phi(t^2) = \cos\theta - is$ with $s \geq T > \sin\theta$, which by (11.10) is $H = e^{i\theta - \alpha}$ with $\coth(\alpha/2) = s/\sin\theta$, so $\alpha > 0$, and $\alpha \to 0$ at the band edges, where $s \to \infty$, $F \to 0$ and $H^\sharp \to e^{i\theta}$. Hence $H^\sharp$ is continuous and nonzero on a neighbourhood of $[a, b]$, that is, regular on the band, with argument θ there. By Lemma 11.3(iii) its zeros are the points with $1/F = e^{-i\theta}$, a point of C; Φ takes this boundary value at exactly one point of $(-\infty, 0)$, and analytically, since C is an analytic arc, with nonzero derivative. So $H^\sharp$ has exactly one zero in $\mathbb{C}_+$, a simple one on $i\mathbb{R}_+$. Finally $\Phi(0) = -1$ gives $F(0) = -1$ and $H^\sharp(0) = 1$, and $\Phi(\sigma) \to 1$ as $\sigma \to -\infty$ gives $H^\sharp(iy) \to -1$. □

Lemma 11.7. *Let $H^\sharp$ be admissible for θ, with (11.13), lossless on $\mathbb{R} \setminus S$, with exactly one zero in $\mathbb{C}_+$, a simple one at $i\eta$, and with $H^\sharp(0) = 1$ and $H^\sharp(iy) \to -1$ as $y \to \infty$; for instance the system of Proposition 11.6. Then $H^\sharp = -B_\eta e^{ih_g}$ with $g(\omega) = \theta - 2\arctan(\omega/\eta)$, $\beta_g = 0$, and the attenuation of $H^\sharp$ on the band is v_g. Consequently η is the root $\eta_\star$ of (11.1).*

Proof. The quotient $E = H^\sharp/(-B_\eta)$ is analytic and zero-free in $\mathbb{C}_+$, unimodular where $H^\sharp$ is, and bounded, hence bounded by 1 by the maximum principle, since $|B_\eta| \to 1$ at $\mathbb{R} \cup \{\infty\}$. It has the real symmetry and $E(0) = H^\sharp(0)/(-B_\eta(0)) = 1$, so by Lemma 4.2 $E = e^{ih}$ with an odd Herglotz function h, $h(0) = 0$. On $\mathbb{R} \setminus S$, $|E| = 1$, so $\operatorname{Im} h = 0$ there: the measure of h is carried by S, and by (4.1), h being odd, $h(z) = \beta z + O(z^{-1})$ at infinity. Along $[0, a]$ the function F of $H^\sharp$ is negative, by (11.13), with $F(0) = -1$ and $F(a) = 0$, so the argument of $H^\sharp$ moves continuously from 0 to θ inside $(-\theta, \theta]$,[29] and that of $-B_\eta$ from 0 to $2\arctan(a/\eta)$ by (10.2). So on the band $\operatorname{Re} h = \theta - 2\arctan(\omega/\eta) = g$, with no multiple of 2π. The argument of Proposition 8.4(iv) now applies to $h - h_g$: it is real on $\mathbb{R} \setminus S$, has real part 0 on the band from above, is continuous up to S (since E is continuous

[27] By scaling, $L(\rho)$ is the extremal distance between $(-\infty, 0)$ and $[1, \rho]$. It decreases because the curve family grows with ρ; it tends to ∞ as $\rho \to 1$, since every curve crosses the half-annulus $\rho - 1 < |\sigma - 1| < 1$, of extremal distance $\frac{1}{\pi} \log \frac{1}{\rho - 1}$; and to 0 as $\rho \to \infty$, since the half-circles $|\sigma| = t$, $1 < t < \rho$, join the two sets and have length πt, which gives modulus at least $\frac{1}{\pi} \log \rho$. Strictness and continuity follow from the explicit value as a ratio of complete elliptic integrals.

[28] It moves the slit along itself and fixes the rest of the boundary; it is injective once $|T' - T|\,|\nabla\chi| < 1$. A diffeomorphism whose differential has ratio of singular values at most K changes an extremal distance by a factor in $[1/K, K]$, by transporting the admissible metrics [28, Ch. 4].

[29] The inverse Möbius map of (11.8) sends $(-\infty, 0]$ to the arc of the unit circle from $e^{-i\theta}$ to $e^{i\theta}$ through 1, with $-1 \mapsto 1$ and $0 \mapsto e^{i\theta}$.

and nonzero near the band), so $(h-h_g)/R$ is entire, and it is $O(z^{-1})$ at infinity. Hence $h=h_g$ and $\beta=\beta_g$. Finally $E(iy)\to(-1)/(-1)=1$, so $h(iy)\to 0$ and $\beta=0$. The attenuation is $\operatorname{Im} h_g=v_g$ on the band. The left side of (11.1) decreases strictly in η from $\pi\int \mathrm{d}t/r$ to 0, so it has exactly one root, and $\beta_g=0$ says that η is this root. □

Proof of Theorem 11.1. The system $H_\theta^\sharp=-B_{\eta_\star}e^{ih_{g_\sharp}}$ is the system of Proposition 11.6, by Lemma 11.7. This gives (i), with $\alpha^\sharp{}_\theta=v_{g_\sharp}>0$ on the open band. Since $g_\sharp\ge 0$ and decreasing for $\eta\ge\eta_\theta$, $\beta_{g_\eta}>0$ there (Lemma 8.5); and β_{g_η} increases with η; so $\eta_\star<\eta_\theta$. Part (ii) is Proposition 11.4. For (iii), by Proposition 8.4(iii) with $v_{g_\sharp}\ge 0$ and $\beta_{g_\sharp}=0$, $\int_a^b\alpha^\sharp{}_\theta=\frac12\int_a^b g_\sharp\,p$, and the computation in the proof of Proposition 10.2 gives $\theta\Lambda-\delta(\eta_\star)$. It is positive since $\alpha^\sharp{}_\theta>0$. The systems H_η of Proposition 10.2 are admissible and differ from $H_\theta^\sharp$, since $\beta_{g_\eta}>0$, so by (ii) their mean is strictly larger. The case $\pi<\theta<2\pi$ follows by passing to $-H$. □

11.6. Consequences

Corollary 11.8. *Let $0<\theta<\pi$ and let H be admissible for θ.*

(i) For every measurable $w\ge 0$ on $[a,b]$, $\int w\alpha\ge\int w\alpha^\sharp{}_\theta$; in particular $\max_{[a,b]}\alpha\ge\max_{[a,b]}\alpha^\sharp{}_\theta$. The system $H_\theta^\sharp$ attains all of these bounds at once.

(ii) A phase lead costs at least as much as no delay: if H is passive, regular on the band, and $\arg H\equiv\theta-\tau\omega \pmod{2\pi}$ on $[a,b]$ for some $\tau\ge 0$, then $\alpha\ge\alpha^\sharp{}_\theta$ on $[a,b]$.

(iii) Every zero-free system with phase θ on the band, in the sense of Theorem 6.2, satisfies $\alpha\ge\alpha^\sharp{}_\theta$ pointwise, in addition to $\frac{1}{b-a}\int\alpha\ge\theta\Lambda$.

Proof. (i) is (11.2). (ii) The system $e^{i\tau z}H$ is passive, has the same modulus on $\mathbb{R}$, and has argument θ on the band. (iii) is a special case of (ii) with $\tau=0$. □

Part (i) answers, for systems with zeros allowed, the worst-frequency question of the zero-free case. Part (ii) does not extend to a delay, $\arg H=\theta+\tau\omega$ with $\tau>0$: the system $e^{i\tau z}H_\theta^\sharp$ still has the loss $\alpha^\sharp{}_\theta$, but it is not optimal: for $\theta=\pi/2$, $\tau=5$ on $[1,2]$ the least mean loss of a zero-free system is about forty times smaller (Theorems 12.2 and 12.6), and with zeros the least loss falls exponentially in τ (Corollary 12.12).

Corollary 11.9. *Fix $0<\theta<\pi$, and let $f=(a+b)/2$, $\Delta f=b-a$. As $\Delta f/f\to 0$,*

$$\Lambda_{\mathrm{all}}(\theta)=\frac{\pi}{8}\sin\theta\,\frac{\Delta f}{f}+O\Big(\Big(\frac{\Delta f}{f}\Big)^2\Big),$$

so that $\Lambda_{\mathrm{all}}(\theta)/\theta\Lambda\to\sin\theta/\theta$.

Proof. Write $\epsilon=\Delta f/f$. By (1.2), $\Lambda=\frac{\pi}{8}\epsilon+O(\epsilon^3)$. By the proof of Proposition 10.2, $\int_a^b p=J=2(b-a)\Lambda$, and by Lemma 5.3(i), $\int_a^b\omega p=0$, so $\frac{1}{b-a}\int_a^b(\omega-f)p=-2f\Lambda$. Taylor's formula for $\arctan(\omega/\eta)$ at $\omega=f$, with a remainder $O((\omega-f)^2/\eta^2)$ that is integrated against $|p|$, whose mean over the band is $O(1)$,[30] gives

$$\delta(\eta)=2\Lambda\arctan\frac{f}{\eta}-2\Lambda\frac{f\eta}{f^2+\eta^2}+O\Big(\frac{\epsilon^2f^2}{\eta^2}\Big).$$

The root of (11.1) satisfies $2\arctan(f/\eta_\star)=\theta+O(\epsilon)$, since the weight $\mathrm{d}t/r$ is carried by an interval of length ϵf around f. With $\phi=2\arctan(f/\eta_\star)$ and $2f\eta/(f^2+\eta^2)=\sin\phi$,

$$\begin{aligned}\Lambda_{\mathrm{all}}&=\theta\Lambda-\delta(\eta_\star)=\Lambda\,(\theta-\phi+\sin\phi)+O(\epsilon^2)\\&=\Lambda\sin\theta+O(\epsilon^2),\end{aligned}$$

because $\Lambda=O(\epsilon)$ and $\theta-\phi=O(\epsilon)$; also $f/\eta_\star$ stays bounded. □

In the computations (Appendix D) the relative error is of the order of $(\Delta f/f)^2$: the ratio of Λ_{all} to the leading term is 1.0021 for $b/a=1.2$ and 1.000149 for $b/a=1.05$ at $\theta=\pi/2$. For small phases zeros hardly help: the gain is a factor $\sin\theta/\theta=1-\theta^2/6+\dots$. For the 1° shift at 5 MHz of Section 13, the least mean attenuation with zeros allowed is 1.37588×10^{-3} Np, against 1.37595×10^{-3} Np without zeros.

For general θ the profile $\alpha^\sharp{}_\theta$ is elliptic, not elementary (Section 11.7), and a simple bound with the right shape survives, for the equilibrium weight of the band

$$w_0(\omega)=\frac{1}{I_0\,r(\omega)},\qquad I_0=\int_a^b\frac{\mathrm{d}t}{r(t)}=\frac{K(k)}{b},$$

$k^2=1-a^2/b^2$.

Corollary 11.10. *Let $0<\theta<\pi$ and let H be admissible for θ. Then*

$$\int_a^b w_0\log\tanh\frac{\alpha}{2}\,\mathrm{d}\omega\;\ge\;\log\sin\theta-\pi\frac{K(a/b)}{K(k)},\tag{11.16}$$

that is, the w_0-geometric mean of $\tanh(\alpha/2)$ is at least $q\sin\theta$, where $q=e^{-\pi K(a/b)/K(k)}$ is the nome of the modulus k. Equality holds if and only if $\theta=\pi/2$ and $H=H^\sharp_{\pi/2}$. For an octave $q=0.085796$, for a decade $q=0.26220$.

Proof. By Lemma 11.3(iv), $\tanh(\alpha/2)=e^u\sin\theta/\sin v\ge e^u\sin\theta$, with equality only where $v=\pi/2$, which requires $\theta=\pi/2$. Let ζ_0 be the function in the proof of Theorem 11.2 and $d=\zeta-\zeta_0$, so that $\operatorname{Im}d=g-g_0$, which is ≤ 0 on $(0,a)$ and ≥ 0 on (b,∞). Both functions have the form (11.9), so $\operatorname{Re}d(x)=\frac{1}{\pi}\,\mathrm{p.v.}\int_0^\infty\operatorname{Im}d(t)\,\frac{2x}{t^2-x^2}\,\mathrm{d}t$

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[30] $\ln\omega=f+\frac{\Delta f}{2}s$, $p\,\mathrm{d}\omega/(b-a)=-s\,\mathrm{d}s/\sqrt{1-s^2}\,(1+O(\epsilon))$.

almost everywhere. Integrating against w_0 and exchanging the integrals, by Tonelli's theorem for $t \notin (a,b)$ and by the antisymmetry of the Hilbert transform between L^p and $L^{p'}$ for $t \in (a,b)$ ($\operatorname{Im} d$ is bounded and $w_0 \in L^{p'}$ for $p' < 2$), gives

$$\int_a^b w_0 \operatorname{Re} d = \frac{1}{\pi}\int_0^\infty \operatorname{Im} d(t)\, k_0(t)\,\mathrm{d}t,$$
$$k_0(t) = \text{p. v.}\int_a^b \frac{2x\, w_0(x)}{t^2 - x^2}\,\mathrm{d}x.$$

In $\sigma = x^2$,

$$k_0(t) = \frac{1}{I_0}\ \text{p. v.}\int_{a^2}^{b^2} \frac{\mathrm{d}\sigma}{(t^2-\sigma)\sqrt{(\sigma-a^2)(b^2-\sigma)}},$$

which vanishes for $a < t < b$ by Glauert's integral (proof of Lemma 8.5), is negative for $t < a$ and positive for $t > b$. So $\int w_0 \operatorname{Re} d \ge 0$, with equality only if $g = g_0$ off the band. For ζ_0 the same computation, with g_0 explicit, gives

$$\int_a^b w_0 \operatorname{Re}\zeta_0 = \int_0^a k_0(t)\,\mathrm{d}t$$
$$= -\frac{\pi}{I_0}\int_0^a \frac{\mathrm{d}t}{\sqrt{(a^2-t^2)(b^2-t^2)}},$$

using, for $s < A$,

$$\int_A^B \frac{\mathrm{d}\sigma}{(\sigma-s)\sqrt{(\sigma-A)(B-\sigma)}} = \frac{\pi}{\sqrt{(A-s)(B-s)}};$$

the last integral is $K(a/b)/b$.[31] Adding up gives (11.16). □

11.7. The optimum in closed form

The profile $\alpha^\sharp{}_\theta$ is elliptic for every θ.[32] Write sn, cn and am for the Jacobi functions of modulus k, $k^2 = 1 - a^2/b^2$ [32, Ch. 22], $F(\psi,k)$ for the incomplete integral of the first kind, and

$$\Pi(n;\psi,k) = \int_0^\psi \frac{\mathrm{d}v}{(1-n\sin^2 v)\sqrt{1-k^2\sin^2 v}},$$

with $\Pi(n,k) = \Pi(n;\pi/2,k)$. For $u \ge 0$ put $Q_u = \sqrt{(u^2+a^2)(u^2+b^2)}$, so that $R(iu) = -Q_u$ by Lemma 8.1 and continuity from $R(0) = -ab$.

Lemma 11.11. *For $u > 0$ let $j_u(t) = t/(t^2+u^2)$. Then*

$$h_{j_u}(z) = \frac{z}{z^2+u^2}\Big(1 + \frac{R(z)}{Q_u}\Big), \qquad \beta_{j_u} = \frac{1}{Q_u},$$

and $v_{j_u}(x) = x\, r(x)/((x^2+u^2)Q_u)$ on the band.

Proof. Call the right side ϕ_u. It is odd, real on $\mathbb{R}\setminus S$, analytic off S, since its apparent poles at $\pm iu$ cancel ($1 + R(\pm iu)/Q_u = 0$), continuous up to S, and $\phi_u(z) = z/Q_u + O(z^{-1})$. On the band $\operatorname{Re}\phi_u(x+i0) = j_u(x)$, because $R(x+i0) = ir(x)$. By the substitution $\sigma = t^2$ and the integral displayed in the proof of Corollary 11.10, $\beta_{j_u} = \frac{1}{\pi}\int_{a^2}^{b^2} \mathrm{d}\sigma/((\sigma+u^2)\sqrt{(\sigma-a^2)(b^2-\sigma)}) = 1/Q_u$. So $\phi_u - h_{j_u}$ has real part 0 on the band and is $O(z^{-1})$ at infinity, and the argument of the proof of Proposition 8.4(iv) gives $\phi_u = h_{j_u}$. □

Since $\arctan(t/\eta) = \frac{\pi}{2} - \int_0^\eta j_u(t)\,\mathrm{d}u$, the phase of Theorem 11.1 is $g_\sharp = \theta - \pi + 2\int_0^{\eta_\star} j_u\,\mathrm{d}u$, and by linearity

$$h_{g_\sharp} = (\theta-\pi)h_\circ + 2\arctan\frac{\eta_\star}{z} + 2zR(z)\int_0^{\eta_\star}\frac{\mathrm{d}u}{(z^2+u^2)Q_u}, \tag{11.17}$$

with $h_\circ$ and A as in Section 9. In particular

$$\alpha^\sharp{}_\theta(x) = 2x\, r(x)\int_0^{\eta_\star}\frac{\mathrm{d}u}{(x^2+u^2)Q_u} - (\pi-\theta)A(x). \tag{11.18}$$

A zero of H at iu thus acts on the band as a measure at the negative squared frequency $-u^2$; this is the mechanism behind Section 12.

Theorem 11.12. *Let $0 < \theta < \pi$. The root of (11.1) is*

$$\eta_\star = b\,\frac{\operatorname{cn}(u_\theta,k)}{\operatorname{sn}(u_\theta,k)}, \qquad u_\theta = \frac{\theta K(k)}{\pi}. \tag{11.19}$$

Proof. By (11.17), $\beta_{g_\sharp} = (\theta-\pi)\tau_\circ + 2\int_0^{\eta_\star}\mathrm{d}u/Q_u$, and (11.1) says $\beta_{g_\sharp} = 0$. The substitution $u = b\cot\phi$ gives $\mathrm{d}u/Q_u = -\,\mathrm{d}\phi/(b\sqrt{1-k^2\sin^2\phi})$, since $u^2 + a^2 = b^2(1 - k^2\sin^2\phi)/\sin^2\phi$, so

$$\int_\eta^\infty \frac{\mathrm{d}u}{Q_u} = \frac{1}{b}F\Big(\arctan\frac{b}{\eta}, k\Big), \qquad \int_0^\infty \frac{\mathrm{d}u}{Q_u} = \frac{K(k)}{b}.$$

With $\tau_\circ = 2K(k)/(\pi b)$ the equation becomes

$$F\big(\arctan(b/\eta_\star), k\big) = \theta K(k)/\pi,$$

that is, $\arctan(b/\eta_\star) = \operatorname{am}(u_\theta,k)$. □

At $\theta = \pi/2$, cn / sn at $K/2$ is $\sqrt{k'} = \sqrt{a/b}$, and $\eta_\star = \sqrt{ab}$ as in Theorem 11.2. Put $\eta = \eta_\star$ and

$$Q = Q_\eta, \qquad n = \frac{b^2-a^2}{b^2+\eta^2}, \qquad c_\theta = \frac{\Pi(n,k)}{(b^2+\eta^2)K(k)}. \tag{11.20}$$

[31] Substitute $t = a\sin\phi$: then $\sqrt{(a^2-t^2)(b^2-t^2)}$ is $a\cos\phi\cdot b\sqrt{1-(a/b)^2\sin^2\phi}$ and $\mathrm{d}t = a\cos\phi\,\mathrm{d}\phi$. Similarly $I_0 = K(k)/b$ by $t^2 = b^2 - (b^2-a^2)\sin^2\phi$.

[32] The explicit zero (11.19) and the elliptic profile of Theorem 11.13 were found with OpenAI's ChatGPT. The derivation through Lemma 11.11 is ours.

Theorem 11.13. *Let $0<\theta<\pi$. On the band,*

$$\alpha^\sharp{}_\theta(x)=\frac{2\eta Q}{b}\Big(c_\theta F(\psi,k)-\frac{\Pi(n;\psi,k)}{b^2+\eta^2}\Big),\qquad(11.21)$$

$$\sin^2\psi=\frac{b^2-x^2}{b^2-a^2},\qquad 0\le\psi\le\frac{\pi}{2},$$

$$\frac{\mathrm{d}\alpha^\sharp{}_\theta}{\mathrm{d}x}=\frac{2\eta Q}{r(x)}\Big(\frac{1}{x^2+\eta^2}-c_\theta\Big).\qquad(11.22)$$

The profile has a single maximum, at $x_\theta=\sqrt{1/c_\theta-\eta^2}$, and

$$\Lambda_{\mathrm{all}}(\theta)=\frac{\pi\eta}{b-a}\,(Qc_\theta-1).\qquad(11.23)$$

Near $z=0$, with $z=a\sin\phi$ and $\phi(0)=0$,

$$H^\sharp_\theta(z)=\exp\Big[\frac{2iQ}{b\eta}\Big(\Pi\Big(-\frac{a^2}{\eta^2};\phi,\frac{a}{b}\Big)-\eta^2c_\theta F\Big(\phi,\frac{a}{b}\Big)\Big)\Big].\qquad(11.24)$$

Proof. Let $h=h_{g_\sharp}$ and $\Psi(z)=R(z)\big(h'(z)+2\eta/(z^2+\eta^2)\big)$. The function $g_\sharp$ extends analytically to the strip $|\operatorname{Im}z|<\eta$, and $h-g_\sharp$ has real part 0 on the band, so by Schwarz reflection h continues analytically across (a,b) and $h'(x+i0)=g_\sharp'(x)+i\alpha^{\sharp\prime}(x)$ with $g_\sharp'=-2\eta/(x^2+\eta^2)$. Hence $\Psi(x+i0)=-r(x)\alpha^{\sharp\prime}(x)$ is real, and by the symmetry $h(\bar z)=\overline{h(z)}$ it equals $\Psi(x-i0)$; so Ψ has no jump across the band, nor, being even, across $(-b,-a)$. Near the ends of S, (11.17) gives

$$\Psi=(\theta-\pi)\tau_\circ(z^2-m^2)+2R^2I+2zRR'I+2zR^2I',$$

with $I(z)=\int_0^\eta \mathrm{d}u/((z^2+u^2)Q_u)$, which is analytic near $\pm a,\pm b$, and $RR'=z(2z^2-a^2-b^2)$; so Ψ is bounded there and the ends are removable. Since h is analytic off S, the only other singularities of Ψ are at $\pm i\eta$, from the term $2\eta R/(z^2+\eta^2)$, and since $R(\pm i\eta)=-Q$, the function $\Psi+2\eta Q/(z^2+\eta^2)$ is entire. At infinity $h=O(z^{-1})$, because $\beta_{g_\sharp}=0$, so $h'=O(z^{-2})$ and Ψ is bounded. Hence $\Psi=d-2\eta Q/(z^2+\eta^2)$ for a constant d, and

$$h'(z)=-\frac{2\eta}{z^2+\eta^2}+\frac{d-2\eta Q/(z^2+\eta^2)}{R(z)}.$$

On the band, $\alpha^{\sharp\prime}=-\Psi/r$ gives (11.22) with $d=2\eta Qc$ for some c. Since $\alpha^\sharp$ vanishes at a and b (Theorem 11.1), $\int_a^b\alpha^{\sharp\prime}=0$ and $c=\int_a^b\mathrm{d}t/((t^2+\eta^2)r)/\int_a^b\mathrm{d}t/r$. The substitution $t=b\sqrt{1-k^2\sin^2v}$, for which $\mathrm{d}t/r=-\,\mathrm{d}v/t$ and $t^2+\eta^2=(b^2+\eta^2)(1-n\sin^2v)$, gives $c=c_\theta$, and integrating (11.22) from x to b gives (11.21). As c_θ is a strict average of $(t^2+\eta^2)^{-1}$ over the band, $\alpha^{\sharp\prime}$ changes sign once, at x_θ. At infinity $h'=2\eta(Qc_\theta-1)z^{-2}+O(z^{-4})$, so $h=-2\eta(Qc_\theta-1)/z+O(z^{-3})$, while by (4.4) with no measure off S, $h=-\frac{2}{\pi}\int_a^b\alpha^\sharp/z+O(z^{-3})$; this gives (11.23). Finally $H^\sharp_\theta=-B_\eta e^{ih}$, $B_\eta'/B_\eta=2i\eta/(z^2+\eta^2)$, so

$$\frac{(H^\sharp_\theta)'}{H^\sharp_\theta}=\frac{2i\eta Q}{R(z)}\Big(c_\theta-\frac{1}{z^2+\eta^2}\Big),$$

and $H^\sharp_\theta(0)=1$. Near 0 put $z=a\sin\phi$; then $R=-ab\cos\phi\sqrt{1-(a/b)^2\sin^2\phi}$ and $z^2+\eta^2=\eta^2(1+(a/\eta)^2\sin^2\phi)$, and integrating gives (11.24). □

The formula (11.24) holds near the origin and is continued analytically along paths in $\mathbb{C}\setminus S$; the residue 1 of the logarithmic derivative at $i\eta$ makes the continuation single-valued with a simple zero there. At $\theta=\pi/2$, Theorem 11.13 reduces to Theorem 11.2: $c_{\pi/2}=1/(2ab)$, $x_\theta=\sqrt{ab}$ and $\Lambda_{\mathrm{all}}(\pi/2)=\frac{\pi}{2}(\sqrt b-\sqrt a)/(\sqrt b+\sqrt a)$. The values of Table IV follow from (11.19), (11.23) and x_θ without any root finding.

12. DELAY, SEVERAL CHANNELS AND SEVERAL BANDS

This section takes up the questions left by Sections 9 and 11: a lead combined with a delay, a delay combined with zeros, matrix phases and several bands. For an affine phase without zeros the least mean loss is found exactly for every slope. A delay can beat the delayed optimum $e^{i\tau z}H^\sharp_\theta$, and for a delay there is no system that is optimal at every frequency at once. For matrix phases there is in general no least attenuation in the Loewner order, but the least trace is explicit. With zeros allowed and a delay, the optimum exists and is unique, and its loss falls off exponentially in the delay, while without zeros it stays above a fixed positive floor.[33]

12.1. A lead with a delay

Fix $s>0$ and $\tau\in\mathbb{R}$, and let $\mathcal{M}_s(\tau)$ be the infimum of $\frac{1}{b-a}\int_a^b\alpha$ over the zero-free passive systems, regular on the band, with phase $-s+\tau\omega$. For $\tau\le0$ it is given by Theorem 9.6. For $y\ge0$ put

$$q(y)=\frac{2}{\pi}\int_a^b\frac{\sqrt y}{\sqrt y+t}\,\frac{\mathrm{d}t}{r(t)}.\qquad(12.1)$$

Lemma 12.1. *$q(0)=0$, $q(y)\to\tau_\circ$ as $y\to\infty$, $q'>0$ and $q''<0$ for $y>0$, and $q(x^2)=xA(x)/r(x)$ for $a<x<b$.*

Proof. The function $y\mapsto\sqrt y/(\sqrt y+t)$ is increasing and strictly concave for $t>0$, and $\frac{2}{\pi}\int_a^b\mathrm{d}t/r=\tau_\circ$. By Proposition 8.4(ii) for $g=1$ and (9.4), $xA(x)/r(x)=\frac{2x^2}{\pi}\,\mathrm{p.v.}\int_a^b\mathrm{d}t/((x^2-t^2)r(t))$. Glauert's integral in $\sigma=t^2$ (proof of Lemma 8.5) says $\mathrm{p.v.}\int_a^b 2t\,\mathrm{d}t/((x^2-t^2)r)=0$, that is, $\mathrm{p.v.}\int\mathrm{d}t/((x-t)r)=\int\mathrm{d}t/((x+t)r)$; with $2x^2/(x^2-t^2)=x/(x-t)+x/(x+t)$ this gives $q(x^2)$. □

[33] Theorems 12.2, 12.6, 12.13, 12.14 and 12.16 and Lemmas 12.3 and 12.15 were found with OpenAI's ChatGPT, starting from Theorems 9.2 and 11.1; every number was recomputed (Appendix D).

So $A = (r/x)\,q(x^2)$ on the band. Put

$$f = \frac{a+b}{2}, \quad \ell = \frac{b-a}{2}, \quad \Gamma_0 = \pi\ell^2 = \int_a^b \frac{r(x)}{x}\,\mathrm{d}x,$$

$$\tau_c = s\,\big(\tau_\circ - q(b^2)\big) = \frac{2s}{\pi}\int_a^b \frac{t\,\mathrm{d}t}{(b+t)\,r(t)} > 0.$$

For $u < a^2$ and $v > b^2$ let

$$\begin{aligned}\Gamma_-(u) &= \int_a^b \frac{x\,r(x)}{x^2-u}\,\mathrm{d}x \\ &= \frac{\pi}{2}\Big(\frac{a^2+b^2}{2} - u - \sqrt{(a^2-u)(b^2-u)}\Big), \\ \Gamma_+(v) &= \int_a^b \frac{x\,r(x)}{v-x^2}\,\mathrm{d}x \\ &= \frac{\pi}{2}\Big(v - \frac{a^2+b^2}{2} - \Delta(v)\Big),\end{aligned}$$

$\Delta(v) = \sqrt{(v-a^2)(v-b^2)}$; both follow from $\sigma = x^2$ and $\sigma = \frac{a^2+b^2}{2} - \frac{b^2-a^2}{2}\cos\phi$, as in the proof of Corollary 9.7(iv), and $\Gamma_-(0) = \Gamma_0$. Finally, for $v > b^2$,

$$x(v) = \frac{v - \Delta(v) + ab}{a+b},$$

$$\Phi(v) = \tau_\circ - q\big(x(v)^2\big) + q'\big(x(v)^2\big)\big(v - x(v)^2\big).$$

Theorem 12.2. *Let $s > 0$.*

(i) If $\tau \le \tau_c$, every zero-free passive system regular on the band with phase $-s + \tau\omega$ satisfies

$$\alpha(x) \ge \frac{r(x)}{x}\big(s\tau_\circ - \tau - s\,q(x^2)\big) \ge 0 \qquad (a \le x \le b),$$

with equality throughout the band for $h = -sh_\circ + \tau z + (s\tau_\circ - \tau)R/z$, and equality at one point of (a,b) forces this h. Hence $\mathcal{M}_s(\tau) = -s\Lambda + \frac{\pi}{4}(b-a)(s\tau_\circ - \tau)$.

(ii) If $\tau > \tau_c$, there is exactly one $v > b^2$ with $s\Phi(v) = \tau$. With $x_* = x(v)$, $y_* = x_*^2$,

$$D = s\,q'(y_*)\,\frac{(v-y_*)^2}{v},$$

$$C = s\Big(q(y_*) - q'(y_*)\,\frac{y_*(v-y_*)}{v}\Big),$$

one has $C > 0$, $D > 0$, $f < x_* < b$, the function

$$h_{\mathrm{opt}} = -sh_\circ + \tau z + R(z)\Big(\frac{C}{z} + \frac{Dz}{v-z^2}\Big) \tag{12.2}$$

is the Herglotz function of an admissible system, and

$$\mathcal{M}_s(\tau) = -s\Lambda + \frac{C\,\Gamma_0 + D\,\Gamma_+(v)}{b-a}. \tag{12.3}$$

Its attenuation vanishes in (a,b) only at x_*, its linear coefficient is 0, and off the band its measure consists of the atom abC at 0 and the atoms $\frac{1}{2}DR(\sqrt{v})$ at $\pm\sqrt{v}$. Every other admissible system has a larger mean.

Part (i) extends Theorem 9.6 from $\tau \le 0$ to $\tau \le \tau_c$: a small delay is still free for a lead, and the optimum is pointwise. In (ii) the origin atom alone would leave a negative linear coefficient, and the cheapest compensation is one lossless resonance above the band, placed by the equation $s\Phi(v) = \tau$.

The cost of the measure. Let H be admissible, with h, μ and β as in Theorem 9.2. Put $c_0 = \mu(\{0\})/(ab)$ and let L and U be the positive measures in the variable t^2 given by $2\,\mathrm{d}\mu(t)/|R(t)|$ on $(0,a)$ and $2\,\mathrm{d}\mu(t)/R(t)$ on (b,∞). Then (9.2) and (9.3) read, with $y = x^2$,

$$\alpha(x) = \frac{r(x)}{x}\big(M(y) - s\,q(y)\big), \tag{12.4}$$

$$M(y) = c_0 + \int \frac{y\,\mathrm{d}L(u)}{y-u} + \int \frac{y\,\mathrm{d}U(w)}{w-y},$$

$$\beta = \tau - s\tau_\circ + c_0 + L(\mathbb{R}) - U(\mathbb{R}) \ge 0, \tag{12.5}$$

and integrating (12.4) by Tonelli's theorem,

$$\int_a^b \alpha = -s(b-a)\Lambda + c_0\Gamma_0 + \int \Gamma_-\,\mathrm{d}L + \int \Gamma_+\,\mathrm{d}U. \tag{12.6}$$

Lemma 12.3. *Let $f < x_* < b$, $y_* = x_*^2$, $\lambda_1 = \pi(x_* - a)(b - x_*)$ and $\lambda_2 = \pi(x_* - f)^2$. Then $\lambda_1, \lambda_2 > 0$, $\lambda_1 + \lambda_2 = \Gamma_0$,*

$$\Gamma_-(u) > \lambda_1\frac{y_*}{y_* - u} + \lambda_2 \qquad (0 < u < a^2), \tag{12.7}$$

$$\Gamma_+(w) + \lambda_2 \ge \lambda_1\frac{y_*}{w - y_*} \qquad (w > b^2), \tag{12.8}$$

and equality holds in (12.8) only at the $w > b^2$ with $x(w) = x_*$.

Proof. The sum is $(x_* - a)(b - x_*) + (x_* - f)^2 = \ell^2$. For $0 \le u < a^2$, $\Gamma_-'/\Gamma_- = 1/\sqrt{(a^2-u)(b^2-u)} \ge 1/(ab-u)$, so $\Gamma_-(u) \ge \Gamma_0\,ab/(ab-u)$, and, since $y_* > f^2 > ab$,

$$\Gamma_-(u) - \lambda_1\frac{y_*}{y_* - u} - \lambda_2 \ge \frac{\lambda_2 u}{ab-u} + \frac{\lambda_1 u\,(y_* - ab)}{(ab-u)(y_* - u)} > 0.$$

For $w > b^2$ put $\rho = x(w) - f$ and $\varepsilon = x_* - f$. Then $0 < \rho < \ell$,

$$w = f^2 + \ell^2 + f\Big(\rho + \frac{\ell^2}{\rho}\Big), \qquad \Gamma_+(w) = \pi f\rho,$$

and a direct expansion gives

$$(w - y_*)\Big(\Gamma_+(w) + \lambda_2 - \lambda_1\frac{y_*}{w - y_*}\Big) = \pi f\Big(f + \frac{\ell^2}{\rho}\Big)(\rho - \varepsilon)^2. \tag{12.9}$$

Here $w > b^2 > y_*$, so the sign is that of the right side, which vanishes only for $\rho = \varepsilon$. □

Proof of Theorem 12.2. (i) Here $\beta_g = -s\tau_\circ + \tau < 0$, since $\tau_c < s\tau_\circ$, and $v_g = -sA$. Corollary 9.3 gives the bound, which is nonnegative because q increases and $s\tau_\circ - \tau \geq s\,q(b^2)$. The stated h has linear coefficient 0, the atom $(s\tau_\circ - \tau)ab > 0$ at the origin and the band density given by the bound, so it is admissible; equality at one point forces it, as in the proof of Theorem 9.4. Integrating, with $\int_a^b A = (b-a)\Lambda$ and $\Gamma_0 = \frac{\pi}{4}(b-a)^2$, gives the value.

(ii) As v increases from b^2 to ∞, $x(v)$ decreases strictly from b to f: $x' = (1 - (v - \frac{a^2+b^2}{2})/\Delta)/(a+b) < 0$. With $y = x(v)^2$,

$$\Phi'(v) = q'(y)(1-2y') + q''(y)\,y'\,(v-y) > 0,$$

$\Phi(b^2+) = \tau_\circ - q(b^2) = \tau_c/s$ and $\Phi(\infty) = \infty$, which gives v. Strict concavity and $q(0) = 0$ give $q(y) > yq'(y)$, so $C > 0$; and $D > 0$. The function $C + Dy/(v-y) - s\,q(y)$ is strictly convex on $[a^2, b^2]$ with value and derivative 0 at y_*, hence positive elsewhere. By Lemma 9.1, h_{opt} has the band density $\pi^{-1}(r/x)(C + Dy/(v-y) - sq(y)) \geq 0$, the atoms stated, and the linear coefficient

$$\tau - s\tau_\circ + C - D = \tau - s\big(\tau_\circ - q(y_*) + q'(y_*)(v - y_*)\big) = 0.$$

So it is admissible. Now let H be any admissible system. By (12.6), Lemma 12.3 with this x_*, $\lambda_1 + \lambda_2 = \Gamma_0$, (12.5) and $M(y_*) \geq s\,q(y_*)$ (from $\alpha(x_*) \geq 0$),

$$\begin{aligned}\int_a^b \alpha &\geq -s(b-a)\Lambda + \lambda_1 M(y_*) + \lambda_2\big(c_0 + L(\mathbb{R}) - U(\mathbb{R})\big)\\ &\geq -s(b-a)\Lambda + \lambda_1 s\,q(y_*) + \lambda_2(s\tau_\circ - \tau). \quad (12.10)\end{aligned}$$

For h_{opt} every inequality is an equality: $L = 0$, U is the atom D at v, where (12.8) is an equality, $\alpha(x_*) = 0$ and $\beta = 0$. This proves (12.3). If H has the same mean, then $L = 0$ by (12.7), U is an atom D' at v, and $c_0 + D'y_*/(v-y_*) = s\,q(y_*)$, $c_0 - D' = s\tau_\circ - \tau$. These two equations have the unique solution (C, D), and (9.1) gives $h = h_{\text{opt}}$. □

The bound (12.10) holds for every admissible measure, and (12.3) can be written $(b-a)\mathcal{M}_s(\tau) = -s(b-a)\Lambda + \lambda_1 s\,q(y_*) + \lambda_2(s\tau_\circ - \tau)$.

Corollary 12.4. *Let $0 < \vartheta < \pi$ and $\tau \geq 0$. Among zero-free passive systems regular on the band with $\arg H \equiv \vartheta + \tau\omega$ modulo 2π on $[a,b]$, the least mean attenuation is $\min\{\vartheta\Lambda,\ \mathcal{M}_{\pi-\vartheta}(\tau)\}$, and it is attained.*

Proof. As in Theorem 9.8 the phase is $\theta' + \tau\omega$ with $\theta' \equiv \vartheta \pmod{\pi}$. For $\theta' > 0$, Theorem 9.4 gives the mean $\theta'\Lambda \geq \vartheta\Lambda$. For $\theta' = -s < 0$, $s \geq \pi - \vartheta$. Scaling h gives $\mathcal{M}_s(\tau) = s\,\mathcal{M}_1(\tau/s)$; multiplying by $e^{i\delta z}$, $\delta \geq 0$, shows that $\mathcal{M}_1$ is nonincreasing; and $\mathcal{M}_1 > 0$ by Theorem 12.2. So $s \mapsto \mathcal{M}_s(\tau)$ is increasing, and the least value is at $s = \pi - \vartheta$. □

For $\tau = 0$ this is Theorem 9.8, since $\mathcal{M}_s(0) = s\Lambda_-$.

Without zeros a delay never makes the loss small. Put

$$m_\infty = \frac{1}{b-a}\int_a^b \frac{r(x)}{x}\Big(q\big(f^2\big) - q\big(x^2\big)\Big)\,\mathrm{d}x.$$

Proposition 12.5. *$m_\infty > 0$. For $s > 0$ and every $\tau \in \mathbb{R}$, $\mathcal{M}_s(\tau) \geq s\,m_\infty$, and $\mathcal{M}_s(\tau)$ decreases to $s\,m_\infty$ as $\tau \to \infty$. So the least mean of Corollary 12.4 is at least $\min\{\vartheta\Lambda, (\pi-\vartheta)m_\infty\}$ for every $\tau \geq 0$, and tends to this value as $\tau \to \infty$.*

Proof. With $\sigma = x^2$, $\int_a^b x\,r(x)\,\mathrm{d}x = \frac{\pi}{16}(b^2-a^2)^2 = f^2\Gamma_0$, so $\int_a^b (r/x)(x^2 - f^2)\,\mathrm{d}x = 0$. By Lemma 12.1, q is strictly concave, so $q(x^2) < q(f^2) + q'(f^2)(x^2 - f^2)$ for $x \neq f$, and integrating against r/x gives $m_\infty > 0$. Since $(r/x)\,q(x^2) = A$ and $\int_a^b A = (b-a)\Lambda$,

$$(b-a)\,s\,m_\infty = -s(b-a)\Lambda + \Gamma_0\,s\,q(f^2).$$

Let H be zero-free, regular on the band, with phase $-s + \tau\omega$, and use the notation of (12.4)–(12.6). For $0 < u < a^2$ the proof of Lemma 12.3 gives $\Gamma_-(u) \geq \Gamma_0 ab/(ab-u) \geq \Gamma_0 f^2/(f^2-u)$, since $ab < f^2$. For $w > b^2$, the identity (12.9) with $\varepsilon = 0$, where $\lambda_1 = \Gamma_0$ and $\lambda_2 = 0$, gives $\Gamma_+(w) \geq \Gamma_0 f^2/(w - f^2)$. So (12.6) gives

$$\int_a^b \alpha \geq -s(b-a)\Lambda + \Gamma_0 M(f^2) \geq -s(b-a)\Lambda + \Gamma_0\,s\,q(f^2),$$

the last step because $\alpha(f) \geq 0$ in (12.4). This is the lower bound; β was not used, so it holds for every τ. By the proof of Corollary 12.4, $\mathcal{M}_s$ is nonincreasing. For the limit let $\tau > \tau_c$ and let v, x_*, λ_1, λ_2 be those of Theorem 12.2(ii). Since q' decreases and $y_* > a^2$, $\tau/s = \Phi(v) \leq \tau_\circ + q'(a^2)v$, so $v \to \infty$ as $\tau \to \infty$. Then $x_* - f = \Gamma_+(v)/(\pi f) \to 0$, by the parametrisation in the proof of Lemma 12.3, so $\lambda_1 \to \Gamma_0$ and $\lambda_2 \to 0$, and the identity after (12.10) gives

$$\begin{aligned}(b-a)\mathcal{M}_s(\tau) &\leq -s(b-a)\Lambda + \lambda_1 s\,q(y_*) + \lambda_2 s\tau_\circ\\ &\to (b-a)\,s\,m_\infty.\end{aligned}$$

The last statement follows from Corollary 12.4. □

On the octave $\frac{\pi}{2}m_\infty = 0.0060483$ nepers. So for the argument $\pi/2 + \tau\omega$ no zero-free system has a mean loss below this value, however long the delay; at $\tau = 5$ the optimum of Theorem 12.2 is less than six per cent above it.

12.2. A delay can beat the delayed optimum

Take the octave $[a,b] = [1,2]$, the argument $\pi/2 + 5\omega$, $s = \pi/2$, and $T_0 = 16$. For $1 < y_0 < 4$ put

$$d = q'(y_0)\frac{(T_0 - y_0)^2}{T_0}, \qquad c = q(y_0) - q'(y_0)\frac{y_0(T_0 - y_0)}{T_0}.$$

Theorem 12.6. *For every $y_0 \in (1,4)$,*

$$h_{y_0} = -sh_\circ + 5z + sR(z)\Big(\frac{c}{z} + \frac{dz}{T_0 - z^2}\Big)$$

is the Herglotz function of a zero-free passive system $H_{y_0} = -e^{ih_{y_0}}$, regular on the band, with $\arg H_{y_0} = \pi/2 + 5\omega$ on $[1,2]$, whose attenuation vanishes at $\sqrt{y_0}$. Hence, for this argument, with or without zeros, the infimum of $\alpha(x)$ is 0 at every x, and no system attains it at every point at once. For $y_0 = 5/2$,

$$\int_1^2 \alpha_{5/2} \le \frac{21\pi^2}{1024} < \frac{\pi}{2}\bigl(3 - 2\sqrt{2}\bigr), \tag{12.11}$$

so $e^{5iz}H^\sharp_{\pi/2}$, whose mean is $\frac{\pi}{2}(3-2\sqrt{2})$, is not optimal for this argument.

Proof. The function $f_0(y) = c + dy/(T_0 - y) - q(y)$ is strictly convex on $[1,4]$ with $f_0(y_0) = f_0'(y_0) = 0$, so $f_0 \ge 0$ with a single zero, and the band attenuation of h_{y_0} is $s(r/x)f_0(x^2)$. Also $c > 0$ and $d > 0$, as in the proof of Theorem 12.2. On $[1,4]$, $q' \le \tau_\circ/8$ and $-q'' \le \tau_\circ/8$: in q' and $-q''$ the integrands, normalised by $\frac{2}{\pi}\,\mathrm{d}t/r$, are $t/(2x(x+t)^2)$ and $t(3x+t)/(4x^3(x+t)^3)$ with $x = \sqrt{y}$, which are at most $1/8$ for $1 \le x, t \le 2$. And $\tau_\circ \le 1$, since $K(k) \le \pi/(2k')$ with $k' = a/b$. So $d \le 225/128$, and the linear coefficient is

$$5 - s\tau_\circ + s(c-d) \ge 5 - \frac{\pi}{2}\cdot\frac{353}{128} > 0,$$

by $\pi < 22/7$. With the atoms $sabc$ at 0 and $\frac12 sdR(4)$ at ±4 and no other measure off S, h_{y_0} is a Herglotz function, and $\arg H_{y_0} = \pi + \operatorname{Re} h_{y_0} = \pi/2 + 5\omega$ on the band. If $\alpha = 0$ on all of $[1,2]$, then $H = ie^{5i\omega}$ there, and by the boundary uniqueness theorem [25, Ch. II] $H = ie^{5iz}$, which is not real on $i\mathbb{R}_+$; so no admissible system has zero attenuation everywhere. For $y_0 = 5/2$, the identity

$$f_0(y) = q(y_0) + q'(y_0)(y - y_0) - q(y) + q'(y_0)\frac{(y-y_0)^2}{16 - y}$$

and Taylor's theorem give $0 \le f_0 \le \frac{7}{96}(y-y_0)^2 \le \frac{21}{128}$. With $\int_1^2 r/x = \pi/4$ this gives the first inequality in (12.11). The second follows from $21\pi/512 < 33/256 < 1/7 < 3 - 2\sqrt{2}$, by $\pi < 22/7$ and $\sqrt{2} < 10/7$. □

The mean of $H_{5/2}$ is 0.0084888 nepers, against 0.26951 for $e^{5iz}H^\sharp_{\pi/2}$. By Corollary 12.4 the least zero-free mean is $\mathcal{M}_{\pi/2}(5)$, because $\mathcal{M}_{\pi/2}(5) \le 21\pi^2/1024 < \pi^2/24 \le \frac{\pi}{2}\Lambda$, by $K(\kappa) - E(\kappa) \ge \pi\kappa^2/4$; by Theorem 12.2 it is attained with

$$\sqrt{v} = 8.870977, \quad x_* = 1.504922,$$
$$C = 0.426778, \quad D = 4.348520,$$

and equals 0.00639032 nepers, about forty-two times less than the delayed optimum with zeros (Figure 15).

12.3. A delay together with zeros

Lemma 11.11 shows how a zero enters. A zero at $i\varepsilon$ adds $p_\varepsilon(t) = 2\arctan(\varepsilon/t) = 2\int_0^\varepsilon j_u(t)\,du$ to the phase that the zero-free factor must carry, since $\arg(-B_\varepsilon) = \pi - p_\varepsilon$ on the real axis; and p_ε costs, in the bound (12.10), what a measure at the negative squared frequencies $-u^2$ would cost.

Proposition 12.7. *On $[1,2]$, for the argument $\pi/2 + 5\omega$, there is a passive system with one zero, at $i/10$, whose mean attenuation is less than $\mathcal{M}_{\pi/2}(5)$. So the least zero-free mean is not the least mean with zeros allowed.*

Proof. Let $s = \pi/2$, $\varepsilon = 1/10$, $g_0 = -s + 5t$ and $g_\varepsilon = g_0 + p_\varepsilon$. By Lemma 11.11,

$$-\frac{x\,v_{g_\varepsilon}(x)}{r(x)} = q_\varepsilon(x^2), \qquad q_\varepsilon = sq - P,$$
$$P(y) = 2\int_0^\varepsilon \frac{y\,du}{(y+u^2)Q_u},$$
$$\beta_{g_\varepsilon} = 5 - s\tau_\circ + 2\int_0^\varepsilon \frac{du}{Q_u} > 0.$$

On $[1,4]$, the integrands of Theorem 12.6 are at least $1/64$ and $1/512$, and $\tau_\circ \ge 1/2$, as $K(k) \ge \pi/2$, so $q' \ge 1/128$ and $-q'' \ge 1/1024$; and $0 \le P' \le \varepsilon^3/3$, $0 \le -P'' \le 2\varepsilon^3/3$. Hence $q_\varepsilon' > 0$ and $q_\varepsilon'' < 0$ there. For $\frac94 \le y \le 4$ the same integral gives $q - yq' \ge 3/32$, and $P \le \varepsilon$, so $q_\varepsilon - yq_\varepsilon' \ge \frac{3\pi}{64} - \frac{1}{10} > 0$; also $q_\varepsilon(4) \ge \frac{\pi}{8} - \varepsilon > 0$, since $q(4) \ge \tau_\circ/2$. The proof of Theorem 12.2(ii), with sq replaced by q_ε and $\tau - s\tau_\circ$ by $\beta_{g_\varepsilon} > -q_\varepsilon(4)$, gives $v > 4$ and $C, D > 0$ with $x_* = x(v) \in (\frac32, 2)$, and the Herglotz function $h_\varepsilon = h_{g_\varepsilon} + R(C/z + Dz/(v - z^2))$ with linear coefficient 0. The system $H_\varepsilon = -B_\varepsilon e^{ih_\varepsilon}$ is passive, has one zero, at $i\varepsilon$, and $\arg H_\varepsilon = \pi - p_\varepsilon + g_\varepsilon = \pi/2 + 5\omega$ on the band.

For a phase g and this x_*, let $\mathcal{L}(g) = \int_a^b v_g - \lambda_1 x_* v_g(x_*)/r(x_*) - \lambda_2\beta_g$. The proof of (12.10) shows that $\int_a^b \alpha \ge \mathcal{L}(g)$ for every zero-free admissible system with phase g, with equality for h_ε. By Lemma 11.11,

$$\mathcal{L}(j_u) = \frac{1}{Q_u}\Bigl(\Gamma_-(-u^2) - \lambda_1\frac{y_*}{y_* + u^2} - \lambda_2\Bigr),$$

For $w < 0$ put $x(w) = (w + \sqrt{(a^2 - w)(b^2 - w)} + ab)/(a + b)$, which lies in $(ab/f, f)$, and $\rho = x(w) - f \in (-\ell^2/f, 0)$. Then again $w = f^2 + \ell^2 + f(\rho + \ell^2/\rho)$, and $\pi f\rho = -\Gamma_-(w)$. The identity (12.9) is a polynomial identity in ρ and ε, so with $\Gamma_+(w)$ replaced by $-\Gamma_-(w)$ it reads

$$(y_* - w)\Bigl(\Gamma_-(w) - \lambda_1\frac{y_*}{y_* - w} - \lambda_2\Bigr) = \pi f\Bigl(f + \frac{\ell^2}{\rho}\Bigr)(\rho - x_* + f)^2 < 0,$$

because $f + \ell^2/\rho < 0$. With $w = -u^2$ this gives $\mathcal{L}(j_u) < 0$, so $\mathcal{L}(p_\varepsilon) = 2\int_0^\varepsilon \mathcal{L}(j_u)\,du < 0$. The Blaschke factor has modulus 1 on $\mathbb{R}$, and

$$\int_1^2 \alpha_{H_\varepsilon} = \mathcal{L}(g_0) + \mathcal{L}(p_\varepsilon) < \mathcal{L}(g_0) \le \mathcal{M}_{\pi/2}(5),$$

the last step by (12.10) for g_0. □

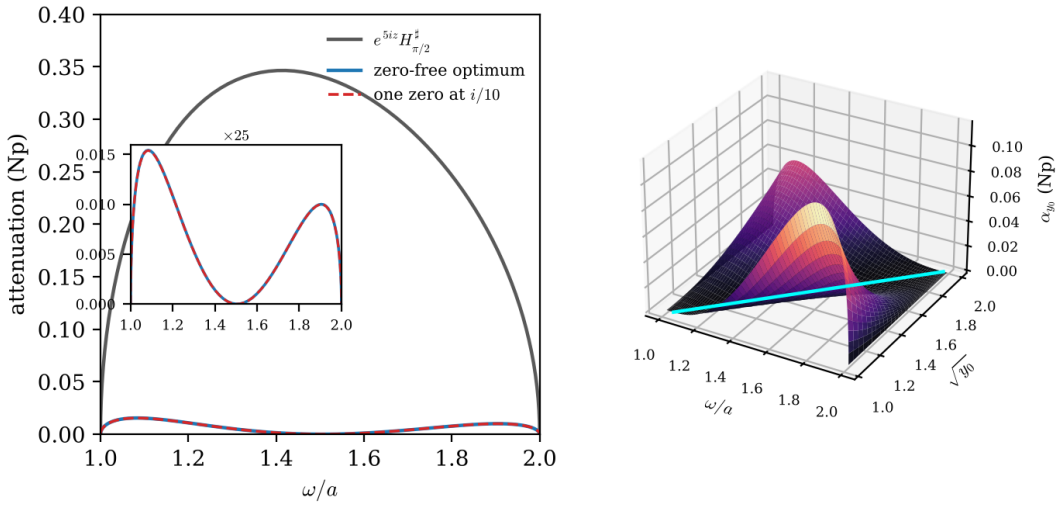


FIG. 15. The argument $\pi/2 + 5\omega$ on an octave. Left: the attenuation of the delayed optimum $e^{5iz}H^{\sharp}_{\pi/2}$ (Theorem 11.1), of the zero-free optimum of Theorem 12.2, which touches zero at $x_* = 1.5049$, and of the system with one zero of Proposition 12.7. Right: the zero-free systems H_{y_0} of Theorem 12.6, each lossless at a different frequency $\sqrt{y_0}$.

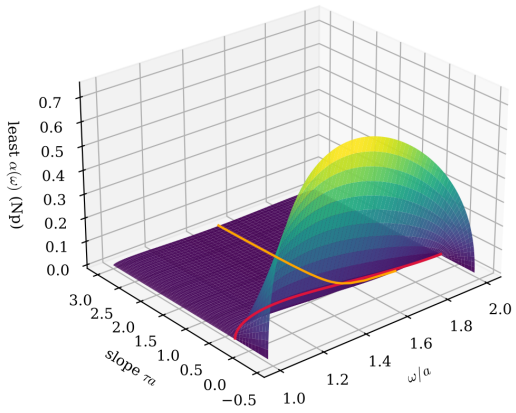


FIG. 16. The least attenuation of Theorem 12.2 on an octave for the phase $-1 + \tau\omega$, as a function of τ. Up to $\tau_c = 0.2852$ it is pointwise optimal and decreases linearly in τ; beyond it the resonance above the band moves in from infinity and the profile touches zero at the contact x_*.

The mean of $H_{1/10}$ is

$$-s\Lambda + 2\int_0^{1/10} \frac{\Gamma_-(-u^2)}{Q_u}\,\mathrm{d}u + C\,\Gamma_0 + D\,\Gamma_+(v) = 0.0063752$$

nepers, against $\mathcal{M}_{\pi/2}(5) = 0.0063903$.

Remark 12.8 (computed, not proved). The construction of Proposition 12.7 works for every height ε of the zero up to $\varepsilon_0 = 0.47247$, where the origin coefficient C vanishes, and the mean then decreases to 0.0051493 nepers. With a pair of zeros at $\pm 1.3654 + 1.2608i$ instead, a linear program for the measure off the band, with the positivity of the attenuation imposed at 399 points of the band, gives about 4.3×10^{-4} nepers; without zeros the same program gives 0.00658 in place of 0.0063903, so it is accurate to a few per cent. These numbers come from computations only.

The optimum exists. The admissible systems are not a convex set, but a weaker class is. For a measurable ψ on $[a, b]$ let $\mathcal{P}_\psi$ be the set of passive systems H with $e^{-i\psi}H \geq 0$ almost everywhere on $[a, b]$, and put

$$J(H) = \int_a^b -\log|H(\omega)|\,\mathrm{d}\omega \in [0, \infty].$$

Systems in $\mathcal{P}_\psi$ may have zeros and need not be regular on the band. Every system admissible for θ lies in $\mathcal{P}_\theta$, and the systems of Proposition 12.7 and Theorems 12.10 and 12.11 lie in $\mathcal{P}_{\theta+\tau\omega}$.

Theorem 12.9. *Let ψ be measurable on $[a, b]$, and suppose that $J(G) < \infty$ for some $G \in \mathcal{P}_\psi$.*

(i) There is exactly one $H_\psi \in \mathcal{P}_\psi$ with $J(H_\psi) = \inf_{\mathcal{P}_\psi} J$.

(ii) An $H \in \mathcal{P}_\psi$ with $J(H) < \infty$ is H_ψ if and only if

$$\int_a^b \frac{|G|}{|H|}\,\mathrm{d}\omega \leq b - a \qquad \text{for every } G \in \mathcal{P}_\psi.$$

Proof. The conditions $|H| \leq 1$, $H(-\bar{z}) = \overline{H(z)}$ and $e^{-i\psi}H \geq 0$ almost everywhere are kept by convex combinations, so $\mathcal{P}_\psi$ is convex. On $\mathcal{P}_\psi$, $J(H) = \int_a^b -\log(e^{-i\psi}H)$, and $-\log$ is convex on $[0, \infty)$, so J is convex.

(i) Let J_* be the infimum and $J(H_k) \to J_*$. The boundary functions are bounded by 1 in $L^\infty(\mathbb{R})$, so a

subsequence converges weak-$*$ to some ϕ. Since H_k is the Poisson integral of its boundary function [25, Ch. I], $H_k(z) \to H(z)$, the Poisson integral of ϕ, at every $z \in \mathbb{C}_+$; so H is analytic, $|H| \le 1$, $H(-\bar z) = \overline{H(z)}$, and $H = \phi$ almost everywhere on $\mathbb{R}$. On (a,b) the boundary functions converge weakly in L^2, so by Mazur's lemma there are convex combinations G_k of $H_k, H_{k+1}, \dots$ with $G_k \to H$ in $L^2(a,b)$, and almost everywhere along a subsequence. Each G_k lies in $\mathcal{P}_\psi$, and $J(G_k) \le \sup_{j\ge k} J(H_j) \to J_*$ by convexity. The limit of the nonnegative functions $e^{-i\psi}G_k$ is nonnegative, so $H \in \mathcal{P}_\psi$, and Fatou's lemma, with $-\log \ge 0$ on $[0,1]$ and lower semicontinuous, gives $J(H) \le J_*$. If H and H' are both minimisers, then $J_* < \infty$, and $J(\frac12(H+H')) < J_*$ unless $|H| = |H'|$ almost everywhere on (a,b), by strict convexity of $-\log$. Then $H = H'$ almost everywhere on (a,b), since both have argument ψ where they do not vanish, and $H = H'$ by the boundary uniqueness theorem [25, Ch. II].

(ii) Here $H \ne 0$ almost everywhere on (a,b). Put $\rho = |G|\,/\,|H|$. If the inequality holds, then $-\log|G| \ge -\log|H| - (\rho - 1)$, and integrating gives $J(G) \ge J(H) + (b-a) - \int \rho \ge J(H)$. Conversely, let $H = H_\psi$ and $0 < t \le \frac12$. Then $H_t = (1-t)H + tG \in \mathcal{P}_\psi$ and $|H_t| = |H|\,(1 + t(\rho-1))$, so

$$0 \le \frac{J(H_t) - J(H)}{t} = \int_a^b \frac{-\log\big(1 + t(\rho-1)\big)}{t}\,\mathrm{d}\omega.$$

The integrand is a difference quotient of a convex function of t, so it decreases as $t \downarrow 0$, to $1-\rho$, and it is at most $2\log 2$ at $t = \frac12$. By monotone convergence $\int(1-\rho) \ge 0$. □

For a constant phase the minimiser is the system of Theorem 11.1. The proof of Proposition 11.4 uses the continuity of F only to pass from almost every point of the band to every point; without it, it shows $\alpha \ge \alpha^\sharp{}_\theta$ almost everywhere for every nonconstant $H \in \mathcal{P}_\theta$. Hence $H_\theta = H_\theta^\sharp$ for $0 < \theta < \pi$, and by Theorem 12.9(ii) every $G \in \mathcal{P}_\theta$ has $\int_a^b |G|\,/\,\big|H_\theta^\sharp\big| \le b - a$. For $\psi = \theta + \tau\omega$ with $\tau > 0$ we have not found H_ψ in closed form. The rest of this subsection shows how small its loss is.

There is still a positive floor. For $y > 0$ let $P_y(x) = y/(\pi(x^2+y^2))$ and $\omega_y = \int_a^b P_y$.

Theorem 12.10. *Let $\theta \notin \pi\mathbb{Z}$, $\tau \ge 0$ and $H \in \mathcal{P}_{\theta+\tau\omega}$. For every $y > 0$,*

$$\frac{1}{b-a}\int_a^b \alpha \;\ge\; \frac{\omega_y}{(b-a)P_y(a)}\,\log\frac{1}{1-\varepsilon_y} > 0,$$
$$\varepsilon_y = \Big(\frac{|\sin\theta|\;e^{-\tau y}}{2^{1-\omega_y}}\Big)^{1/\omega_y} \in (0,1).$$

Proof. Let $J = H - e^{i\theta + i\tau z}$, analytic in $\mathbb{C}_+$ with $|J| \le 2$. On the imaginary axis $H(iy)$ is real, so $|J(iy)| \ge |\sin\theta|\,e^{-\tau y} > 0$. Almost everywhere on the band $|J(x)| = 1 - e^{-\alpha(x)}$. As $\log|J|$ is subharmonic and bounded above, it lies below the Poisson integral of its boundary values [25, Ch. II], so

$$\log\big(\,|\sin\theta|\,e^{-\tau y}\big) \le \int_a^b P_y \log\big(1 - e^{-\alpha}\big) + (1-\omega_y)\log 2.$$

Since $u \mapsto \log(1-e^{-u})$ is increasing and concave, Jensen's inequality for the probability density P_y/ω_y on $[a,b]$ gives $\omega_y^{-1}\int_a^b P_y\alpha \ge \log(1/(1-\varepsilon_y))$, and $P_y \le P_y(a)$ on the band. □

The floor is exponentially small in τ, and so is the loss that zeros make possible. Let $\tau_\circ$ be as in Lemma 12.1.

Theorem 12.11. *Let $\theta \in \mathbb{R}$ and $a\tau \ge 12$. There is a passive system H, regular on the band, with $\arg H \equiv \theta + \tau\omega$ on $[a,b]$ and*

$$\frac{1}{b-a}\int_a^b \alpha \le e^{3-a\tau/e}\Big(2 + \frac{\pi}{4}(b-a)\tau_\circ + \frac{b(b^2-a^2)}{2e\,a^2}\,\tau\Big).$$

The idea is to turn $\cos\theta + i\sin\theta$ into a function that is close to $e^{i\theta}$ on the band and is cancelled by the delay $e^{i\tau z}$ in the upper half-plane: a smoothed sign function of exponential type τ.

Proof. The kernel. Let $m = 2\lfloor a\tau/(2e)\rfloor$, so that m is even, $4 \le m \le a\tau/e$ and $m > a\tau/e - 2$, and let $\kappa = \tau/m \ge e/a$. Put

$$I_m = \int_{\mathbb{R}} \Big(\frac{\sin u}{u}\Big)^m \mathrm{d}u, \qquad T(z) = 2\int_0^z K,$$
$$K(\zeta) = \frac{\kappa}{I_m}\Big(\frac{\sin\kappa\zeta}{\kappa\zeta}\Big)^m.$$

K is entire, even, nonnegative on $\mathbb{R}$, with $\int_{\mathbb{R}} K = 1$. So T is entire, odd, real and increasing on $\mathbb{R}$, with $|T| \le 1$ there. Since $\sin w/w = \int_0^1 \cos(tw)\,\mathrm{d}t$, $|\sin w/w| \le e^{|\mathrm{Im}\,w|}$, so $|K(x+iy)| \le (\kappa/I_m)e^{\tau|y|}$ and $|T(x+iy)| \le 1 + 2\kappa e^{\tau|y|}/(\tau I_m)$.

The factor F. Let $S = \cos\theta + i\sin\theta\,T$ and $F = e^{i\tau z}S$. By the last bound F is bounded in $\mathbb{C}_+$; it is continuous up to $\mathbb{R}$ and $|F(x)|^2 = \cos^2\theta + \sin^2\theta\,T(x)^2 \le 1$ there. So F is the Poisson integral of its boundary values and $|F| \le 1$ in $\mathbb{C}_+$. As T has real coefficients and is odd, $F(-\bar z) = \overline{F(z)}$.

The band. For $x \ge a$,

$$0 \le 1 - T(x) = 2\int_x^\infty K \le \frac{2(\kappa x)^{1-m}}{(m-1)I_m} \le \varepsilon = \frac{2(\kappa a)^{1-m}}{(m-1)I_m},$$

and $0 \le T'(x) = 2K(x) \le (m-1)\varepsilon/a$. Since $\sin u/u \ge 1 - u^2/6 \ge 0$ for $|u| \le \sqrt6$ and $(1-v^2)^m \ge 1 - mv^2$,

$$I_m \ge \sqrt6\int_{-1}^1 (1-v^2)^m\,\mathrm{d}v$$
$$\ge \sqrt6\int_{|v|\le m^{-1/2}} (1 - mv^2)\,\mathrm{d}v = \frac{4\sqrt6}{3\sqrt m} > \frac{3}{\sqrt m}.$$

With $2\sqrt{m} \le 3(m-1)$, $\kappa a \ge e$ and $m > a\tau/e - 2$, this gives $\varepsilon \le (\kappa a)^{1-m} \le e^{3-a\tau/e}$, and $\varepsilon \le e^{3-12/e} < \frac14$. On $[a,b]$, $T \in [1-\varepsilon, 1]$, so $|S| \ge 1-\varepsilon$ and $-\log|S| \le -\log(1-\varepsilon) \le 2\varepsilon$. Write $\arg S = \theta + \delta$ there, with δ continuous. The derivative of $\arg(\cos\theta + i\sin\theta\, T)$ in T is $\sin\theta\cos\theta/(\cos^2\theta + \sin^2\theta\, T^2)$, at most $\frac12(1-\varepsilon)^{-2} \le \frac89$ in modulus, so $|\delta| \le \varepsilon$ and $|\delta'| \le T' \le (m-1)\varepsilon/a$.

The phase correction. Let $g = -\delta$ and $G(t) = g(t)/t$ on $[a,b]$. Then $|G'| \le |\delta'|/a + |\delta|/a^2 \le m\varepsilon/a^2$. By Glauert's integral (proof of Lemma 12.1), the function v_g of Proposition 8.4 is

$$v_g(x) = \frac{2x\, r(x)}{\pi} \int_a^b \frac{t\,(G(t) - G(x))}{r(t)\,(x^2 - t^2)}\,\mathrm{d}t,$$

and here $|G(t) - G(x)| \le |x^2 - t^2| \max|G'|/(2a)$, while $\int_a^b t\,\mathrm{d}t/r(t) = \pi/2$ and $x\,r(x) \le b(b^2-a^2)/2$. So

$$|v_g| \le \eta = \frac{b(b^2-a^2)\,m\,\varepsilon}{4a^3} \le \frac{b(b^2-a^2)\,\tau\,\varepsilon}{4e\,a^2}.$$

Also $|\beta_g| \le \tau_\circ\varepsilon$. Put $c = \max(0, -\beta_g)$ and $h = h_g + cR/z$. As in the proof of Proposition 8.4(iii), $\operatorname{Im}(h_g - \beta_g z)$ is harmonic in $\mathbb{C}_+$, continuous up to $\mathbb{R}$, zero on $\mathbb{R}\setminus S$, equal to $v_g \ge -\eta$ on the band, and tends to 0 at infinity, so $\operatorname{Im} h_g \ge \beta_g \operatorname{Im} z - \eta$. By Lemma 8.1, $R(0) = -ab$, R is even and $R(z) = z^2 + O(1)$, so $(R+ab)/z - z$ is analytic in $\mathbb{C}\setminus S$, real on $\mathbb{R}\setminus S$, $O(z^{-1})$ at infinity, and has imaginary part $r(x)/x \ge 0$ on the upper side of the band; hence its imaginary part is nonnegative in $\mathbb{C}_+$, and $\operatorname{Im}(R/z) \ge \operatorname{Im} z + ab\operatorname{Im}(-1/z) \ge \operatorname{Im} z$. So

$$\operatorname{Im} h \ge (\beta_g + c)\operatorname{Im} z - \eta \ge -\eta \qquad \text{in } \mathbb{C}_+.$$

The system. Let $H = e^{-\eta} F\, e^{ih}$. Then $|H| \le 1$ in $\mathbb{C}_+$, and $H(-\bar z) = \overline{H(z)}$ because h is odd and real on $\mathbb{R}\setminus S$. It is continuous up to an open interval around $[a,b]$ and has no zeros there. On the band $\operatorname{Re} h = g = -\delta$, so $\arg H = \tau\omega + \theta + \delta - \delta = \theta + \tau\omega$, and

$$\alpha = \eta - \log|S| + v_g + c\,\frac{r}{x}.$$

With $\int_a^b r/x = \frac{\pi}{4}(b-a)^2$ the mean is at most $2\eta + 2\varepsilon + \frac{\pi}{4}(b-a)\tau_\circ\varepsilon$, which is the bound. □

For $\theta = \pi/2$ the factor $S = iT$ already has the exact argument $\pi/2$, so $h = 0$, $\eta = 0$ and $\alpha = -\log T$. On the octave its mean is 1.7×10^{-3} nepers at $\tau = 12$, 4.9×10^{-5} at $\tau = 20$ and 1.4×10^{-7} at $\tau = 30$, against the zero-free floor 0.0060483 of Proposition 12.5.

Corollary 12.12. *Let $\theta \notin \pi\mathbb{Z}$, and let $E_\theta(\tau) = J(H_{\theta+\tau\omega})/(b-a)$ be the least mean attenuation for the argument $\theta + \tau\omega$, with or without zeros. Then E_θ is nonincreasing in $\tau \ge 0$, and*

$$-\frac{\pi ab}{b-a} \le \liminf_{\tau\to\infty} \frac{\log E_\theta(\tau)}{\tau} \le \limsup_{\tau\to\infty} \frac{\log E_\theta(\tau)}{\tau} \le -\frac{a}{e}.$$

Proof. Multiplication by $e^{i\delta z}$, $\delta \ge 0$, maps $\mathcal{P}_{\theta+\tau\omega}$ into $\mathcal{P}_{\theta+(\tau+\delta)\omega}$ and keeps $|H|$ on $\mathbb{R}$, which gives the monotonicity; the system of Theorem 12.11 gives the upper bound. For the lower bound take $y = \tau^{-1/2}$ in Theorem 12.10. As $y \to 0$, $\omega_y/y = \frac{1}{\pi}\int_a^b \mathrm{d}x/(x^2+y^2) = \frac{b-a}{\pi ab} + O(y^2)$ and $\omega_y/P_y(a) \to a(b-a)/b$. With $\log(1/(1-\varepsilon)) \ge \varepsilon$,

$$\log E_\theta(\tau) \ge \log\varepsilon_y + O(1) = -\frac{\tau y}{\omega_y} + O\Big(\frac{1}{y}\Big)$$
$$= -\frac{\pi ab}{b-a}\,\tau + O\big(\tau^{1/2}\big). \qquad \square$$

So with zeros the least loss for a delay falls off exponentially in the delay, while without zeros it stays above $(\pi - \vartheta)m_\infty$ or $\vartheta\Lambda$. On the octave the exponent lies between -2π and $-1/e$. The optimal system $H_{\theta+\tau\omega}$ itself, and the exact exponent, remain open.

12.4. Several channels

A *generator* on the band is an $n\times n$ matrix function F, analytic in $\mathbb{C}_+$, with $\operatorname{Im} F = (F - F^*)/(2i) \succeq 0$ there, $F(-\bar z) = -F(z)^*$, and continuous up to an open interval $I \supset [a,b]$; its *phase* and *attenuation* on the band are $\operatorname{Re} F = (F+F^*)/2$ and $\operatorname{Im} F$ at $x + i0$. For every $u \in \mathbb{C}^n$, $h_u = u^*Fu$ is a Herglotz function with the symmetry of (4.3), continuous up to I, so e^{ih_u} is a zero-free passive system regular on the band, with phase $u^*(\operatorname{Re} F)u$ and attenuation $u^*(\operatorname{Im} F)u$. For a Hermitian matrix B, B_+ denotes its positive part.

Theorem 12.13. *Let Θ and T be Hermitian, $B = \tau_\circ\Theta + T$. Every generator with phase $\Theta + \omega T$ on the band satisfies*

$$\operatorname{tr}\operatorname{Im} F(x) \ge A(x)\operatorname{tr}\Theta + \frac{r(x)}{x}\operatorname{tr}(-B)_+ \qquad (a \le x \le b).$$

If $\Theta \succeq 0$, equality holds at every point for $F_{\mathrm{opt}} = h_\circ\Theta + zT + (-B)_+R/z$, so that the least mean of $\operatorname{tr}\operatorname{Im} F$ is $\Lambda\operatorname{tr}\Theta + \frac{\pi}{4}(b-a)\operatorname{tr}(-B)_+$.

Proof. Let $u_1, \dots, u_n$ be an orthonormal basis of eigenvectors of B. The scalar system of u_j has phase $u_j^*\Theta u_j + \omega\, u_j^*Tu_j$, so by Corollary 9.3 and (9.4) its attenuation is at least $(u_j^*\Theta u_j)A + (r/x)\max\{0, -u_j^*Bu_j\}$. Summing over j gives the bound. For $\Theta \succeq 0$, F_{opt} has the linear coefficient $B + (-B)_+ = B_+ \succeq 0$, the origin atom $ab(-B)_+$ and the band density $\pi^{-1}(A\Theta + (r/x)(-B)_+) \succeq 0$ (Lemma 9.1), so $\operatorname{Im} F_{\mathrm{opt}} \succeq 0$ in $\mathbb{C}_+$, and its trace attains the bound. □

No commutation of Θ and T is needed. The trace is the right objective: the stronger statement, a single generator below all others in the Loewner order, is false.

Theorem 12.14. *Let $A_0 = \operatorname{diag}(1,-1)$, $T = -\tau_\circ I - A_0$, and consider the phase $I + \omega T$ on any band. There are generators with this phase, but no generator F with this phase satisfies $\operatorname{Im} F \preceq \operatorname{Im} G$ on the band for every generator G with this phase.*

Proof. For $C \succeq 0$ with $C - A_0 \succeq 0$, $F_C = h_\circ I + zT + CR/z$ is a generator with this phase: its linear coefficient is $C - A_0$, its origin atom abC and its band density $\pi^{-1}(AI + (r/x)C)$. This holds for

$$C_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad C_2 = \frac{1}{4}\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix},$$

since C_2 and $C_2 - A_0$ have positive diagonals and determinant $1/16$. The scalar systems of e_1 and e_2 have phases $1 - (\tau_\circ + 1)\omega$ and $1 - (\tau_\circ - 1)\omega$, so by Theorem 9.4 the diagonal entries of $\operatorname{Im} F$ are at least $A + r/x$ and A, the diagonal of $\operatorname{Im} F_{C_1}$. If $\operatorname{Im} F \preceq \operatorname{Im} F_{C_1}$, the difference is positive semidefinite with zero diagonal, hence zero, and $\operatorname{Im} F = \operatorname{Im} F_{C_1}$. But $\operatorname{Im} F_{C_2} - \operatorname{Im} F_{C_1} = (r/x)(C_2 - C_1)$, and $C_2 - C_1$ has the eigenvalues $3/4$ and $-1/4$. □

12.5. Several bands

Let $0 < a_1 < b_1 < a_2 < \cdots < a_m < b_m$, choose $c_j \in (b_j, a_{j+1})$ in each gap, and let R_m be the branch of $\big(\prod_j (z^2 - a_j^2)(z^2 - b_j^2)\big)^{1/2}$ in the complement of the bands $\pm[a_j, b_j]$ with $R_m \sim z^{2m}$ at infinity. Put

$$p(z) = \prod_{j<m} (z^2 - c_j^2), \qquad D_m = \frac{R_m}{p}, \qquad Q_m = \frac{D_m}{z}.$$

On the j-th positive band $R_m(x + i0)/i$ and $p(x)$ are real with the sign $(-1)^{m-j}$, so $D_m(x + i0) = i\rho(x)$ with $\rho > 0$ on every band.

Lemma 12.15. *Q_m is an odd Herglotz function with linear coefficient 1, the atom $\nu_0 = -D_m(0) = \prod_j a_j b_j / \prod_j c_j^2$ at 0, the atoms $\nu_j = -\operatorname{Res}_{c_j} Q_m > 0$ at $\pm c_j$, the density $\rho(x)/(\pi x)$ on the bands, and no other measure.*

Proof. At c_j the residue is $R_m(c_j)/(2c_j^2 \prod_{l \neq j}(c_j^2 - c_l^2))$; its numerator has the sign $(-1)^{m-j}$ and its denominator $(-1)^{m-1-j}$. The sign of $D_m(0)$ is $(-1)^m(-1)^{m-1} = -1$. Subtracting from Q_m the corresponding terms of (4.4) leaves an odd function without jumps across the bands, bounded near their ends and $O(z^{-1})$ at infinity, hence zero, as in the proof of Lemma 9.1. □

For Hermitian $G_j \in C^1([a_j, b_j])$ define

$$F_G(z) = \frac{2zD_m(z)}{\pi} \sum_j \int_{a_j}^{b_j} \frac{G_j(t)\,\mathrm{d}t}{\rho(t)\,(z^2 - t^2)},$$

with linear coefficient B_G, gap residues $-M_j$ at c_j, and band density $\pi^{-1}V_G$, $V_G(x) = \operatorname{Im} F_G(x + i0)$, all explicit Cauchy integrals.

Theorem 12.16. *The number*

$$c_* = \max\Big\{0, -\lambda_{\min}(B_G), \max_j \frac{-\lambda_{\min}(M_j)}{\nu_j}, \sup \lambda_{\max}\Big(-\frac{xV_G(x)}{\rho(x)}\Big)\Big\},$$

the supremum over the open bands, is finite, and for $c \geq c_*$, $F = F_G + cQ_mI$ is a generator on every band with $\operatorname{Re} F = G_j$ on the j-th band; for $c < c_*$ it is not. For $n = 1$, e^{iF} is a zero-free passive system with phase G_j on each band.

Proof. By the Plemelj formula, as in Proposition 8.4(ii), $\operatorname{Re} F_G = G_j$ on the j-th band. There $\rho = r_j L_j$ with $r_j = \sqrt{(t^2 - a_j^2)(b_j^2 - t^2)}$ and $L_j > 0$ smooth on $[a_j, b_j]$; in $\sigma = t^2$ the own-band density is C^1, and subtracting its value at $\sigma = x^2$ with Glauert's integral bounds V_G/ρ up to the ends. The other bands contribute smooth terms. So the supremum is finite. As in Lemma 12.15, F_G has the representation (4.4) with Hermitian coefficients B_G, M_j at $\pm c_j$, $\pi^{-1}V_G$ on the bands, and nothing at 0. Adding cQ_mI gives $B_G + cI$, $c\nu_0 I$, $M_j + c\nu_j I$ and $\pi^{-1}(V_G + c\rho I/x)$, all positive semidefinite exactly when $c \geq c_*$; the coefficients of a Herglotz representation are unique. □

So every smooth phase on finitely many bands, scalar or matrix, can be carried by a passive system, at the price of resonances at 0 and in the gaps. The theorem minimises only within the family $F_G + cQ_mI$; the least loss over all measures on several bands is not known.

13. WAVES IN PASSIVE MEDIA

13.1. A slab of a passive medium

Consider a plane wave crossing a slab of thickness L of a linear, homogeneous, isotropic and temporally dispersive medium with relative permittivity $\epsilon(\omega)$ and permeability $\mu_r(\omega)$. We consider the propagation factor of a single pass and ignore reflections at the faces, which is exact for an impedance-matched slab and otherwise describes the field inside the medium. In our convention the factor is

$$H(\omega) = \exp\Big(\frac{i\omega n(\omega) L}{c}\Big), \qquad n = \sqrt{\epsilon\mu_r} = n_r + i n_i.$$

Passivity of the medium means that $z\epsilon(z)$ and $z\mu_r(z)$ are Herglotz functions and that ϵ and μ_r tend to 1 as $z \to i\infty$ [11, 17, 18]. Reality of the fields makes ϵ and μ_r real on $i\mathbb{R}_+$, so that $z\epsilon(z)$ and $z\mu_r(z)$ lie on the closed positive imaginary axis there.[34]

Lemma 13.1. *$h(z) = zn(z)L/c$ is a Herglotz function with $h(iy) \in i\mathbb{R}$ and $h(-\bar{z}) = -\overline{h(z)}$, and its linear coefficient in (4.4) is $\beta = L/c$. Hence $H = e^{ih}$ is a zero-free passive system with $\sigma = 1$, phase $\varphi = \omega n_r L/c$ and attenuation $\alpha = \omega n_i L/c$.*

---

[34] Analyticity in $\mathbb{C}_+$ is causality of the constitutive relations; $\operatorname{Im}(\omega\epsilon(\omega)) \geq 0$ for real ω is the absence of gain; the limit expresses that every medium is transparent at very high frequency. The Herglotz property in $\mathbb{C}_+$ follows from these boundary properties by the Poisson representation of the harmonic function $\operatorname{Im}(z\epsilon(z))$.

Proof. Put $f(z) = z\epsilon(z)$ and $g(z) = z\mu_r(z)$, so that $zn(z) = \sqrt{f(z)g(z)}$. For $z \in \mathbb{C}_+$, $\arg f$ and $\arg g$ lie in $[0,\pi]$; we take the square root with argument $\frac{1}{2}(\arg f + \arg g) \in [0,\pi]$, which is analytic and is the branch with $n \to 1$. So $zn(z)$ maps $\mathbb{C}_+$ to the closed upper half-plane. On $i\mathbb{R}_+$, f and g lie on $i\mathbb{R}_+$, so their product is negative and zn lies on $i\mathbb{R}_+$. The symmetry follows from that of f and g, and $\beta = \lim h(iy)/(iy) = L/c$ because $n(iy) \to 1$. □

The coefficient $\beta = L/c$ is the vacuum delay: no signal crosses the slab faster than light. It drops out of the bound, as every delay does.

Theorem 13.2. *Let the medium be passive, with n continuous and nonzero on a neighbourhood of $[a,b]$ in $\overline{\mathbb{C}_+}$, and suppose that on the band*

$$n_r(\omega) = n_0 + \frac{C}{\omega}, \qquad a \le \omega \le b, \tag{13.1}$$

with constants $n_0 \in \mathbb{R}$ and $C \ge 0$. Then

$$\frac{1}{b-a}\int_a^b \omega\, n_i(\omega)\,\mathrm{d}\omega \;\ge\; C\,\Lambda, \tag{13.2}$$

with equality if and only if the Herglotz measure of $zn(z)$ is carried by S.

Proof. By Lemma 13.1, H is zero-free, passive and regular on the band, with phase $\varphi = \omega n_r L/c = CL/c + (n_0 L/c)\,\omega$. Theorem 6.2 with $\theta = CL/c$ and $\tau = n_0 L/c$ gives $\frac{1}{b-a}\int \omega n_i L/c \ge (CL/c)\Lambda$, and L/c cancels. □

The thickness of the slab has dropped out: (13.2) is a property of the material. The condition (13.1) says that the phase index exceeds the group index $n_g = \mathrm{d}(\omega n_r)/\mathrm{d}\omega = n_0$ by C/ω, or equivalently that $\mathrm{d}n_r/\mathrm{d}\omega = -C/\omega^2 < 0$. So Theorem 13.2 is a quantitative form of the old observation that anomalous dispersion requires absorption [8, 9, 11]: an index falling like C/ω across a band requires the frequency-weighted extinction ωn_i to be at least $C\Lambda$ on average. In terms of the intensity absorption coefficient $a_{\mathrm{abs}} = 2\omega n_i/c$ the bound reads $\overline{a_{\mathrm{abs}}} \ge 2C\Lambda/c$.

13.2. Approximate constant phase

The sum rule is linear in the phase, so the bound is stable under perturbations. Put $P = \int_a^b |p(\omega)|\,\mathrm{d}\omega$.

Corollary 13.3. *Let H be zero-free, passive and regular on the band, and suppose $|\varphi(\omega) - \theta - \tau\omega| \le \varepsilon$ on $[a,b]$. Then*

$$\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega \;\ge\; \theta\Lambda - \frac{P}{2(b-a)}\,\varepsilon.$$

In particular a zero-free system that is lossless on the band cannot approximate a constant phase lag $\theta > 0$ to within $\varepsilon < \theta J/P$.

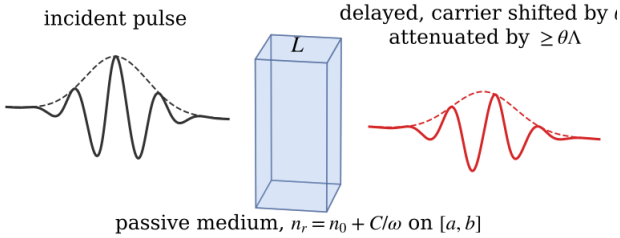


FIG. 17. Schematic of Theorem 13.2. A pulse crosses a slab of a passive medium whose index falls like $n_0 + C/\omega$ on the band. The envelope is delayed by $n_0 L/c$, the carrier is shifted under it by $\theta = CL/c$, and the pulse leaves attenuated (the factor is exaggerated). By Theorem 13.2 the mean of ωn_i over the band is at least $C\Lambda$, whatever the thickness L.

Proof. By Theorem 6.1 and Lemma 5.3(i), $\int_a^b \alpha \ge \frac{1}{2}\int_a^b \varphi p = \frac{1}{2}\theta J + \frac{1}{2}\int_a^b (\varphi - \theta - \tau\omega)p$, and the last term is at least $-\frac{1}{2}\varepsilon P$. If $\alpha = 0$ on the band, then $0 \ge \frac{1}{2}(\theta J - \varepsilon P)$. □

For an octave $J/P = 0.2584$, so a lossless zero-free system cannot hold its phase within $0.258\,\theta$ of any $\theta + \tau\omega$ with $\theta > 0$ across the octave; for the decade $[1,10]$ the ratio is 0.617. The ratio J/P does not depend on the scale of the band.

Remark 13.4 (Broadband phase shifters). Broadband "90° phase shifters" are pairs of lossless all-pass networks whose phases differ by approximately $\pi/2$ across a band [15]. Each phase is increasing (Proposition 4.5); only their difference is nearly constant. This is consistent with Corollary 13.3: what the theorem forbids is a lossless network whose own phase is nearly constant on the band, and a pair of networks never needs one. A device that shifts a single signal by θ without a reference channel must lose transmitted power.

13.3. What the bound means for energy

On the band the power transmission is $T(\omega) = |H(\omega)|^2 = e^{-2\alpha(\omega)}$.

Proposition 13.5. *Let H be as in Theorem 6.2 with $\theta > 0$.*

(i) There is $\omega_1 \in [a,b]$ with $T(\omega_1) \le e^{-2\theta\Lambda}$, and for every $\varepsilon > 0$ there is a signal with spectrum in $[-b,-a]\cup[a,b]$ that loses at least the fraction $1 - e^{-2\theta\Lambda} - \varepsilon$ of its energy.

(ii) The geometric mean of T over the band is at most $e^{-2\theta\Lambda}$.

(iii) In a quantised description, a single quantum of frequency ω_1 fails to be transmitted with probability at least $1-e^{-2\theta\Lambda}$, and the mean energy it removes from the transmitted channel is at least $(1-e^{-2\theta\Lambda})\hbar\omega_1$.

(iv) For a narrow band of relative width $\Delta f/f$ and θ in degrees, $1 - e^{-2\theta\Lambda} = \frac{\pi^2}{720}\,\theta\,\frac{\Delta f}{f}\,\big(1 + O(\theta\Delta f/f) + O((\Delta f/f)^2)\big)$.

Proof. (i) The mean of the continuous function α is at least $\theta\Lambda$, so $\alpha(\omega_1) \ge \theta\Lambda$ at some ω_1. A signal whose spectrum is concentrated near $\pm\omega_1$ transmits a fraction of its energy arbitrarily close to $T(\omega_1)$ by Parseval's identity and the continuity of T. (ii) is Corollary 6.7. (iii) A linear lossy element acts on a single quantum as a beam splitter coupling the transmitted mode to the other modes of the device; the quantum is transmitted with probability $T(\omega_1)$. (iv) By (7.3), $\Lambda = \frac{\pi}{8}\frac{\Delta f}{f}(1 + O((\Delta f/f)^2))$, and θ degrees are $\pi\theta/180$ radians, so $2\theta\Lambda = \frac{\pi^2}{720}\theta\frac{\Delta f}{f}(1+\cdots)$; finally $1 - e^{-x} = x(1 + O(x))$. □

The bound is on transmission. The power that is not transmitted may be absorbed, reflected or routed to other ports, and Proposition 13.5 does not say which: a scalar transfer function sees only the output channel. For the impedance-matched slab of Theorem 13.2, which reflects nothing and has no other port, all of it is absorbed. The energy that is lost is not the "energy of the phase shift". A phase factor is energy neutral (Proposition 3.1). The loss is the price that causality and passivity attach to producing that factor on a band.

13.4. The 5 MHz example done correctly

Take the example of [1]: a 1° phase shift of a wave at 5 MHz, and give it a band, 4.5 to 5.5 MHz, so $\Delta f/f = 0.2$. Then $\kappa = 0.1$ and

$$\Lambda = 0.078836,$$
$$\theta\Lambda = \frac{\pi}{180}\Lambda = 1.3760 \times 10^{-3}\,\mathrm{Np} = 0.011951\,\mathrm{dB}.$$

A passive device that shifts the carrier–envelope phase by 1° across this band therefore fails to transmit at least 0.275% of the power at its worst frequency, $1 - e^{-2\theta\Lambda} = 2.748 \times 10^{-3}$, against 2.742×10^{-3} from Proposition 13.5(iv). A quantum of $hf = 3.313 \times 10^{-27}$ J at that frequency removes from the output on average at least 9.10×10^{-30} J. This happens to be close to the value $hf/360 = 9.20 \times 10^{-30}$ J that a phase slipping by 1° per period would remove (Proposition 3.2), but the agreement is accidental: the ratio of the two numbers is 0.989, close to $\frac{\pi^2}{2}\frac{\Delta f}{f} = 0.987$ by Proposition 13.5(iv), and it tends to zero with the bandwidth. For a single sinusoid, $\Delta f = 0$ and a delay of 555.6 ps does the job without loss.

14. DISCUSSION AND OPEN PROBLEMS

14.1. What has been shown

A passive system can delay a band for free, but it cannot shift the phase of the whole band by a constant without loss. The least mean attenuation is $\theta\Lambda$, with $\Lambda = (K(\kappa) - E(\kappa))/\kappa$, and it is attained. The proof is one sum rule, (6.1), built from the test function W; the lead bound, the worst-frequency and geometric-mean bounds, the bound for anomalous dispersion and the stability under perturbations of the phase follow from it or from the extremal system.

For an affine phase every zero-free system is the extremal function plus terms that each raise the attenuation at every point of the band (Theorem 9.2), so one system is optimal at every frequency at once. With zeros the sum rule is lost, but after the change of variables of Lemma 11.3 a comparison argument gives the same kind of pointwise optimum, attained by a system with one zero, whose profile is elliptic (Theorem 11.13). At a right angle the system is algebraic and $\left|H(\sqrt{ab})\right|^2 \le a/b$.

With a delay the picture changes. Without zeros the least mean loss is known exactly for every slope and stays above a positive floor. With zeros the optimum exists, is unique and is characterised by the inequality of Theorem 12.9(ii), and its loss decays exponentially with the delay.

14.2. Open problems

(1) *The optimum for a delay with zeros.* For $\arg H \equiv \theta + \tau\omega$ with $\tau > 0$, find $H_{\theta+\tau\omega}$ of Theorem 12.9, or at least the exact exponent in Corollary 12.12; we only know that it lies between $-\pi ab/(b-a)$ and $-a/e$. The change of variables of Lemma 11.3 needs a constant phase.

(2) *General phase without zeros.* Corollary 9.3 is sharp whenever $h_g + \max\{0, -\beta_g\}R/z$ is a Herglotz function. For an affine phase Theorem 12.2 finds the cheapest measure off the band in every case; for a phase that is not affine and for which v_g changes sign it is not known.

(3) *Several channels.* Theorem 12.13 is sharp for $\Theta \succeq 0$. For indefinite Θ the least trace is not known,

and by Theorem 12.14 there is in general no least attenuation in the Loewner order.

(4) *Several bands.* Theorem 12.16 decides when a given compensation works. The sharp constant, a period of a hyperelliptic differential in place of Λ, has not been computed.

(5) *Reflection and matching.* The Fano bound [14] and the present bound constrain different parts of a reflection coefficient. A joint bound on its modulus and argument over a band would combine them.

(6) *Formalisation.* The finite arithmetic is checked in Lean (Appendix D). The analytic part, which needs boundary values of Cauchy integrals, Hardy spaces and the complete elliptic integrals, is not machine-checked.

Appendix A: Complete elliptic integrals

This appendix derives every property of K and E used in the paper. The modulus is $k \in [0,1)$ and the complementary modulus is $k' = \sqrt{1-k^2}$; we write $K' = K(k')$ and $E' = E(k')$.

1. Power series

For $|u|<1$, $(1-u)^{-1/2} = \sum_{n\ge 0} c_n u^n$ and $(1-u)^{1/2} = \sum_{n\ge 0} c_n u^n/(1-2n)$, with $c_n = \binom{2n}{n}4^{-n}$; the first is the binomial series, $\binom{-1/2}{n}(-1)^n = c_n$, and the second follows by integrating the first term by term. The Wallis integral $\frac{2}{\pi}\int_0^{\pi/2}\sin^{2n}\phi\,\mathrm{d}\phi = c_n$ follows by induction from the reduction formula $\int_0^{\pi/2}\sin^{2n} = \frac{2n-1}{2n}\int_0^{\pi/2}\sin^{2n-2}$ and the recurrence $c_n = \frac{2n-1}{2n}c_{n-1}$. With $u = k^2\sin^2\phi$ and uniform convergence for $k<1$,

$$K(k) = \frac{\pi}{2}\sum_{n\ge 0} c_n^2 k^{2n}, \qquad E(k) = \frac{\pi}{2}\sum_{n\ge 0}\frac{c_n^2}{1-2n}k^{2n}. \tag{A.1}$$

Subtracting, $K-E = \frac{\pi}{2}\sum_{n\ge 1} c_n^2\frac{2n}{2n-1}k^{2n}$, which is the series of Proposition 7.2(i).[35]

2. Derivatives

Differentiating under the integral sign,

$$\frac{\mathrm{d}E}{\mathrm{d}k} = \frac{E-K}{k}, \qquad \frac{\mathrm{d}K}{\mathrm{d}k} = \frac{E-k'^2K}{k\,k'^2}. \tag{A.2}$$

The first is immediate: $\frac{\mathrm{d}}{\mathrm{d}k}\sqrt{1-k^2\sin^2\phi} = -k\sin^2\phi/\sqrt{\cdot}$, and $-k^2\sin^2\phi = (1-k^2\sin^2\phi)-1$. For the second, the derivative is $\int_0^{\pi/2} k\sin^2\phi\,(1-k^2\sin^2\phi)^{-3/2}\,\mathrm{d}\phi$, and the identity

$$\frac{\mathrm{d}}{\mathrm{d}\phi}\frac{\sin\phi\cos\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{1}{k^2}\Big(\sqrt{1-k^2\sin^2\phi} - \frac{k'^2}{(1-k^2\sin^2\phi)^{3/2}}\Big),$$

integrated over $[0,\pi/2]$, gives $\int(1-k^2\sin^2)^{-3/2} = E/k'^2$; then $k^2\sin^2\phi(1-k^2\sin^2\phi)^{-3/2} = (1-k^2\sin^2)^{-3/2} - (1-k^2\sin^2)^{-1/2}$ yields (A.2).[36]

3. Legendre's relation

Lemma A.1. *$E(k)K' + E'K(k) - K(k)K' = \pi/2$ for $0<k<1$.*

Proof. Let $L(k)$ be the left side. Since $\mathrm{d}k'/\mathrm{d}k = -k/k'$, (A.2) applied at k' gives

$$\frac{\mathrm{d}E'}{\mathrm{d}k} = \frac{k(K'-E')}{k'^2}, \qquad \frac{\mathrm{d}K'}{\mathrm{d}k} = -\frac{E'-k^2K'}{kk'^2}.$$

Multiply $\mathrm{d}L/\mathrm{d}k$ by kk'^2 and insert the four derivatives:

$$\begin{aligned} kk'^2L' &= k'^2(E-K)K' - E(E'-k^2K') + k^2K(K'-E') \\ &\quad + E'(E-k'^2K) - K'(E-k'^2K) + K(E'-k^2K'). \end{aligned}$$

Expanding, the terms in KK' and in EE' cancel in pairs, and what remains is

$$kk'^2L' = (k^2+k'^2-1)\,(EK'-KE') = 0.$$

So L is constant. As $k\to 0$, $K,E\to\pi/2$ and $E'\to 1$, while $K-E = O(k^2)$ by (A.1) and $K' = O(\log(1/k))$ by (A.3) below; hence $L = E'K + (E-K)K' \to \pi/2$. □

4. The arithmetic–geometric mean and Landen's transformation

For $p,q>0$ let

$$I(p,q) = \int_0^{\pi/2}\frac{\mathrm{d}\phi}{\sqrt{p^2\cos^2\phi + q^2\sin^2\phi}},$$

so that $K(k) = I(1,k')$.

[35] The coefficients $d_n = c_n^2\,2n/(2n-1)$ are $\frac{1}{2}, \frac{3}{16}, \frac{15}{128}, \frac{175}{2048}, \dots$; the ratio $d_{n+1}/d_n = (2n+1)(2n-1)/((2n+2)2n)$ follows from $c_{n+1}/c_n = (2n+1)/(2n+2)$.

[36] To check the displayed identity, put $s=\sin\phi$, $c=\cos\phi$, $\Delta = 1-k^2s^2$. The derivative of $sc\,\Delta^{-1/2}$ is $(c^2-s^2)\Delta^{-1/2} + k^2s^2c^2\Delta^{-3/2} = \Delta^{-3/2}\big((c^2-s^2)\Delta + k^2s^2c^2\big)$, and with $c^2 = 1-s^2$ the bracket is $1-2s^2+k^2s^4 = \frac{1}{k^2}\big(\Delta^2 - k'^2\big)$.

Lemma A.2 (Gauss). *$I(p,q) = I\big(\frac{p+q}{2}, \sqrt{pq}\big)$. Consequently $I(p,q) = \pi/(2M(p,q))$, with M the arithmetic–geometric mean of p and q.*

Proof. With $x = q\tan\phi$, $I(p,q) = \frac12\int_{\mathbb R}\big((p^2+x^2)(q^2+x^2)\big)^{-1/2}\,\mathrm dx$. Substitute $x = \frac12(t - pq/t)$, $t>0$, which maps $(0,\infty)$ increasingly onto $\mathbb R$, with $\mathrm dx = \frac12(1+pq/t^2)\,\mathrm dt$. Then $4x^2+(p+q)^2 = (t^2+p^2)(t^2+q^2)/t^2$ and $x^2+pq = \frac14(t+pq/t)^2$, so

$$\big((p_1^2+x^2)(q_1^2+x^2)\big)^{-1/2}\,\mathrm dx = \frac{2\,\mathrm dt}{\sqrt{(t^2+p^2)(t^2+q^2)}},$$

with $p_1 = \frac{p+q}{2}$, $q_1 = \sqrt{pq}$. Integrating over $t\in(0,\infty)$ and using that the integrand is even in t,

$$\begin{aligned} 2I(p_1,q_1) &= 2\int_0^\infty \frac{\mathrm dt}{\sqrt{(t^2+p^2)(t^2+q^2)}} \\ &= \int_{\mathbb R}\frac{\mathrm dt}{\sqrt{(t^2+p^2)(t^2+q^2)}} = 2I(p,q). \end{aligned}$$

The iterates converge to $M(p,q)$, and $I(M,M) = \pi/(2M)$. □

Proposition A.3 (Landen). *Let $0<k'<1$ and $\kappa = (1-k')/(1+k')$. Then*

$$K(k) = (1+\kappa)K(\kappa), \qquad E(k) = (1+k')E(\kappa) - k'K(k).$$

Proof. By Lemma A.2, $K(k) = I(1,k') = I\big(\frac{1+k'}{2}, \sqrt{k'}\big) = \frac{2}{1+k'}I\big(1, \frac{2\sqrt{k'}}{1+k'}\big)$, and $\frac{2\sqrt{k'}}{1+k'} = \sqrt{1-\kappa^2} = \kappa'$, so $K(k) = \frac{2}{1+k'}K(\kappa) = (1+\kappa)K(\kappa)$. For E, regard k as a function of κ, $k = 2\sqrt\kappa/(1+\kappa)$, $k' = (1-\kappa)/(1+\kappa)$, and differentiate the relation for K with (A.2). Since

$$\frac{\mathrm dk}{\mathrm d\kappa} = \frac{1-\kappa}{\sqrt\kappa(1+\kappa)^2}, \qquad kk'^2 = \frac{2\sqrt\kappa(1-\kappa)^2}{(1+\kappa)^3},$$
$$\text{so} \qquad \frac{\mathrm dk/\,\mathrm d\kappa}{kk'^2} = \frac{1+\kappa}{2\kappa(1-\kappa)},$$

the derivative of $K(k)$ is $\big(E(k) - k'^2K(k)\big)\frac{1+\kappa}{2\kappa(1-\kappa)}$, and that of $(1+\kappa)K(\kappa)$ is $K(\kappa) + \big(E(\kappa) - (1-\kappa^2)K(\kappa)\big)/(\kappa(1-\kappa))$. Equating, multiplying by $2\kappa(1-\kappa)/(1+\kappa)$ and using $k'^2(1+\kappa) = (1-\kappa)^2/(1+\kappa)$,

$$\begin{aligned} E(k) &= \frac{2E(\kappa)}{1+\kappa} + \frac{K(\kappa)}{1+\kappa}\Big((1-\kappa)^2 + 2\kappa(1-\kappa) \\ &\qquad - 2(1-\kappa)(1+\kappa)\Big) \\ &= \frac{2E(\kappa)}{1+\kappa} - (1-\kappa)K(\kappa). \end{aligned}$$

Since $2/(1+\kappa) = 1+k'$ and $(1-\kappa)K(\kappa) = k'K(k)$, this is the claim. □

5. Behaviour at the ends

As $k\to0$, (A.1) gives $K, E = \frac\pi2 + O(k^2)$. As $k\to1$, $E(k)\to1$ and

$$K(k) = \log\frac{4}{k'} + O\big(k'^2\log k'\big). \tag{A.3}$$

The limit of E is dominated convergence. For K, the substitution $x = k'\tan\phi$ of Lemma A.2 gives $K(k) = \int_0^\infty\big((1+x^2)(k'^2+x^2)\big)^{-1/2}\,\mathrm dx$. Split at $x=1$:

$$\int_0^1\frac{\mathrm dx}{\sqrt{k'^2+x^2}} = \operatorname{arsinh}\frac{1}{k'} = \log\frac{2}{k'} + o(1),$$

and, by dominated convergence,

$$\int_0^1\frac{(1+x^2)^{-1/2}-1}{\sqrt{k'^2+x^2}}\,\mathrm dx \to \log\frac{2}{1+\sqrt2},$$
$$\int_1^\infty\frac{\mathrm dx}{\sqrt{(1+x^2)(k'^2+x^2)}} \to \log(1+\sqrt2),$$

the first limit being $\int_0^1\big(\frac{1}{x\sqrt{1+x^2}} - \frac1x\big)\,\mathrm dx = \log\frac{2}{1+\sqrt2}$, both from the primitive $-\operatorname{arsinh}(1/x)$ of $1/(x\sqrt{1+x^2})$. The three limits add up to $\log(4/k')$. The error term is [32, 19.12.1].

Appendix B: The elementary integrals

1. Moments of the arcsine weight

For $A<B$ put $m = \frac{A+B}{2}$, $d = \frac{B-A}{2}$ and $\sigma = m - d\cos\phi$, $0\le\phi\le\pi$. Then $(\sigma-A)(B-\sigma) = d^2\sin^2\phi$ and $\mathrm d\sigma = d\sin\phi\,\mathrm d\phi$, so $\mathrm d\sigma/\sqrt{(\sigma-A)(B-\sigma)} = \mathrm d\phi$, and

$$\begin{aligned} \int_A^B\frac{\mathrm d\sigma}{\sqrt{(\sigma-A)(B-\sigma)}} &= \pi, \\ \int_A^B\frac{\sigma\,\mathrm d\sigma}{\sqrt{\cdot}} &= \pi m = \frac\pi2(A+B), \\ \int_A^B\frac{\sigma^2\,\mathrm d\sigma}{\sqrt{\cdot}} &= \pi\Big(m^2+\frac{d^2}{2}\Big) \\ &= \frac\pi8\big(3A^2+2AB+3B^2\big), \\ \int_A^B\frac{\mathrm d\sigma}{\sigma\sqrt{\cdot}} &= \int_0^\pi\frac{\mathrm d\phi}{m-d\cos\phi} \\ &= \frac{\pi}{\sqrt{m^2-d^2}} = \frac{\pi}{\sqrt{AB}}. \end{aligned}$$

The last integral is the classical $\int_0^\pi\mathrm d\phi/(m-d\cos\phi) = \pi/\sqrt{m^2-d^2}$ for $m>|d|$, which follows from $u = \tan(\phi/2)$.[37]

[37] With $u=\tan(\phi/2)$, $\cos\phi = (1-u^2)/(1+u^2)$ and $\mathrm d\phi = 2\,\mathrm du/(1+u^2)$, so the integral is $\int_0^\infty 2\,\mathrm du/\big((m-d)+(m+d)u^2\big) = \pi/\sqrt{(m-d)(m+d)}$.

These give Lemma 5.3(iii),(iv): with $x\,p(x)\,\mathrm{d}x = \frac12\varpi(\sigma)\,\mathrm{d}\sigma$ and $p(x)\,\mathrm{d}x/x = \frac12\varpi(\sigma)\,\mathrm{d}\sigma/\sigma$,

$$\begin{aligned}\int_a^b \frac{p}{x}\,\mathrm{d}x &= \frac12\Big((A+B)\frac{\pi}{\sqrt{AB}} - 2\pi\Big)\\ &= \frac{\pi(b-a)^2}{2ab},\\ \int_a^b x^3 p\,\mathrm{d}x &= \frac12\Big(\frac{\pi}{2}(A+B)^2\\ &\qquad - \frac{\pi}{4}(3A^2+2AB+3B^2)\Big)\\ &= -\frac{\pi}{8}(B-A)^2.\end{aligned}$$

2. The integrals I_0, I_2 and J

With $t^2 = b^2 - (b^2-a^2)\sin^2\phi$ (Proposition 7.1), $t\,\mathrm{d}t = -(b^2-a^2)\sin\phi\cos\phi\,\mathrm{d}\phi$ and $r(t) = (b^2-a^2)\sin\phi\cos\phi$, so $\mathrm{d}t/r = -\,\mathrm{d}\phi/t$ and

$$\begin{aligned}I_0 &= \int_0^{\pi/2} \frac{\mathrm{d}\phi}{b\sqrt{1-k^2\sin^2\phi}} = \frac{K(k)}{b},\\ I_2 &= \int_0^{\pi/2} b\sqrt{1-k^2\sin^2\phi}\,\mathrm{d}\phi = bE(k).\end{aligned}$$

Hence $J = (a^2+b^2)I_0 - 2I_2 = b\big((1+k'^2)K(k) - 2E(k)\big)$. By Proposition A.3, $-2E(k) = -2(1+k')E(\kappa) + 2k'K(k)$, and $(1+k'^2)+2k' = (1+k')^2$, so

$$\begin{aligned}J &= b(1+k')^2K(k) - 2b(1+k')E(\kappa)\\ &= b(1+k')^2(1+\kappa)K(\kappa) - 2b(1+k')E(\kappa).\end{aligned}$$

With $1+k' = (a+b)/b$ and $1+\kappa = 2b/(a+b)$ the coefficients are $2(a+b)$ and $2(a+b)$, which is (7.1).

3. The integrals on $(0,a)$

With $s = a\sin\phi$,

$$\sqrt{(a^2-s^2)(b^2-s^2)} = a\cos\phi\, b\sqrt{1-k'^2\sin^2\phi},$$

so

$$\begin{aligned}\int_0^a \frac{\mathrm{d}s}{\sqrt{(a^2-s^2)(b^2-s^2)}} &= \frac{K(k')}{b},\\ \int_0^a \frac{s^2\,\mathrm{d}s}{\sqrt{(a^2-s^2)(b^2-s^2)}} &= b\big(K(k')-E(k')\big),\end{aligned}$$

the second because $a^2\sin^2\phi = b^2k'^2\sin^2\phi$ and

$$\int_0^{\pi/2} \frac{k'^2\sin^2\phi\;\mathrm{d}\phi}{\sqrt{1-k'^2\sin^2\phi}} = K(k') - E(k').$$

These are the integrals in the proof of Theorem 8.2(ii), where

$$\frac{m^2}{b}K' - b(K'-E') = \frac{b}{K}\big(EK' - KK' + E'K\big) = \frac{\pi b}{2K}$$

by $m^2 = b^2E/K$ and Lemma A.1.

Appendix C: The extremal profile in closed form

1. Three formulas for $\alpha_\star$

Lemma C.1. *For $a<x<b$,*

$$\begin{aligned}\alpha_\star(x) &= \tau_\star\int_a^x \frac{m^2-s^2}{r(s)}\,\mathrm{d}s\\ &= \frac{2x\,r(x)\theta}{\pi}\,\mathrm{p.v.}\int_a^b \frac{\mathrm{d}t}{r(t)(x^2-t^2)}\\ &= \frac{2\theta\,r(x)}{\pi}\int_a^b \frac{\mathrm{d}t}{r(t)(x+t)}.\end{aligned}$$

The last form shows at once that $\alpha_\star>0$ on (a,b) and that it vanishes like $\sqrt{x-a}$ and $\sqrt{b-x}$ at the band edges.

Proof. The first equality is Theorem 8.2(ii), the second Proposition 8.4(ii),(iv) with $g=\theta$. For the third, apply Cauchy's formula to $1/R$, which is analytic off S, $O(z^{-2})$ at infinity, and has the jumps $1/R(t+i0) - 1/R(t-i0) = -2i/r(t)$ on (a,b) and $+2i/r(t)$ on $(-b,-a)$ by Lemma 8.1. Then

$$\frac{1}{R(z)} = \frac{1}{\pi}\int_a^b \frac{\mathrm{d}t}{r(t)}\Big(\frac{1}{z-t} - \frac{1}{z+t}\Big).$$

At $z = x+i0$ the left side is $-i/r(x)$, purely imaginary, so the real part of the right side vanishes: $\mathrm{p.v.}\int_a^b \frac{\mathrm{d}t}{r(t)(x-t)} = \int_a^b \frac{\mathrm{d}t}{r(t)(x+t)}$. Since $\frac{2x}{x^2-t^2} = \frac{1}{x-t} + \frac{1}{x+t}$, the second form equals the third. □

2. The mean of $\alpha_\star$

By Fubini's theorem applied to the first form,

$$\begin{aligned}\int_a^b \alpha_\star\,\mathrm{d}x &= \tau_\star\int_a^b \frac{(m^2-s^2)(b-s)}{r(s)}\,\mathrm{d}s\\ &= \tau_\star\big(b(m^2I_0 - I_2) - m^2I_1 + I_3\big),\end{aligned}$$

with $I_j = \int_a^b s^j\,\mathrm{d}s/r(s)$. The first bracket vanishes by the choice of m. With $\sigma = s^2$, $I_1 = \frac{\pi}{2}$ and $I_3 = \frac{\pi}{4}(a^2+b^2)$ by Appendix B, so

$$\begin{aligned}\int_a^b \alpha_\star &= \tau_\star\frac{\pi}{4}\big(a^2+b^2-2m^2\big)\\ &= \frac{\theta K(k)}{2b}\big(a^2+b^2\big) - \theta b\,E(k) = \frac{\theta}{2}J,\end{aligned}$$

in agreement with Theorem 6.2.[38]

3. The lead extremal

The system $H_\star(-ab/z)$ of Theorem 6.6 has, in the representation (4.4), linear coefficient 0, measure on S given by its attenuation, and an atom $ab\,\tau_\star$ at the origin (Theorem 6.6, proof). Its phase on the band is $-|\theta|$. The sum rule (6.1), with the atom costing $\frac{\pi}{4}W(0)\,ab\,\tau_\star$ and $W(0)=(b-a)^2/(ab)$, gives

$$\begin{aligned}\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega &= -|\theta|\,\Lambda+\frac{\pi}{4}(b-a)\tau_\star\\ &= -|\theta|\,\frac{K-E}{\kappa}+|\theta|\,\kappa K\\ &= |\theta|\,\frac{E(\kappa)-\kappa'^2K(\kappa)}{\kappa},\end{aligned}$$

where K, E are taken at κ and $\frac{\pi}{4}(b-a)\tau_\star = |\theta|\,(b-a)K(\kappa)/(a+b) = |\theta|\,\kappa K(\kappa)$. This is Remark 6.8.

Appendix D: Verification

No statement of the paper rests on a computation: every result has a proof in the text. Two kinds of checks accompany the source files.

Numerical checks. The script `anc/checks.py` re-evaluates the identities of the paper in high-precision arithmetic: the boundary values of W and the kernel identity (5.3) at sample points; the moments of Lemma 5.3; the three forms of J and the Landen relations; the bounds on Λ and Table II; m^2, Legendre's relation and $\tau_\star$; the three formulas of Lemma C.1 and the mean of $\alpha_\star$; the octave values of Remark 8.3; the lead extremal; the single-zero reduction, its asymptotics and Table III; the ratios J/P of Corollary 13.3; for Section 11, the root $\eta_\star=\sqrt{ab}$ at a right angle, the agreement of the Cauchy integral $v_{g_\sharp}$ with both closed forms (11.5), the integral (11.6), the value at the geometric centre and the symmetry, every entry of Table IV, the positivity of $\alpha^\sharp{}_\theta$ on the band for three band ratios and three phases, the mean of $\alpha^\sharp{}_\theta$ and the pointwise comparison with $\alpha_\star$ at $\theta=1$, the narrow-band ratios after Corollary 11.9 and the nomes and the case of equality of Corollary 11.10; for Section 9, the branch of R, both identities of Lemma 9.1 at nine points for three bands, the positivity of k_t, (9.3) for a single resonance below and above the band, Lemma 9.5 in the plane and on the band, the integral of r/x, the means Λ and Λ_- of the two profiles, the Herglotz property, the linear coefficient and the mean of the compensated optimum of Theorem 9.4, and the octave values $A(m)$, Λ_- and ϑ_c; the closed forms, the floor and the decay bounds of Section 12; and the numbers of Sections 2 and 13. It reports 421 checks passed and none failed.

Lean. The file `lean/PhaseShiftCost.lean` is checked by the kernel of Lean 4 (version 4.34.0, core library only). It contains 87 theorems and no `sorry`. It verifies the dimensional analysis and the arithmetic of Section 2; the coefficients d_n of the series of Λ, their identification with $\binom{2n}{n}^2 4^{-2n}\,2n/(2n-1)$ for $n\le 60$, and their monotonicity for every n; enclosures of Λ/π to nine decimals for six band ratios, from partial sums of the series and the coefficient bound behind the tail estimate; and the polynomial identities used in Sections 5 to 7, 10 and 11 and in the appendices; for Section 11 these are the identities behind (11.4), both forms of (11.5), the symmetry $\omega\mapsto ab/\omega$, the value at $\sqrt{ab}$ and (11.6), and for Section 9 the signs of the kernels, the comparison of Corollary 9.3, the step of Corollary 9.7(iv), the vanishing linear coefficients of the compensated optima and the threshold ϑ_c, and for Section 12 the identities and constants of Proposition 12.5, Theorem 12.11, and Corollary 12.12 and the determinants of Theorem 12.14. Of the 87 theorems, 34 depend on no axiom, two on `propext` only, three on `propext` and `Quot.sound`, and the others on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), through the tactics `grind` and `omega`.

The analytic theorems are not formalised. The sum rule (Theorem 6.1), the identification of Λ with elliptic integrals and with its series, the extremal system, the reconstruction of Theorem 9.2, and the results on zeros rest on boundary values of Cauchy integrals, the M. Riesz theorem, Hardy spaces, conformal maps and extremal length, Landen's transformation and the summation of a geometric series, none of which is formalised in the file. They are proved in the text and in Appendices A to C.

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Data availability. The LaTeX source, the script that re-evaluates every numerical value of the paper (421 checks) and the Lean 4 certificate of its finite and algebraic parts (87 theorems, no `sorry`) are in the repository `https://github.com/creelie/gsc`, release v1.2.0,

[38] Here $\tau_\star\pi/4=\theta K(k)/(2b)$ and $\tau_\star\pi m^2/2=\theta bE(k)$; the last expression is $\frac{\theta}{2}\big((a^2+b^2)I_0-2I_2\big)$.

archived on Zenodo under doi:10.5281/zenodo.23768958. No statement of the paper depends on them. The text is a mix of our own writing and writing produced with AI assistants. Anthropic Claude assisted with exposition, figures, the numerical checks and the Lean certificate, and proposed the proofs of Proposition 12.5 and Theorems 12.9 and 12.11 and Corollary 12.12. OpenAI's ChatGPT found the positive reconstruction of Section 9, the pointwise bounds derived from it and the results credited in Sections 11.7 and 12. The authors are responsible for the text.

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